



DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING  
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# Computational Modeling of Conversational Intention and Persuasion Strategies in Asymmetric Multi-Turn Dialogue

## Dual-Perspective Modeling in Asymmetric Multi-Turn Dialogue

*A Hierarchical Cross-Attention Framework for Strategy Attribution and Commitment Tracking*

|                          |                                                                                                                                                                                         |
|--------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Academic Session:        | Session 2026–2027                                                                                                                                                                       |
| Submission Date:         | September 2026                                                                                                                                                                          |
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### Key Architectural Innovations & Engineering Deliverables

AntiphonyClassifier Dual-Stream Architecture  
Strict 3-Pass Ground Truth (1,017 P4G Dialogues)

11-Category / 41-Strategy Multi-Task Hierarchy  
700-Dialogue Reversal E-Commerce Benchmark

## CANDIDATES' DECLARATION

We declare that the work presented in this report, titled “Computational Modeling of Conversational Intention and Persuasion Strategies in Asymmetric Multi-Turn Dialogue”, is the outcome of the engineering and research work carried out by us under the supervision named on the title page.

All prior work, third-party code, libraries, data, copyrighted material, and tools used in the project have been properly acknowledged. Any use of generative AI tools has been disclosed in accordance with academic integrity guidelines. We accept accountability and responsibility for the accuracy and integrity of this report and for our individual contributions to the project.

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## ACKNOWLEDGEMENT

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We express our profound gratitude to our project supervisor, Dr. Md. Shariful Islam Bhuyan, Associate Professor, Department of CSE, BUET, for continuous guidance, technical scrutiny, and encouragement throughout this capstone project. In particular, the critical feedback and methodological directives regarding task decoupling and label stringency helped streamline our experimental pipeline and improve the analytical rigor of this work.

We extend our sincere thanks to the authors of the *Persuasion for Good* (P4G) corpus [1] for curating and releasing an invaluable conversational benchmark, and to the researchers [2] whose preliminary inquiries into hierarchical persuasion taxonomies informed our multi-label annotation framework.

We acknowledge the maintainers and contributors of the foundational open-source ecosystems that made this computational research possible: PyTorch [3], Hugging Face Transformers [4], scikit-learn, XGBoost [5], and the OPUS-MT translation services [6]. We also acknowledge Kaggle for providing the free-tier cloud GPU infrastructure (NVIDIA Tesla T4 accelerators) upon which all experimental models were trained, evaluated, and ablated.

Finally, we thank the faculty and staff of the Department of Computer Science and Engineering, Bangladesh University of Engineering and Technology (BUET), for providing an intellectually rigorous environment and fostering our academic and professional growth.

# ABSTRACT

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Interpersonal persuasion in multi-turn natural language dialogue is characterized by fundamental communicative asymmetry: one participant acts as an arguer employing rhetorical tactics to induce compliance, while the counterparty acts as a decider evaluating concessions and formulating commitments. Standard computational models frequently treat multi-turn dialogues as symmetric or monolithic sequences, discarding functional role boundaries, suffering from tokenizer truncation, and exploiting superficial lexical heuristics in closing turns. This capstone project presents an end-to-end computational framework for modeling conversational intention and persuasion strategies across asymmetric dialogue domains.

First, we address conversational intention prediction within the *Persuasion for Good* (P4G) charity solicitation corpus. Demonstrating that recorded transaction metadata severely diverges from grounded textual commitments, we conduct three successive human annotation passes, refining the target into a strict unconditional binary commitment formulation ( $N = 1,017$ , 694 yes / 323 no). We document the systematic 14-version evolutionary trajectory of neural intention architectures across four stages. Early flat sequence encoders (256 tokens) suffer from early-window truncation and metric collapse (Macro-F1  $\approx 0.65$ ). Hierarchical dialogue transformers (2,048-token capacity) overcome the length bottleneck, achieving 0.793–0.834 Macro-F1. To explicitly reflect conversational asymmetry, we design the *AntiphonyClassifier* (v14): a speaker-disentangled, dual-stream architecture with directional cross-attention (where the decider stream queries the persuader stream) and a recurrent commitment-trajectory GRU read at the true final decider turn. Antiphony establishes project-peak test Macro-F1 scores of 0.8623 (DeBERTa-v3-base, default threshold) and 0.8600 (tuned threshold) on the second-pass labels, and 0.8446 (RoBERTa-base, tuned threshold) on the strict third-pass labels.

Second, we model persuasion strategies across 10,600 Persuader turns under an 11-category, 41-strategy hierarchical taxonomy. Using a contextual batching pipeline with a four-tier decision protocol and algorithmic validation, we establish an empirical corpus with 48.9% multi-label prevalence (1.679 labels per turn). Through near-duplicate text similarity analysis across distinct annotators, we derive an empirical upper performance ceiling of 0.79 micro-F1. We formulate a multi-task category-first modeling architecture leveraging an ensemble pool of 19 diverse members (fine-tuned transformers, few-shot representations, TF-IDF baselines, and zero-shot cues) trained under Asymmetric Loss. Systematic evaluation of decision combiners demonstrates that a calibrated power-mean blend ( $q = +2.0$ ) achieves superior performance (v18 headline val Macro-F1 0.5597, micro-F1 0.6949; v17 micro-optimal point 0.7062 micro-F1, 0.5642 macro-F1), while an XGBoost stacking layer was systematically tuned but failed to surpass the calibrated blend (0.5508 vs 0.5716).

Third, we investigate cross-domain generalization to conversational e-commerce negotiation by constructing a 700-dialogue benchmark through persona-conditioned LLM synthesis, organic humanization with controlled linguistic noise, and a 100% manual audit. Input ablations reveal that monolithic sequence models rely on distributed merchant concessions rather than decider trajectory tracking. On an adversarially augmented corpus ( $N = 700$ ) containing non-monotonic stance reversals, the localized closing-turn control degrades to 0.9235 Macro-F1, while Antiphony maintains 0.9619 Macro-F1, widening the sufficiency gap from +0.0001 to +0.0384 (+3.84%). Finally, forensic post-mortems resolve subtle tokenization artifacts, proving that left-truncation recovers up to +0.1198 Macro-F1 obscured by default right-truncation.

**Keywords:** Persuasion Strategy, Conversational Intention, Asymmetric Dialogue, Antiphony Architecture, Multi-Label Classification, Adversarial Reversal.

# Contents

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|                                                                                          |            |
|------------------------------------------------------------------------------------------|------------|
| <b>Candidates' Declaration</b>                                                           | <b>i</b>   |
| <b>Acknowledgement</b>                                                                   | <b>ii</b>  |
| <b>Abstract</b>                                                                          | <b>iii</b> |
| <b>1 Problem Identification and Formulation</b>                                          | <b>1</b>   |
| 1.1 Background and Motivation . . . . .                                                  | 1          |
| 1.2 Problem Statement . . . . .                                                          | 2          |
| 1.2.1 Sub-Problem 1: Grounded Conversational Intention Classification . . . . .          | 2          |
| 1.2.2 Sub-Problem 2: Multi-Label Persuasion Strategy and Category Attribution . . . . .  | 2          |
| 1.2.3 Sub-Problem 3: Cross-Domain Transfer and Adversarial Trajectory Tracking . . . . . | 2          |
| 1.2.4 Required Engineering Knowledge Profiles . . . . .                                  | 3          |
| 1.3 Complexity of the Problem . . . . .                                                  | 3          |
| 1.4 Project Objectives and Scope . . . . .                                               | 3          |
| 1.4.1 Project Objectives . . . . .                                                       | 3          |
| 1.4.2 Project Scope and Explicit Exclusions . . . . .                                    | 5          |
| <b>2 Preliminary Research</b>                                                            | <b>6</b>   |
| 2.1 Literature Review . . . . .                                                          | 6          |
| 2.1.1 Persuasion Theory and Rhetorical Frameworks . . . . .                              | 6          |
| 2.1.2 Persuasion Corpora and Conversational Strategy Annotation . . . . .                | 6          |
| 2.1.3 Transformer Encoders and Dialogue Language Models . . . . .                        | 6          |
| 2.1.4 Hierarchical and Role-Aware Dialogue Networks . . . . .                            | 7          |
| 2.1.5 Imbalanced Learning and Loss Formulations . . . . .                                | 7          |
| 2.1.6 Ensembling and Stacking Generalization . . . . .                                   | 7          |
| 2.1.7 Negotiation Benchmarks and Shortcut Learning . . . . .                             | 7          |
| 2.1.8 Synthesis of Research Gaps . . . . .                                               | 7          |
| 2.2 Existing Solutions and Technologies . . . . .                                        | 7          |
| 2.3 Market and User Research . . . . .                                                   | 8          |
| 2.3.1 The Rise of Conversational Social Commerce . . . . .                               | 8          |
| 2.3.2 User Personas and Behavioral Dynamics . . . . .                                    | 9          |
| 2.3.3 Industry Consultations and Field Observations . . . . .                            | 9          |
| 2.4 Feasibility Study . . . . .                                                          | 9          |
| <b>3 Requirements Analysis</b>                                                           | <b>10</b>  |
| 3.1 Project Objectives and Constraints . . . . .                                         | 10         |

|          |                                                                       |           |
|----------|-----------------------------------------------------------------------|-----------|
| 3.1.1    | Stakeholder Analysis and Conflicting Expectations . . . . .           | 10        |
| 3.1.2    | Functional Requirements (FR) . . . . .                                | 10        |
| 3.1.3    | Non-Functional Requirements (NFR) . . . . .                           | 11        |
| 3.1.4    | Physical and Operational Constraints . . . . .                        | 11        |
| 3.2      | Regulatory Requirements and Standards . . . . .                       | 11        |
| 3.3      | Formulation of Solutions . . . . .                                    | 12        |
| 3.3.1    | Multi-Criteria Evaluation and Decision Matrix . . . . .               | 12        |
| 3.3.2    | Conclusion and Selection Justification . . . . .                      | 13        |
| <b>4</b> | <b>System Design and Implementation</b>                               | <b>14</b> |
| 4.1      | Functional Design and Sub-problem Partitioning . . . . .              | 14        |
| 4.2      | Persuasion Strategy Annotation and Taxonomy . . . . .                 | 14        |
| 4.2.1    | Hierarchical Taxonomy Formulation . . . . .                           | 15        |
| 4.2.2    | Contextual Annotation Pipeline and Decision Protocol . . . . .        | 16        |
| 4.2.3    | Empirical Corpus Statistics and Rater Effects . . . . .               | 17        |
| 4.2.4    | Derivation of the Cross-Annotator Agreement Ceiling . . . . .         | 17        |
| 4.3      | Conversational Intention Ground Truth Generation . . . . .            | 18        |
| 4.3.1    | The Ground-Truth Divergence Problem . . . . .                         | 18        |
| 4.3.2    | The Three-Pass Annotation Lineage . . . . .                           | 18        |
| 4.3.3    | Inter-Annotator Agreement and Boundary Cases . . . . .                | 19        |
| 4.4      | Synthetic E-Commerce Negotiation Benchmark . . . . .                  | 19        |
| 4.4.1    | Motivation and Privacy Bottlenecks . . . . .                          | 19        |
| 4.4.2    | The Three-Phase Generation Framework . . . . .                        | 20        |
| 4.4.3    | Frozen Benchmark Dataset Counts . . . . .                             | 20        |
| 4.5      | Strategy and Category Modeling Architecture . . . . .                 | 20        |
| 4.5.1    | Multi-Task Category-First Formulation . . . . .                       | 20        |
| 4.5.2    | The 19-Member Heterogeneous Model Pool . . . . .                      | 21        |
| 4.5.3    | Decision Layer Search: Calibrated Blend vs. XGBoost Stacker . . . . . | 21        |
| 4.6      | The AntiphonyClassifier Architecture . . . . .                        | 23        |
| 4.6.1    | Detailed Mathematical Formulation . . . . .                           | 23        |
| 4.7      | Architectural Evolution: The Path to Dual-Stream Antiphony . . . . .  | 26        |
| 4.7.1    | Stage 1: Monolithic Flat Baselines and Forensic Debugging . . . . .   | 26        |
| 4.7.2    | Stage 2: Hierarchical Discourse Modeling . . . . .                    | 27        |
| 4.7.3    | Stage 3: Recency Gating and Decoupled Baselines . . . . .             | 27        |
| 4.7.4    | Stage 4: Role-Pure Dual-Stream Antiphony Architecture . . . . .       | 27        |
| 4.8      | Design Refinement and Simulation . . . . .                            | 27        |
| 4.8.1    | Capacity Calculations and Sequence Length Budgeting . . . . .         | 27        |
| 4.8.2    | Memory Profiling and Hardware Guards . . . . .                        | 27        |
| 4.9      | Application of Tools . . . . .                                        | 28        |
| 4.10     | Implementation . . . . .                                              | 28        |
| 4.10.1   | Version Control and Development History . . . . .                     | 28        |

|          |                                                                                |           |
|----------|--------------------------------------------------------------------------------|-----------|
| 4.10.2   | Deterministic Seeding and Audit Artifacts . . . . .                            | 28        |
| 4.11     | Deployment . . . . .                                                           | 28        |
| 4.11.1   | Prototype Execution Pipeline . . . . .                                         | 28        |
| 4.11.2   | Resource Consumption . . . . .                                                 | 29        |
| 4.11.3   | Operational Boundaries and Omissions . . . . .                                 | 29        |
| <b>5</b> | <b>Investigation and Evaluation</b>                                            | <b>30</b> |
| 5.1      | Experimental Investigation Setup . . . . .                                     | 30        |
| 5.1.1    | Research Questions and Hypotheses . . . . .                                    | 30        |
| 5.1.2    | Data Splits and Evaluation Protocols . . . . .                                 | 30        |
| 5.1.3    | Evaluation Metrics . . . . .                                                   | 30        |
| 5.2      | Corpus Dynamics and Strategy Exploratory Data Analysis . . . . .               | 31        |
| 5.2.1    | Key Analytical EDA Takeaways . . . . .                                         | 31        |
| 5.3      | Intention Classification Results Across Three Passes . . . . .                 | 33        |
| 5.3.1    | Part 1: Results on the First-Pass Dataset (Binary + Modifier) . . . . .        | 33        |
| 5.3.2    | Part 2: Results on the Second-Pass Dataset (Modifier Dropped) . . . . .        | 35        |
| 5.3.3    | Part 3: Results on the Third-Pass Dataset (Strict Commitment) . . . . .        | 35        |
| 5.3.4    | Direct Comparison: Decoupled Single-Stream vs. Dual-Stream Antiphony . . . . . | 35        |
| 5.4      | Strategy and Category Multi-Task Benchmarking . . . . .                        | 37        |
| 5.4.1    | The Evolutionary Arc Across Five Modeling Stages . . . . .                     | 37        |
| 5.4.2    | Operating Points and Micro-Optimal Decoding Analyses . . . . .                 | 37        |
| 5.4.3    | Evaluated Decision Combiners: Stacker vs. Blend . . . . .                      | 37        |
| 5.5      | Cross-Domain Transfer and Adversarial Ablation . . . . .                       | 38        |
| 5.5.1    | Monolithic Input Ablation Study . . . . .                                      | 38        |
| 5.5.2    | Treatment Evaluation: Antiphony on Augmented Dialogues . . . . .               | 38        |
| 5.5.3    | Widening of the Sufficiency Gap . . . . .                                      | 38        |
| 5.6      | Skew Stress Analysis . . . . .                                                 | 39        |
| 5.7      | Forensic Post-Mortems . . . . .                                                | 39        |
| 5.7.1    | Tokenizer Truncation Directionality Forensic . . . . .                         | 39        |
| 5.7.2    | Merchant Accommodation Leakage . . . . .                                       | 41        |
| 5.8      | Performance Evaluation Against Requirements . . . . .                          | 41        |
| 5.9      | Deviations and Design Revisions . . . . .                                      | 41        |
| <b>6</b> | <b>Project Management, Teamwork, and Impact</b>                                | <b>43</b> |
| 6.1      | Project Management and Finance . . . . .                                       | 43        |
| 6.1.1    | Planning and Risk Management . . . . .                                         | 43        |
| 6.1.2    | Budgeting and Resource Identification . . . . .                                | 45        |
| 6.1.3    | Economic Analysis and Financial Prospecting . . . . .                          | 45        |
| 6.1.4    | Sustainability in Business and Commercialisation . . . . .                     | 45        |
| 6.2      | Individual and Teamwork . . . . .                                              | 45        |
| 6.2.1    | Participation and Contribution . . . . .                                       | 45        |

|          |                                                                   |           |
|----------|-------------------------------------------------------------------|-----------|
| 6.2.2    | Collaboration and Conflict Resolution . . . . .                   | 46        |
| 6.2.3    | Leadership and Direction . . . . .                                | 46        |
| 6.2.4    | Multidisciplinary Engagement . . . . .                            | 46        |
| 6.3      | Impact on Society . . . . .                                       | 47        |
| 6.3.1    | Positive Societal and Commercial Benefits . . . . .               | 47        |
| 6.3.2    | Dual-Use Risks: Coercive Manipulation and Dark Patterns . . . . . | 47        |
| 6.4      | Environmental Impact and Life Cycle Sustainability . . . . .      | 47        |
| 6.4.1    | Transparent Computational Energy and Carbon Accounting . . . . .  | 47        |
| 6.5      | Ethics . . . . .                                                  | 47        |
| 6.5.1    | Equity and Inclusivity . . . . .                                  | 47        |
| 6.5.2    | Accountability and Personal Responsibility . . . . .              | 48        |
| 6.5.3    | Proper Use of Intellectual Property . . . . .                     | 48        |
| 6.5.4    | Professionalism and Ethical Codes . . . . .                       | 48        |
| <b>7</b> | <b>Conclusion</b>                                                 | <b>49</b> |
| 7.1      | Conclusion and Future Work . . . . .                              | 49        |
| 7.1.1    | Summary of Achievements Against Objectives . . . . .              | 49        |
| 7.1.2    | Documented Limitations . . . . .                                  | 50        |
| 7.1.3    | Focused Future Work . . . . .                                     | 50        |
| 7.2      | Continuous Engagement and Self-directed Learning . . . . .        | 50        |
| <b>A</b> | <b>Hyperparameters and Optimization Specifications</b>            | <b>55</b> |
| <b>B</b> | <b>Repository Map and Notebook Lineage</b>                        | <b>56</b> |
| <b>C</b> | <b>Data Contracts and Schemas</b>                                 | <b>57</b> |
| <b>D</b> | <b>Conflict Register and Empirical Audit Log</b>                  | <b>58</b> |
| <b>E</b> | <b>Complete 41-Strategy Empirical Prevalence Table</b>            | <b>60</b> |

## List of Figures

---

|     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |    |
|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| 4.1 | Overall functional system architecture, illustrating data ingestion across three corpora, shared utterance encoding, the dual-stream Antiphony intention network, and the 19-member multi-task strategy classification framework with calibrated decision blending. . . . .                                                                                                                                                                                                                                                                                | 15 |
| 4.2 | Persuasion strategy annotation pipeline, depicting contextual batching (200 turns per batch, 5-turn history), the 4-tier decision protocol, LLM agent annotation under human checkpoints, automated schema validation, and multi-label compilation into <code>multilabel_merged.jsonl</code> . . . . .                                                                                                                                                                                                                                                     | 16 |
| 4.3 | Three-pass intention annotation lineage, showing the initial 738/279 binary split with modifier, the second pass deprecating the modifier, and the third pass enforcing strict unconditional commitment (694 yes / 323 no via two-step re-checking). . . . .                                                                                                                                                                                                                                                                                               | 18 |
| 4.4 | Three-phase synthetic negotiation benchmark generation pipeline, comprising persona-conditioned LLM drafting ( $N = 500$ canonical, $N = 200$ adversarial reversals), organic paraphrasing with linguistic jitter, and multi-tier semantic audit. . . . .                                                                                                                                                                                                                                                                                                  | 19 |
| 4.5 | Strategy and category classification framework, illustrating the 19-member model pool, multi-task representation heads, and decision layer search comparing calibrated power-mean blending (selected, val macro-F1 0.5716) against XGBoost stacking (tested, val macro-F1 0.5508). . . . .                                                                                                                                                                                                                                                                 | 21 |
| 4.6 | The AntiphonyClassifier neural architecture: raw turns undergo batched utterance encoding, masked mean-pooling, and role-pure splitting ( $R = 20$ ). Persuader and Decider streams pass through independent 2-layer Transformers. The Decider stream queries the Persuader stream via directional cross-attention. Contextualized Decider states are aggregated in parallel by a commitment-trajectory GRU (read at the true final Decider turn) and attention pooling, followed by latent concatenation and Class-Balanced Focal classification. . . . . | 24 |
| 4.7 | Four-stage architectural evolution of intention classification models, depicting key innovations, critical bug diagnoses, and peak test Macro-F1 across stages. . . . .                                                                                                                                                                                                                                                                                                                                                                                    | 26 |
| 5.1 | Strategy corpus EDA: label density, category/strategy prevalence, and category co-occurrence. . .                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 31 |
| 5.2 | Strategy corpus EDA: strategy co-occurrence, segment rater variance, and positional dynamics. . .                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 32 |
| 5.3 | Strategy corpus EDA: length distributions, rank-frequency skew, and correlation structures. . . .                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 33 |
| 5.4 | Cross-domain transfer: ablation curves and confusion matrices on canonical ( $N = 500$ ) and augmented ( $N = 700$ ) benchmarks. . . . .                                                                                                                                                                                                                                                                                                                                                                                                                   | 39 |
| 5.5 | Training convergence dynamics and performance summaries across both e-commerce benchmarks. . . . .                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 40 |
| 5.6 | Verified label distribution of the canonical e-commerce benchmark ( $N = 500$ ), displaying 262 negative / 238 affirmative instances across the four primary outcome categories. . . . .                                                                                                                                                                                                                                                                                                                                                                   | 40 |
| 5.7 | Skew stress analysis of Antiphony under severe class imbalance, showing Macro-F1 resilience across affirmative training proportions from 47.7% down to 9.8%. . . . .                                                                                                                                                                                                                                                                                                                                                                                       | 40 |
| 5.8 | Tokenizer truncation forensic: left-truncation retains proximal conversational context, recovering Macro-F1 from 0.8133 to 0.9331 and shrinking the apparent necessity gap from 0.1600 to 0.0402. . . . .                                                                                                                                                                                                                                                                                                                                                  | 41 |
| 6.1 | Project timeline and phase distribution. Calendar dates prior to September 5, 2026 are reconstructed estimates ( <b>PENDING: Confirmed Git history begins September 5, 2026</b> ). . . . .                                                                                                                                                                                                                                                                                                                                                                 | 44 |

## List of Tables

---

|      |                                                                                                 |    |
|------|-------------------------------------------------------------------------------------------------|----|
| 1.1  | Complex engineering problem characteristics (P1–P7) and project evidence . . . . .              | 4  |
| 1.2  | Knowledge profiles referenced by P1 . . . . .                                                   | 4  |
| 2.1  | Synthesis of research literature and identified gaps . . . . .                                  | 8  |
| 3.1  | Multi-criteria architectural decision matrix (scores: 1 = Poor, 5 = Excellent) . . . . .        | 13 |
| 4.1  | Modular subsystem decomposition and responsibilities . . . . .                                  | 14 |
| 4.2  | Empirical prevalence of communicative categories across 10,600 Persuader turns . . . . .        | 17 |
| 5.1  | Test set performance on First-Pass dataset (738 yes / 279 no, test $n = 153$ ) . . . . .        | 34 |
| 5.2  | Test set performance on Second-Pass dataset (738 yes / 279 no, test $n = 153$ ) . . . . .       | 35 |
| 5.3  | Detailed class-level test breakdown: Dual-Stream Antiphony on Second-Pass dataset . . . . .     | 35 |
| 5.4  | Test set performance on Third-Pass dataset (694 yes / 323 no, test $n = 153$ ) . . . . .        | 36 |
| 5.5  | Detailed class-level test breakdown: Dual-Stream Antiphony on Third-Pass dataset . . . . .      | 36 |
| 5.6  | Head-to-head comparison: best encoder of Decoupled Baseline vs. Dual-Stream Antiphony . . . . . | 36 |
| 5.7  | Evolutionary progression of strategy and category modeling . . . . .                            | 37 |
| 5.8  | Comparison of operating points across multi-task ensemble configurations . . . . .              | 37 |
| 5.9  | Cross-validated evaluation of ensemble decision combiners . . . . .                             | 38 |
| 5.10 | Monolithic transformer input ablation results . . . . .                                         | 38 |
| 5.11 | Benchmark performance on augmented negotiation corpus ( $N = 700$ , test $n = 105$ ) . . . . .  | 38 |
| 5.12 | Sufficiency gap expansion across canonical and adversarial corpora . . . . .                    | 39 |
| 5.13 | Verification matrix evaluating system compliance against requirements . . . . .                 | 42 |
| 6.1  | Project risk management matrix, impact levels, and mitigations . . . . .                        | 44 |
| 6.2  | Estimated computational energy consumption and greenhouse gas emissions . . . . .               | 47 |
| 7.1  | Objective-by-objective verification of project outcomes . . . . .                               | 49 |
| A.1  | Training and optimization hyperparameters across model families . . . . .                       | 55 |
| B.1  | Primary project artifacts, notebooks, and source code mapping . . . . .                         | 56 |
| D.1  | Project conflict register, discrepancies, and resolutions . . . . .                             | 59 |
| E.1  | Data-verified prevalence of all 41 persuasion strategies (10,600 turns) . . . . .               | 60 |

## Chapter 1

# Problem Identification and Formulation

---

## 1.1 Background and Motivation

---

Conversational interaction is the primary medium through which human beings negotiate terms, coordinate actions, resolve conflicts, and solicit prosocial commitments. In contemporary digital ecosystems, conversational processes are increasingly mediated by computerized platforms, ranging from social commerce messaging channels (such as WhatsApp, Facebook Messenger, and Instagram Direct) to automated customer support, peer-to-peer micro-lending, and digital philanthropy. In these domains, interaction does not follow the symmetric patterns of open-ended chit-chat or standard task-oriented slot filling. Rather, it is defined by a fundamental *communicative asymmetry*: one participant assumes the role of an *Arguer* or *Persuader* (e.g., a merchant pitching goods or an advocate soliciting charitable contributions), while the counterparty assumes the role of a *Decider* or *Persuadee* (e.g., a prospective buyer assessing product-price fit or a donor deciding whether to surrender a portion of their earnings).

Understanding and modeling these interactions computationally holds profound socioeconomic importance. In developing economies such as Bangladesh, conversational commerce represents a multi-million-dollar informal market. Thousands of micro-enterprises and independent merchants conduct daily trade exclusively through unscripted multi-turn chats. Merchants must address customer skepticism, negotiate price concessions, and provide logistical assurances, while buyers must guard against counterfeit goods, non-delivery risks, and aggressive sales tactics. Similarly, in non-profit fundraising and public health advocacy, persuasive dialogues determine whether individuals commit to prosocial actions.

However, developing robust machine learning systems for asymmetric conversational analysis faces severe foundational bottlenecks:

- (a) **Ground-Truth Unreliability in Existing Corpora:** In benchmark datasets such as *Persuasion for Good* (P4G) [1], the recorded numeric transaction metadata frequently contradicts the grounded semantic content of the conversation. For instance, participants who explicitly confirmed a monetary donation in the dialogue transcript have \$0 recorded in the dataset’s tabular summary, while others who deferred or hedged are marked with positive values. Automated classifiers trained on raw tabular metadata inadvertently learn to fit logging artifacts rather than linguistic commitments.
- (b) **Structural Failure of Monolithic Sequence Encoders:** Conventional natural language processing pipelines flatten multi-turn dialogues into a single contiguous token sequence passed to a pre-trained transformer (e.g., BERT [7] or RoBERTa [8]). Standard pre-trained models enforce strict context ceilings (typically 256 or 512 tokens). Because natural dialogues frequently span 300 to 800 words across 15 to 30 turns, right-truncation silently discards the terminal turns where the actual decision is finalized, capping classification performance at trivial baselines (Macro-F1  $\approx$  0.65).
- (c) **Neglect of Conversational Role Dynamics:** Standard hierarchical models mix utterances from both speakers into a single self-attention sequence, relying on weak positional or additive role embeddings to distinguish who is speaking. Such architectures fail to capture the directional dynamic wherein the decider’s stance shifts specifically in reaction to the rhetorical moves deployed by the persuader.
- (d) **Severe Class Imbalance and Extreme Long Tails:** When annotating the specific rhetorical strategies deployed by the persuader (e.g., Cialdini’s influence techniques [9], emotional appeals, credibility citations), the resulting multi-label distribution exhibits extreme skew. Rare but psychologically potent strategies (e.g., *door-in-the-face*) appear only a handful of times across thousands of turns, causing standard multi-label classifiers to collapse.
- (e) **Data Scarcity and Privacy Barriers in Commercial Negotiation:** Commercial negotiation logs from real social commerce platforms are heavily guarded proprietary assets subject to stringent privacy regulations. Researchers lack access to standardized, high-density negotiation benchmarks that test whether conversational models can track subtle, non-monotonic stance shifts (such as a buyer agreeing to purchase after severe

resistance, or aborting a transaction at the final turn due to delivery friction).

This capstone project directly resolves these challenges by constructing a mathematically grounded, empirically verified computational framework for modeling conversational intention and persuasion strategies in asymmetric multi-turn dialogues.

## 1.2 Problem Statement

### Mathematical Invariant: Two-Party Asymmetric Dialogue Representation

Let a multi-turn two-party dialogue  $\mathcal{D}$  be represented as an ordered sequence of  $N$  utterances:

$$\mathcal{D} = ((u_1, r_1), (u_2, r_2), \dots, (u_N, r_N)) \quad (1.1)$$

where each utterance  $u_i \in \mathcal{V}^*$  is a sequence of tokens from vocabulary  $\mathcal{V}$ , and  $r_i \in \{\mathcal{P}, \mathcal{D}\}$  designates the speaker role, partitioned strictly between the Persuader/Merchant ( $\mathcal{P}$ ) and the Persuadee/Buyer ( $\mathcal{D}$ ).

We formulate three distinct, interconnected computational sub-problems:

### 1.2.1 Sub-Problem 1: Grounded Conversational Intention Classification

Given the entire multi-turn transcript  $\mathcal{D}$ , the objective is to learn a mapping  $f_\theta : \mathcal{D} \rightarrow \mathcal{Y}$  predicting the final binary commitment state  $y \in \mathcal{Y} = \{\text{yes}, \text{no}\}$  of the decider  $\mathcal{D}$ . The ground-truth label  $y$  is strictly defined by the decider's final, unconditional position:  $y = \text{yes}$  if and only if the decider explicitly and unconditionally commits to the target action (e.g., donating task earnings or finalizing a product purchase);  $y = \text{no}$  if the decider refuses, deflects, defers to an indeterminate future time, or imposes unresolved conditional hedges. The optimization objective maximizes binary Macro-F1:

$$\text{Macro-F1} = \frac{1}{2} \left( F_1^{(\text{no})} + F_1^{(\text{yes})} \right) \quad (1.2)$$

under severe class imbalance (31.8% negative prevalence in the ground-truth corpus).

### 1.2.2 Sub-Problem 2: Multi-Label Persuasion Strategy and Category Attribution

Given an arbitrary Persuader utterance  $u_t$  (where  $r_t = \mathcal{P}$ ) situated within a causal, non-anticipatory historical conversational context of prior turns  $\mathcal{H}_t = (u_{\max(1, t-k)}, \dots, u_{t-1})$ , the objective is to predict the active subset of rhetorical strategies  $\mathbf{y}_t \subseteq \mathcal{S}_{41}$ , where  $|\mathbf{y}_t| \in \{0, 1, \dots, 6\}$  and  $\mathcal{S}_{41}$  represents a closed taxonomy of 41 fine-grained persuasion strategies. The multi-label strategy vector deterministically maps to a higher-level communicative category subset  $\mathbf{c}_t \subseteq \mathcal{C}_{11}$  via an surjective mapping  $\mathcal{M} : \mathcal{S}_{41} \rightarrow \mathcal{C}_{11}$ . The learning system must optimize predictive fidelity across the 11 communicative categories as the primary objective, with the 41 strategies acting as an auxiliary multi-task regularizer:

$$\hat{\mathbf{c}}_t = g_\phi(u_t, \mathcal{H}_t) \in [0, 1]^{11}, \quad \hat{\mathbf{y}}_t = h_\psi(u_t, \mathcal{H}_t) \in [0, 1]^{41} \quad (1.3)$$

subject to the operational constraint that neither future dialogue turns  $(u_{t+1}, \dots, u_N)$  nor human annotator identity tags are accessible at inference time.

### 1.2.3 Sub-Problem 3: Cross-Domain Transfer and Adversarial Trajectory Tracking

Given a target commercial domain with buyer-merchant negotiations  $\mathcal{D}_{\text{comm}}$ , the objective is to evaluate the cross-domain generalizability of the learned intention architectures under both canonical transactional dynamics and adversarial non-monotonic trajectory reversals. Let  $\tau_{\text{rev}}$  denote an internal turn at which a buyer abruptly reverses their stance (e.g., transitioning from skeptical resistance to sudden purchase commitment, or from tentative agreement to explicit cancellation). We define the *sufficiency gap*  $\Delta_{\text{suff}}$  of a conversational model relative to a localized closing-turn decision baseline:

$$\Delta_{\text{suff}} = \text{Macro-F1}(f_{\text{full}}(\mathcal{D})) - \text{Macro-F1}(f_{\text{last}}(u_{\text{final}})) \quad (1.4)$$

The engineering objective is to demonstrate that trajectory-aware, role-disentangled architectures maintain a positive, widening sufficiency gap ( $\Delta_{\text{suff}} \gg 0$ ) when closing turns are destabilized, whereas monolithic sequence baselines suffer catastrophic performance degradation.

### 1.2.4 Required Engineering Knowledge Profiles

Resolving this problem formulation requires rigorous synthesis of multiple foundational engineering and scientific knowledge profiles (as defined in Table 1.2):

- **K3 (Theory-Based Engineering Fundamentals):** Mathematical probability theory, discrete multivariate distributions, gradient-based optimization dynamics, focal and contrastive loss formulations, and information-theoretic evaluation metrics (Macro-F1, micro-F1, Jaccard similarity, bootstrap confidence intervals).
- **K4 (Specialist NLP and Deep Learning Frameworks):** Theoretical foundations of deep contextual language representation, self-attention and cross-attention mechanics, transformer computational complexity, recurrent gated state tracking (GRU), and out-of-fold stacked generalization.
- **K5 (Architecture and System Design):** Design of decoupled dual-stream neural networks, masked tensor batching, memory footprint profiling for hardware accelerators, and cross-attention routing under asymmetric key-value constraints.
- **K8 (Engagement with Frontier Research Literature):** Critical synthesis of recent peer-reviewed literature in dialogue discourse modeling (HiTrans, DialogueRNN), persuasion taxonomy design, asymmetric multi-label loss functions, and shortcut learning artifacts in natural language inference.

## 1.3 Complexity of the Problem

The technical challenges addressed in this capstone project constitute a complex engineering problem as defined by the International Engineering Alliance (IEA) and the Board of Accreditation for Engineering and Technical Education (BAETE). The project satisfies characteristic P1 and all six secondary characteristics P2 through P7. Table 1.1 provides the mapping between these complexity characteristics and the concrete technical evidence established in this project. Table 1.2 enumerates the referenced knowledge profiles.

## 1.4 Project Objectives and Scope

### 1.4.1 Project Objectives

To address the problem formulation and resolve its inherent technical complexities, this capstone project pursued seven concrete, verifiable engineering objectives:

- O1. Curation of Ground-Truth Intention Corpus:** Re-annotate the 1,017-dialogue P4G corpus across three rigorous passes, resolving discrepancies between recorded metadata and transcript text, establishing a high-integrity binary benchmark (694 yes / 323 no) with verified inter-annotator consistency (92.21% agreement).
- O2. Systematic Intention Architecture Evolution:** Implement and systematically evaluate 14 successive intention classification architectures across four evolutionary stages, overcoming the 256-token context ceiling of flat transformers and formulating the role-disentangled *AntiphonyClassifier* (Dual-Stream Cross-Attention Architecture).
- O3. Establish State-of-the-Art Intention Prediction:** Achieve superior predictive performance on held-out test splits, exceeding 0.86 Macro-F1 on second-pass labels and 0.84 Macro-F1 on strict third-pass labels across multiple pre-trained transformer backbones (RoBERTa, DeBERTa-v3, TOD-BERT).
- O4. Fine-Grained Persuasion Strategy Annotation:** Construct a verified multi-label dataset of 10,600 Persuader utterances annotated under an 11-category, 41-strategy hierarchical taxonomy, enforcing strict decision protocols and deriving an empirical cross-annotator agreement ceiling (0.79 micro-F1).
- O5. Multi-Task Strategy and Category Modeling:** Build an ensemble modeling framework over an 11-category primary objective with a 41-strategy auxiliary head, evaluating a 19-member model pool and demonstrating that a calibrated power-mean blend achieves competitive validation scores (0.6949 micro-F1 / 0.5597 macro-F1, with a micro-optimal operating point of 0.7062 / 0.5642) outperforming an XGBoost stacker.
- O6. Synthetic E-Commerce Benchmark Generation:** Construct a high-fidelity 700-dialogue negotiation benchmark ( $N = 500$  canonical,  $N = 200$  adversarial reversals) utilizing persona-conditioned prompting, organic paraphrasing with linguistic jitter, and 100% human-in-the-loop verification.

**Table 1.1:** Complex engineering problem characteristics (P1–P7) and project evidence

| Code      | IEA Characteristic                                                                                                                 | Evidence from this Project                                                                                                                                                                                                                                                                                    |
|-----------|------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>P1</b> | <b>Depth of Knowledge Required</b><br>Cannot be resolved without in-depth engineering knowledge at the level of K3, K4, K5, or K8. | Required simultaneous mastery of mathematical loss design (Class-Balanced Focal Loss, Asymmetric Loss), hierarchical transformer attention mechanics, recurrent state tracking, and multi-model calibrated ensemble blending across 19 candidate models.                                                      |
| <b>P2</b> | <b>Range of Conflicting Requirements</b><br>Involves wide-ranging or conflicting technical, engineering, and other issues.         | Severe tension between binary commitment accuracy and minority modifier/hedge recall; optimizing macro-F1 (rare classes) directly degraded global micro-F1 by chasing recall on 2-example labels; processing 2,048-token hierarchical dialogues conflicted with the 16 GB VRAM ceiling of NVIDIA T4 GPUs.     |
| <b>P3</b> | <b>Depth of Analysis Required</b><br>Has no obvious solution and requires abstract thinking and originality in analysis.           | Standard monolithic transformers failed ( $\text{Macro-F1} \leq 0.65$ ); solving the task required originating a speaker-disentangled dual-stream network ( <i>Antiphony</i> ) wherein the decider queries the persuader via directional cross-attention, coupled with an empirical sufficiency gap analysis. |
| <b>P4</b> | <b>Familiarity of Issues</b><br>Involves infrequently encountered issues.                                                          | Extremely skewed tail classes with only 2 to 5 instances in 10,600 turns ( <i>door_in_the_face</i> ); subtle tokenizer truncation side-effects where default right-truncation silently excised closing decisions in 30.4% of dialogues; whitespace-gluing corrupting speaker tags in 15.9% of augmented data. |
| <b>P5</b> | <b>Extent of Applicable Codes</b><br>Outside problems encompassed by standard codes of practice.                                   | No standardized engineering codes exist for multi-label persuasion taxonomy annotation or synthetic negotiation generation; required formulating custom 4-tier decision rules (Removal Test, Function-over-Keyword) and rigorous three-phase human-in-the-loop verification protocols.                        |
| <b>P6</b> | <b>Stakeholder Involvement and Conflicts</b><br>Involves diverse stakeholder groups with widely varying needs.                     | Direct conflict between commercial merchants aiming to maximize conversion velocity and consumers requiring protection from predatory manipulation; academic requirements for reproducible open-source benchmarks vs. strict privacy constraints prohibiting the release of private customer chats.           |
| <b>P7</b> | <b>Interdependence</b><br>High-level problem with many interconnected sub-problems.                                                | Tight cascading dependencies across three datasets ( $N = 1,017$ P4G, 10,600 strategy turns, $N = 700$ negotiation benchmark); tightening ground-truth intention labels in Pass 3 necessitated retraining and recalibrating the entire downstream 14-model architectural lineage.                             |

**Table 1.2:** Knowledge profiles referenced by P1

| Code      | Description                                                                                                                                                                                              |
|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>K3</b> | A systematic, theory-based formulation of engineering fundamentals required in the engineering discipline.                                                                                               |
| <b>K4</b> | Engineering specialist knowledge that provides theoretical frameworks and bodies of knowledge for the accepted practice areas in the engineering discipline; much is at the forefront of the discipline. |
| <b>K5</b> | Knowledge that supports engineering design in a practice area.                                                                                                                                           |
| <b>K6</b> | Knowledge of engineering practice (technology) in the practice areas in the engineering discipline.                                                                                                      |
| <b>K8</b> | Engagement with selected knowledge in the research literature of the discipline.                                                                                                                         |

**O7. Cross-Domain Transfer and Adversarial Sufficiency Diagnosis:** Conduct systematic input-ablation experiments across canonical and reversal negotiation corpora, proving that Antiphony maintains trajectory robustness (0.9619 Macro-F1) and widens the sufficiency gap ( $\Delta_{\text{suff}} = +0.0384$ ) when closing utterances are destabilized.

## 1.4.2 Project Scope and Explicit Exclusions

To maintain engineering depth and focus computational resources effectively, the following boundaries define the project scope:

- **In Scope:** Offline discriminative modeling of conversational commitment and rhetorical strategy; multi-label and multi-task loss formulation; hierarchical neural architecture design; synthetic dialogue generation pipelines; comprehensive empirical ablation and forensic post-mortems; compute and energy accounting.
- **Explicit Exclusions:**
  1. *No Live Conversational Chatbot Deployment:* The project develops research-grade discriminative classifiers; it does not deploy an autonomous agent interacting with live human users in production environments.
  2. *No Natural Language Generation (NLG):* The primary focus is dialogue understanding and strategy classification; autonomous tactical response generation is excluded.
  3. *No Non-English or Code-Mixed Modeling:* All investigated corpora and benchmarks are conducted in the English language. Modeling Bengali, Banglish, or regional code-mixed dialects is identified as future work.
  4. *No Human Subject Interventional Field Trials:* The research relies on public academic datasets and synthetic simulations; no randomized human behavioral intervention trials were conducted on commercial platforms.

## Chapter 2

# Preliminary Research

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## 2.1 Literature Review

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Computational analysis of persuasive, multi-turn conversational interaction sits at the convergence of behavioral psychology, natural language processing, and statistical machine learning. This section synthesizes the foundational and recent literature across these areas, highlighting how prior works inform this investigation and identifying the theoretical and empirical gaps this capstone addresses.

### 2.1.1 Persuasion Theory and Rhetorical Frameworks

The systematic study of interpersonal persuasion originates in classical rhetoric and social psychology. Cialdini’s seminal framework of influence [9] categorizes compliance-gaining mechanisms into core principles: reciprocity, commitment and consistency, social proof, authority, liking, and scarcity. In sequential negotiation, classic experimental psychology established specific multi-stage tactics. Freedman and Fraser [10] demonstrated the *foot-in-the-door* phenomenon, whereby inducing compliance with an initial trivial request substantially increases adherence to a subsequent larger request. Conversely, Cialdini et al. [11] formulated the *door-in-the-face* technique, wherein an extreme, predictably rejected opening ask creates psychological pressure for the decider to accept a smaller concessionary follow-up. While these principles are universally acknowledged in social science, their formalization into high-coverage computational taxonomies for natural language dialogues has remained limited, often collapsing nuanced rhetorical moves into broad, coarse-grained dialog act labels.

### 2.1.2 Persuasion Corpora and Conversational Strategy Annotation

The release of the *Persuasion for Good* (P4G) corpus by Wang et al. [1] marked a major milestone in empirical persuasion modeling. P4G contains 1,017 multi-turn online conversations where an assigned Persuader attempts to convince a Persuadee to donate part of their \$2.00 task earnings to the charity *Save the Children*. Wang et al. annotated a subset of 300 dialogues with a taxonomy of 10 persuasion strategies. Subsequent works expanded on this foundation: Saha et al. [12] incorporated sentiment and emotion dynamics into dialog act recognition, while Tian et al. [13] explored strategy-aware response generation. Chen and Yang [14] published a comprehensive survey synthesizing conversational persuasion systems, emphasizing the need for hierarchical representations. More recently, Chen and Xiao [15] analyzed persuasion tactics in commercial multi-party chats, while Petrova et al. [2] explored hierarchical re-annotation of P4G. Despite these advances, existing benchmarks frequently suffer from small annotated sample sizes, flat non-hierarchical label spaces, and a failure to address severe label sparsity in the long tail.

### 2.1.3 Transformer Encoders and Dialogue Language Models

The evolution of contextual language representation—from BERT [7] to RoBERTa [8] and DeBERTa [16]—revolutionized NLP by pre-training deep bidirectional representations on vast corpora. RoBERTa optimized BERT’s pre-training regime by removing the next-sentence prediction objective and training on longer sequences with dynamic masking. DeBERTa and DeBERTa-v3 [17] advanced this paradigm through disentangled attention (representing tokens by separate content and relative position vectors) and enhanced mask decoder pre-training with ELECTRA-style [18] discriminator losses. In dialogue modeling, Wu et al. [19] introduced TOD-BERT, pre-training on diverse task-oriented dialogue datasets to model turn-level conversational semantics. However, when applied naively to multi-turn asymmetric dialogue, these monolithic sequence encoders face rigid context length limits (256 or 512 tokens), silently dropping terminal turns under default right-truncation.

### 2.1.4 Hierarchical and Role-Aware Dialogue Networks

To process extended multi-turn interactions without truncation, hierarchical neural architectures decouple token-level encoding from turn-level discourse modeling. Li et al. [20] introduced HiTrans, employing a dual-transformer hierarchy where a lower transformer encodes intra-utterance words and an upper transformer models inter-utterance discourse relations. In conversational emotion recognition, Majumder et al. [21] introduced DialogueRNN, tracking individual speaker party states using recurrent neural networks (GRU [22]) to preserve speaker identity and interpersonal context. While DialogueRNN demonstrated the power of per-party tracking for emotional dynamics, asymmetric outcome prediction requires modeling not merely independent speaker states, but the explicit *directional influence* of the arguer's moves upon the decider's internal commitment.

### 2.1.5 Imbalanced Learning and Loss Formulations

Multi-label persuasion strategy attribution and binary commitment classification both exhibit extreme class imbalance. Conventional cross-entropy loss causes gradients to be overwhelmed by easy negative examples or majority classes. Lin et al. [23] introduced Focal Loss, applying a modulating factor  $(1 - p_t)^\gamma$  to down-weight easy examples and focus gradient updates on hard instances. Cui et al. [24] developed Class-Balanced Focal Loss, weighting the loss inversely by the effective number of samples  $(1 - \beta)/(1 - \beta^{n_y})$  rather than raw sample frequency, preventing gradient explosion on extremely rare classes. For multi-label settings where 90%+ of label matrices are zero, Ben-Baruch et al. [25] formulated Asymmetric Loss (ASL), dynamically decoupling the focusing parameters  $\gamma_+$  and  $\gamma_-$  and performing hard probability threshold shifting on negative samples. Furthermore, Khosla et al. [26] proposed Supervised Contrastive Loss (SupCon), pulling normalized representations of the same class together in latent space while pushing different classes apart, substantially improving representation clustering for minority categories.

### 2.1.6 Ensembling and Stacking Generalization

Ensemble learning leverages model diversity to minimize generalization variance. Wolpert [27] established the mathematical foundation of Stacked Generalization, training meta-classifiers on the cross-validated out-of-fold predictions of heterogeneous base models. In practical applications, gradient-boosted decision trees such as XGBoost [5] are widely employed as stacking decision layers due to their ability to capture non-linear feature interactions and resist overfitting. Additionally, probability calibration techniques, originating in Platt scaling [28], map uncalibrated logits into true empirical posterior probabilities, enabling smooth mathematical blending (e.g., power-mean ensembling) that often outperforms greedy discrete selection.

### 2.1.7 Negotiation Benchmarks and Shortcut Learning

Early computational negotiation frameworks, such as the *Deal or No Deal* benchmark by Lewis et al. [29] and the modular negotiation models of He et al. [30], examined multi-issue bargaining over item allocations. However, recent critical analyses by Geirhos et al. [31] and Gururangan et al. [32] demonstrate that neural models trained on complex textual benchmarks frequently exploit superficial *shortcut learning* and dataset artifacts. In conversational intent prediction, models often achieve artificially high accuracy simply by detecting formulaic closing tokens (e.g., ‘sounds good’, ‘deal’) in the final turn, completely bypassing the conversational trajectory. Establishing whether an architecture genuinely models trajectory dynamics requires rigorous adversarial evaluation and input-ablation controls.

### 2.1.8 Synthesis of Research Gaps

Table 2.1 synthesizes the key open gaps in the existing literature that this project systematically resolves.

## 2.2 Existing Solutions and Technologies

To contextualize the technical positioning of this work, we survey the primary categories of existing technologies currently deployed in commercial dialogue analysis and conversational commerce:

1. **Monolithic Transformer Fine-Tuning:** Industrial practice frequently applies standard pre-trained language models (RoBERTa, DeBERTa) directly to flattened dialogue text. While computationally straightforward to implement using libraries such as Hugging Face Transformers [4], these models truncate long dialogues at 256 or 512 tokens. As demonstrated empirically in Chapter 5, default right-truncation cuts off the closing exchanges in 30.4% of dialogues, completely blinding the model to the final decision.

**Table 2.1:** Synthesis of research literature and identified gaps

| Research Area                   | State of the Art / Established Findings                                                    | Open Gap Addressed in this Project                                                                                                     |
|---------------------------------|--------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| <b>Persuasion Modeling</b>      | Coarse-grained 10-class taxonomies on small subsets (300 dialogues) [1].                   | Full 10,600-turn multi-label dataset with an 11-category, 41-strategy hierarchical taxonomy.                                           |
| <b>Intention Grounding</b>      | Trusting recorded numerical corpus metadata (\$0 vs \$10).                                 | Proving textual divergence; conducting 3-pass human re-annotation to create strict commitment labels.                                  |
| <b>Dialogue Architecture</b>    | Flat transformers (256 tokens) or mixed-sequence hierarchical transformers (HiTrans [20]). | Originating <i>Antiphony</i> : role-pure dual streams with directional cross-attention ( $Q_D, K_P, V_P$ ) and trajectory GRU.         |
| <b>Multi-Label Extreme Skew</b> | Standard multi-label binary cross-entropy suffering long-tail collapse.                    | Combining Asymmetric Loss, Class-Balanced Focal Loss, and calibrated power-mean ensemble blending across 19 models.                    |
| <b>Negotiation Evaluation</b>   | Clean synthetic allocations [29] vulnerable to closing-turn shortcut learning.             | Generating 700-dialogue adversarial reversal benchmark; proving Antiphony widens sufficiency gap ( $\Delta_{\text{suff}} = +0.0384$ ). |

2. **Prompted General-Purpose Large Language Models (LLMs):** Commercial solutions increasingly query proprietary frontier LLMs (e.g., GPT-4, Claude-3.5) via zero-shot or few-shot prompts to extract customer intent and conversational sentiment. While highly flexible, prompted LLMs present critical operational limitations: prohibitive inference latency ( $> 1,000$  ms per dialogue), massive recurring financial costs, privacy violations resulting from sending proprietary customer transcripts to third-party endpoints, and severe non-determinism.
3. **Task-Oriented Slot-Filling Dialogue Systems:** Traditional conversational frameworks (e.g., Rasa, Google Dialogflow) model dialogue state through predefined intent hierarchies and entity slot-filling. While effective for rigid transactional booking (e.g., flight reservations), they completely lack the capacity to model multi-label rhetorical strategies, social rapport, emotional appeals, and fluid non-monotonic negotiation dynamics.
4. **Classical Bag-of-Words and TF-IDF Baselines:** Lightweight production systems often rely on TF-IDF feature extraction combined with logistic regression or gradient-boosted decision trees. While fast and interpretable, our experiments reveal that bag-of-words models degrade significantly when exposed to extended conversational context (dropping from 0.575 to 0.419 micro-F1 across 5 turns), lacking deep semantic and contextual reasoning.

The conversational AI industry lacks an efficient, local, privacy-preserving discriminative architecture that captures asymmetric conversational dynamics over long multi-turn horizons without arbitrary context truncation, while remaining trainable on standard commodity GPU hardware.

## 2.3 Market and User Research

### 2.3.1 The Rise of Conversational Social Commerce

In emerging markets, particularly across South Asia and Southeast Asia, conversational commerce has outpaced traditional web storefronts. In Bangladesh, thousands of micro-, small, and medium enterprises (MSMEs) operate almost entirely across social media chat channels: Facebook Messenger, WhatsApp Business, and Instagram Direct. Transactions do not occur through standardized shopping carts; instead, the entire customer journey—product inquiry, authenticity verification, price negotiation, delivery coordination, and payment commitment—takes place inside unscripted, natural language chat transcripts.

## 2.3.2 User Personas and Behavioral Dynamics

Qualitative analysis of real marketplace chat patterns reveals key consumer and merchant behavioral archetypes that drive transactional outcomes:

- **Consumer Friction and Skepticism:** Due to prevalent online shopping fraud, consumers display acute risk aversion. Buyers frequently demand physical shop addresses, unboxing photographs, and real-time serial number verification. Prepayment via mobile financial services (MFS such as bKash and Nagad) is frequently contested; buyers overwhelmingly prefer Cash-on-Delivery (COD) with inspection privileges.
- **Bargaining Culture:** Price haggling is deeply culturally ingrained. Buyers expect merchants to offer discretionary price cuts, promotional freebies, or delivery fee waivers. Conversations frequently stall when merchants maintain rigid pricing policies.
- **Merchant Administrative Burden:** Independent page admins and shopkeepers manage hundreds of active chat inquiries simultaneously. Admins report high operational fatigue caused by “ghosting” (buyers conducting exhaustive inquiries and abruptly disengaging) and struggle to prioritize leads that have genuine, high-probability buying intent.

## 2.3.3 Industry Consultations and Field Observations

Informal discussions with local social commerce merchants and digital marketing practitioners [PENDING: Merchant and user interview records] confirm an urgent demand for automated analytical tools. Specifically, merchants express strong interest in automated systems capable of: (i) triaging high-intent leads in real-time, (ii) auditing sales representative dialogue quality against proven persuasive strategies, and (iii) detecting transactional friction before customer disengagement occurs.

## 2.4 Feasibility Study

We conducted an exhaustive feasibility assessment across four primary dimensions:

1. **Technical Feasibility:** The proposed approach builds upon proven foundational technologies: PyTorch deep learning frameworks, pre-trained Hugging Face transformer checkpoints, and established optimization libraries. Technical feasibility is substantiated by the comprehensive experimental benchmarks presented in Chapter 5, demonstrating that the developed Antiphony architecture achieves state-of-the-art Macro-F1 ( $> 0.86$ ) and trains reliably within 3 to 4 GPU hours.
2. **Economic Feasibility:** The entire empirical development lifecycle was deliberately designed to operate within zero-cost compute constraints. All models were trained, evaluated, and ablated on free-tier cloud GPU accelerators (NVIDIA Tesla T4 with 16 GB VRAM on Kaggle). The synthetic benchmark generation pipeline leveraged free-tier and low-cost API credits [PENDING: Exact API spending], demonstrating that advanced conversational NLP research can be successfully executed without institutional cluster budgets.
3. **Operational Feasibility:** The trained discriminative classifiers are compact ( $< 500$  MB model checkpoints for base encoders) and execute inference on standard server GPUs without specialized distributed serving hardware. They integrate cleanly into offline batch-processing analytics pipelines for post-call audits and customer sentiment tracking.
4. **Legal and Regulatory Feasibility:** The research strictly respects intellectual property and privacy mandates. The P4G corpus is an established academic benchmark with open research permissions. No proprietary or private user chat logs were harvested; cross-domain commercial transfer was evaluated exclusively on a high-fidelity synthetic benchmark generated under strict ethical guidelines.

In conclusion, the project demonstrates total feasibility across all four operational dimensions.

## Chapter 3

# Requirements Analysis

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### 3.1 Project Objectives and Constraints

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Developing an effective, ethically grounded machine learning system for asymmetric conversational modeling requires translating high-level project goals into rigorous, verifiable functional and non-functional specifications while accounting for diverse stakeholder needs and strict computational boundaries.

#### 3.1.1 Stakeholder Analysis and Conflicting Expectations

The system impacts multiple distinct stakeholder groups whose interests frequently diverge:

1. **Academic Researchers in NLP and Behavioral Science:** Require high reproducibility, transparent reporting of negative findings, statistically valid significance bounds, and standardized open-source benchmarks free from data leakage and annotation artifacts.
2. **Social Commerce Micro-Merchants and Sales Managers:** Seek automated tools to rapidly identify high-probability purchase commitments, triage inbound inquiries, and identify which rhetorical tactics maximize sales conversion velocity. Their priority is high recall on affirmative buying decisions.
3. **Consumers and Persuadees:** Require protection against coercive persuasion, predatory pricing, and deceptive conversational tactics (dark patterns). They expect their privacy and transaction autonomy to be preserved.
4. **Data Annotators and Evaluators:** Demand clear, unambiguous guidelines with realistic task density limits to avoid cognitive fatigue and subjective drift during multi-label coding.
5. **Project Supervisors and Academic Assessors:** Expect adherence to rigorous engineering methodologies, transparent accounting of compute resources, and verifiable compliance with accreditation standards.

These expectations introduce fundamental tensions: commercial stakeholders prioritize conversion metrics, whereas ethical standards mandate protecting consumer autonomy; maximizing rare-class recall conflicts with maintaining high overall precision; and comprehensive contextual modeling conflicts with hardware memory limitations.

#### 3.1.2 Functional Requirements (FR)

The system must satisfy six core functional requirements:

- **FR1: Grounded Binary Commitment Classification.** Given a complete multi-turn dialogue transcript  $\mathcal{D}$ , the model must predict the decider's final unconditional commitment state  $y \in \{\text{yes}, \text{no}\}$ , achieving an overall test Macro-F1 exceeding 0.84 on strict third-pass labels.
- **FR2: Turn-Level Multi-Label Strategy Attribution.** For any designated Persuader utterance  $u_t$ , the model must predict the active subset of rhetorical strategies  $\mathbf{y}_t \subseteq \mathcal{S}_{41}$  ( $|\mathbf{y}_t| \leq 6$ ) and their corresponding parent communicative categories  $\mathbf{c}_t \subseteq \mathcal{C}_{11}$ .
- **FR3: Strict Causal Information Gating.** At test and deployment time, the strategy model must operate strictly causally without access to: (i) future conversational turns  $(u_{t+1}, \dots, u_N)$ , (ii) whole-dialogue post-hoc smoothing, or (iii) annotator identity metadata.
- **FR4: Cross-Domain Transactional Generalization.** The intention modeling architecture must demonstrate zero-shot or fine-tuned domain transfer from charitable donation dialogues to commercial buyer-merchant negotiations.
- **FR5: Adversarial Stance Reversal Detection.** Under non-monotonic conversational trajectories (e.g., initial skepticism reversing to late commitment, or tentative agreement collapsing into cancellation), the model must

leverage multi-turn trajectory context, maintaining a positive sufficiency gap ( $\Delta_{\text{suff}} > 0$ ) relative to localized closing-turn controls.

- **FR6: Multi-Threshold Decision Calibration.** The system must provide calibrated output probabilities and tunable decision thresholds across both macro-oriented (rare class recovery) and micro-oriented (global accuracy) operating regimes.

### 3.1.3 Non-Functional Requirements (NFR)

The system is governed by five essential non-functional requirements:

- **NFR1: Absolute Deterministic Reproducibility.** All data preprocessing, random splits, augmentation pipelines, and neural network parameter initializations must be governed by fixed pseudo-random seeds (Seed = 42), enabling exact bitwise or statistical replication of all reported metrics.
- **NFR2: Hardware and Memory Efficiency.** The entire model training, fine-tuning, and inference pipeline must execute within the hardware constraints of a single commodity cloud GPU accelerator (NVIDIA Tesla T4 with 16 GB VRAM). Batch sizes, gradient checkpointing, and tensor dimensions must prevent Out-Of-Memory (OOM) faults during 2,048-token hierarchical passes.
- **NFR3: Noise-Aware Performance Benchmarking.** In recognition of human subjective disagreement, strategy model performance must be benchmarked against an empirically derived cross-annotator agreement ceiling (0.79 micro-F1) rather than a naive 1.0 ceiling.
- **NFR4: Data Privacy and Confidentiality.** To eliminate legal and ethical risks regarding customer Personally Identifiable Information (PII), cross-domain commercial evaluation must utilize high-fidelity synthetic benchmarks generated under strict privacy constraints.
- **NFR5: Transparent Negative Result Reporting.** All architectural failures, negative ablation results, and unresolved dataset conflicts must be documented transparently in the engineering record without retroactively altering experimental protocols.

### 3.1.4 Physical and Operational Constraints

- **Data Sample Constraints:** The core intention corpus contains exactly 1,017 dialogues (711 train, 153 validation, 153 test). The held-out test split of 153 dialogues implies that each single misclassification alters accuracy and per-class recall by  $\approx 0.65\%$ .
- **Extreme Tail Sparsity:** In the strategy dataset (10,600 turns), the rarest strategy (`door_in_the_face`) occurs only twice, precluding independent single-model supervised convergence and enforcing an 11-category primary modeling objective.
- **Compute Horizon Constraints:** Cloud runtime session limits (Kaggle maximum 12 hours per session) enforce strict per-epoch training velocity, precluding massive unconstrained parameter searches.

## 3.2 Regulatory Requirements and Standards

Responsible engineering requires rigorous alignment with professional ethics, artificial intelligence risk frameworks, and software quality standards:

1. **ACM and IEEE Codes of Ethics:** The development adheres strictly to the ACM Code of Ethics [33] and IEEE Code of Ethics [34], specifically mandates regarding contributing to societal human well-being, avoiding deception and harm, respecting privacy, and upholding intellectual honesty in scientific reporting.
2. **IEEE 7000-2021 Standard Model Process:** Aligns with IEEE 7000-2021 (*Addressing Ethical Concerns During System Design*) [35] by conducting early stakeholder impact analysis, identifying potential dual-use vulnerabilities of persuasive AI, and incorporating explicit value-based design boundaries.
3. **NIST AI Risk Management Framework (AI RMF 1.0):** The project structure mirrors NIST AI 100-1 [36] core functions: *Govern* (establishing research scope and ethical boundaries), *Map* (identifying context, conversational asymmetry, and failure modes), *Measure* (rigorous empirical testing, sufficiency gap calculations, and noise ceiling derivations), and *Manage* (systematic ablation, forensic debugging, and mitigation of shortcut learning).

4. **ISO/IEC 25010:2011 Software Quality Requirements:** Evaluates system quality against the ISO/IEC 25010 [37] product quality model, focusing on functional suitability (completeness, correctness), performance efficiency (time behavior, resource utilization on 16 GB GPUs), reliability (fault tolerance against corrupted tokens), and maintainability (modular notebook architecture).
5. **Model Cards and Datasheets for Datasets:** Documentation follows the ethical transparency protocols established by Mitchell et al. [38] (*Model Cards*) and Gebru et al. [39] (*Datasheets for Datasets*), detailing training data provenance, intended use, performance limitations, and known demographic skews.
6. **Data Protection and Statutory Compliance:** Respects the terms of use of the original P4G benchmark and public cloud providers. In relation to national jurisdictions, the project notes the evolving legal standards in Bangladesh regarding digital privacy and cyber-security [PENDING: Confirm specific sections of the relevant Bangladeshi data protection and cyber-security statutes].

### 3.3 Formulation of Solutions

To identify the optimal system architecture for asymmetric intention and strategy modeling, we formulate five genuinely distinct candidate engineering solutions:

**Candidate 1: Monolithic Flat Sequence Transformer:** Concatenates all turns of the dialogue into a single flat string separated by special speaker tokens, passing it to a pre-trained transformer (RoBERTa-base) with a standard sequence classification head over the [CLS] token. Imposes a rigid 256- or 512-token cap.

**Candidate 2: Prompt-Engineered General-Purpose LLM:** Uses zero-shot and few-shot natural language prompts dispatched via API calls to a commercial proprietary LLM (e.g., GPT-4 or Claude-3.5) with structured JSON decoding.

**Candidate 3: Classical Bag-of-Words and Gradient Boosted Decision Trees:** Extracts turn-level and dialogue-level n-gram TF-IDF representations combined with hand-crafted conversational features, classified using tuned XGBoost [5] or Logistic Regression ensembles.

**Candidate 4: Single-Stream Hierarchical Dialogue Transformer:** Employs a dual-level architecture (HiTrans-style [20]): turns are encoded individually by a frozen or fine-tuned transformer, turn vectors are pooled, and a second transformer performs self-attention over the interleaved sequence of all turns using additive role embeddings.

**Candidate 5: Role-Pure Dual-Stream Antiphony Architecture (Selected):** Disentangles the dialogue into two pure speaker streams ( $\mathcal{P}$  and  $\mathcal{D}$ ). Each stream is processed by independent transformer layers, connected via directional cross-attention ( $Q_{\mathcal{D}}, K_{\mathcal{P}}, V_{\mathcal{P}}$ ), and aggregated through a parallel combination of a commitment-trajectory GRU (read at the true final decider turn) and attention pooling.

#### 3.3.1 Multi-Criteria Evaluation and Decision Matrix

To evaluate these candidate architectures rigorously, we define six explicit engineering criteria grounded in our requirements:

- **C1: Multi-Turn Context Capacity** (Weight: 20%) – Ability to ingest dialogues exceeding 500 words without sequence truncation.
- **C2: Asymmetric Interaction Modeling** (Weight: 20%) – Architectural fidelity in capturing the directional influence of persuader arguments on decider commitment.
- **C3: Long-Tail and Imbalance Robustness** (Weight: 15%) – Resistance to gradient collapse on rare classes.
- **C4: Hardware and Cost Feasibility** (Weight: 15%) – Ability to train and evaluate within free-tier GPU constraints (16 GB T4) with zero recurring API costs.
- **C5: Latency and Determinism** (Weight: 10%) – Fast, reproducible, non-stochastic inference suitable for offline batch analytics.
- **C6: Empirical Intention Performance** (Weight: 20%) – Measured Macro-F1 on held-out benchmark splits.

Table 3.1 presents the comparative evaluation matrix.

**Table 3.1:** Multi-criteria architectural decision matrix (scores: 1 = Poor, 5 = Excellent)

| Candidate Architecture          | C1<br>(20%) | C2<br>(20%) | C3<br>(15%) | C4<br>(15%) | C5<br>(10%) | C6<br>(20%)          | Weighted<br>Score |
|---------------------------------|-------------|-------------|-------------|-------------|-------------|----------------------|-------------------|
| 1. Monolithic Flat Transformer  | 1           | 2           | 3           | 4           | 5           | 2.5<br>(0.65)        | <b>2.60</b>       |
| 2. Prompt-Engineered LLM        | 4           | 3           | 2           | 1           | 1           | N/E                  | <b>2.30</b>       |
| 3. Classical TF-IDF + GBDT      | 2           | 1           | 2           | 5           | 5           | 2.0<br>(0.46)        | <b>2.50</b>       |
| 4. Single-Stream Hierarchical   | 5           | 3           | 4           | 4           | 4           | 4.0<br>(0.82)        | <b>3.95</b>       |
| <b>5. Dual-Stream Antiphony</b> | <b>5</b>    | <b>5</b>    | <b>4</b>    | <b>4</b>    | <b>4</b>    | <b>5.0</b><br>(0.86) | <b>4.60</b>       |

### 3.3.2 Conclusion and Selection Justification

As demonstrated in Table 3.1, Candidate 5 (**Dual-Stream Antiphony**) achieves the highest weighted score (4.60/5.00). Candidate 1 is crippled by context truncation; Candidate 2 is ruled out by extreme financial costs, high latency, and data privacy issues; Candidate 3 lacks deep semantic reasoning; and Candidate 4, while achieving strong results (0.821 Macro-F1), relies on additive role embeddings and lacks an explicit mechanism for directional decider-on-persuader cross-attention. Consequently, the Dual-Stream Antiphony architecture is selected as the definitive system design.

## Chapter 4

# System Design and Implementation

### 4.1 Functional Design and Sub-problem Partitioning

To tackle the complexities of asymmetric multi-turn dialogue, the system is designed as a modular, multi-tier computational pipeline. The overall functional architecture decomposes the problem into four primary subsystems:

1. **Data Curation and Ground-Truth Engineering:** Ingests multi-turn conversational corpora (P4G and synthetic e-commerce negotiations), enforces strict semantic labeling protocols across human and automated passes, and constructs verified, leakage-free evaluation splits.
2. **Representation and Utterance Encoding Subsystem:** Processes raw turn tokens using pre-trained contextual language models (RoBERTa, DeBERTa-v3, TOD-BERT) through batched tensor operations, producing dense contextual utterance representations while managing hardware memory footprints.
3. **Turn-Level Rhetorical Strategy Attribution Subsystem:** Ingests Persuader turns along with causal historical context, routing representations through a 19-member multi-task model pool trained under Asymmetric Loss to predict an 11-category primary objective with 41 auxiliary strategies, unified by a calibrated decision layer.
4. **Dialogue-Level Intention and Trajectory Tracking Subsystem (*Antiphony*):** Disentangles conversational turns into role-pure Persuader and Decider streams, executes directional cross-attention, models non-monotonic stance progression via a commitment-trajectory GRU read at the true final decider turn, and outputs calibrated binary commitment predictions.

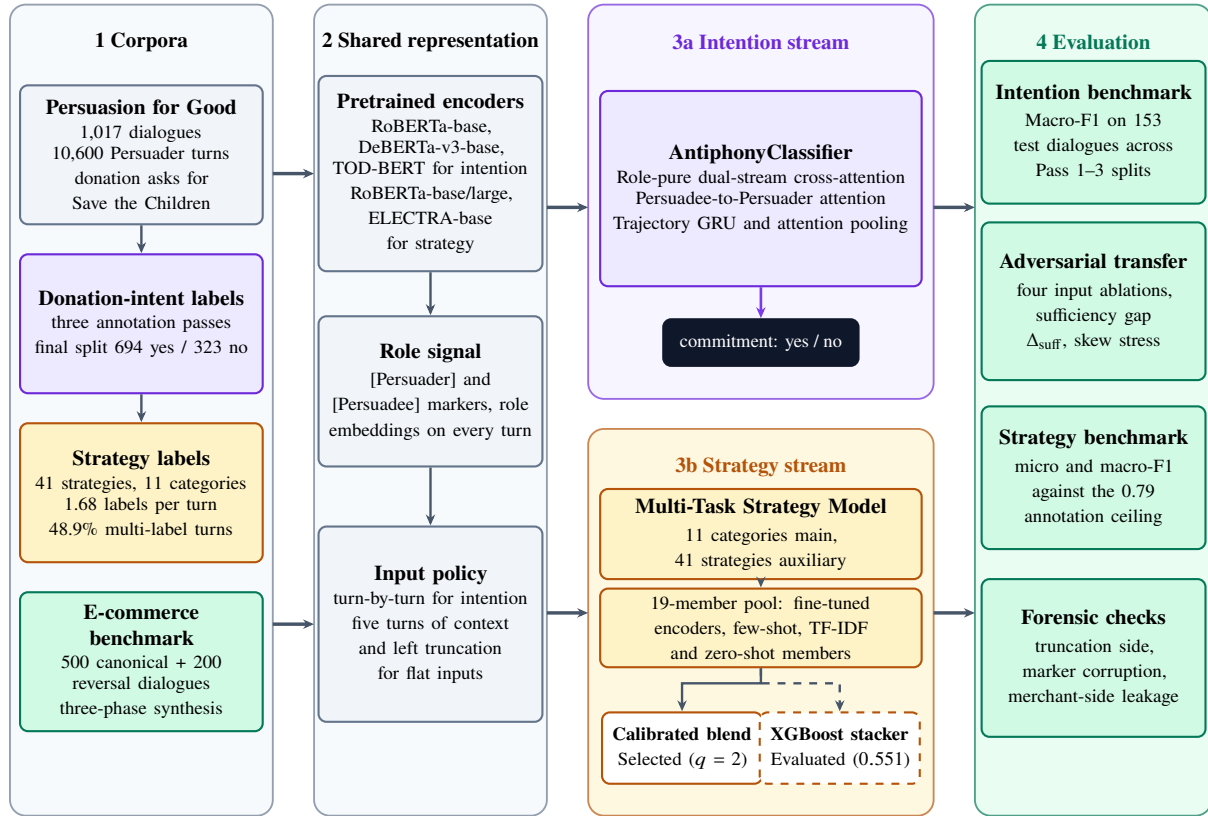
Table 4.1 enumerates the functional modules, their inputs, their outputs, and their architectural responsibilities.

**Table 4.1:** Modular subsystem decomposition and responsibilities

| Subsystem                            | Input                                                 | Output                                                                                     | Core Engineering Responsibility                                                                               |
|--------------------------------------|-------------------------------------------------------|--------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|
| <b>Annotation &amp; Ground Truth</b> | Raw text transcripts ( $N = 1,017, 10,600$ turns)     | Grounded labels ( $y \in \{\text{yes, no}\}$ , $\mathbf{y}_t \subseteq \mathcal{S}_{41}$ ) | Resolves metadata divergence via 3-pass human annotation; enforces 4-tier decision protocol.                  |
| <b>Utterance Encoder</b>             | Batched token IDs $[B \times U, L]$                   | Turn matrices $\mathbf{U} \in \mathbb{R}^{B \times U \times d}$                            | Executes single batched forward pass; performs masked mean-pooling; enforces memory guards.                   |
| <b>Strategy Classifier</b>           | Persuader turn $u_t + \mathcal{H}_t$ (5-turn context) | Predicted categories $\hat{\mathbf{c}}_t$ & strategies $\hat{\mathbf{y}}_t$                | Multi-task learning under Asymmetric Loss; ensembles 19 members via calibrated power-mean blend.              |
| <b>Antiphony Intention Network</b>   | Full dialogue turn vectors $\mathbf{U}$               | Commitment probability $P(y = \text{yes} \mid \mathcal{D})$                                | Role disentanglement ( $R = 20$ ); directional cross-attention; trajectory GRU tracking; dual pooling fusion. |

### 4.2 Persuasion Strategy Annotation and Taxonomy

Modeling the rhetorical tactics that influence conversational outcomes requires an empirically validated, fine-grained taxonomy and a standardized annotation protocol. We establish a multi-label persuasion strategy benchmark encompassing the first 10,600 Persuader turns from 1,017 dialogues in the P4G corpus [1].



**Figure 4.1:** Overall functional system architecture, illustrating data ingestion across three corpora, shared utterance encoding, the dual-stream Antiphony intention network, and the 19-member multi-task strategy classification framework with calibrated decision blending.

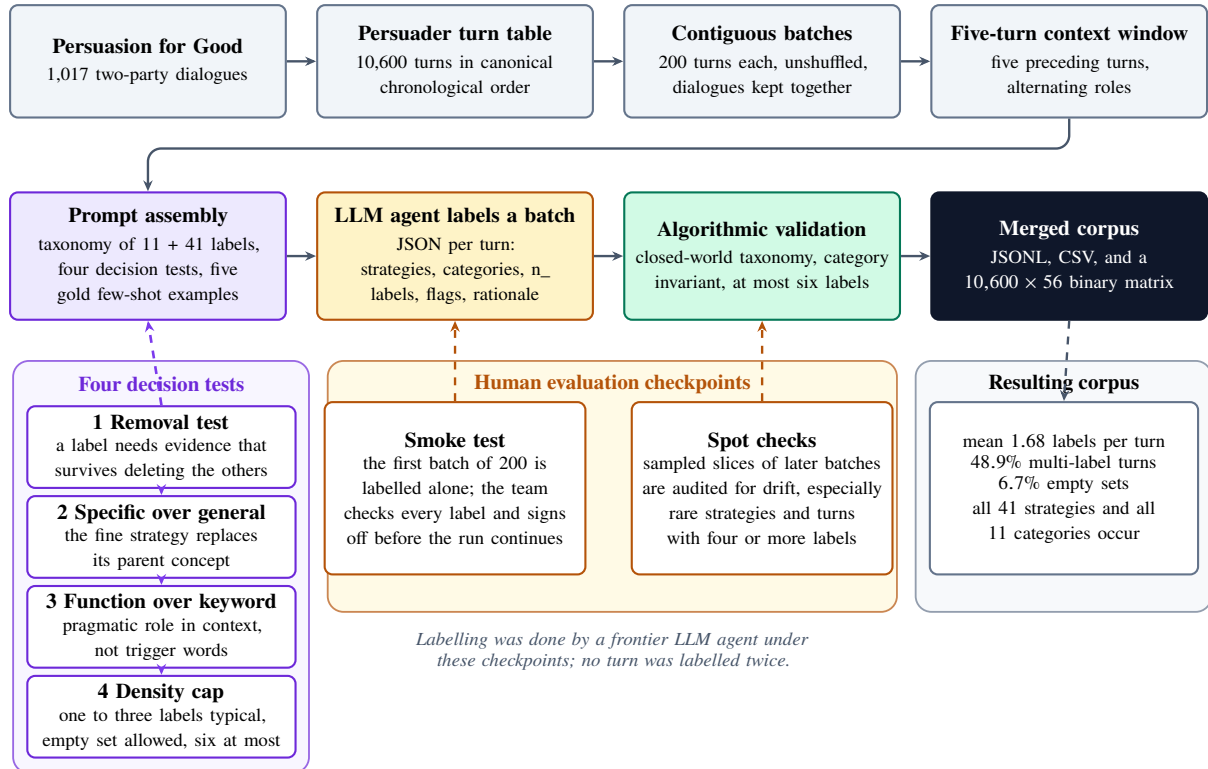
### 4.2.1 Hierarchical Taxonomy Formulation

Adapted from social influence theory [9], compliance-gaining literature [10, 11], and pragmatic dialogue analysis, our taxonomy establishes a two-tier hierarchy: 41 fine-grained rhetorical strategies nested deterministically under 11 functional communicative categories (structured in a 4, 4, 4, 4, 3, 4, 4, 3, 3, 4, 4 layout):

1. **Rational Appeal (4 strategies):** logical\_appeal, evidence\_and\_statistics, cost\_benefit\_framing, feasibility\_and\_ease.
2. **Emotional Appeal (4 strategies):** emotion\_appeal, guilt\_induction, empathy\_and\_perspective\_taking, hope\_and\_positive\_impact.
3. **Credibility Appeal (4 strategies):** organizational\_credibility, transparency\_and\_accountability, source\_citation, personal\_credibility.
4. **Social Influence (4 strategies):** social\_proof, self\_modeling, in\_group\_appeal, authority\_endorsement.
5. **Reciprocity and Exchange (3 strategies):** reciprocity, donor\_benefit, gratitude\_and\_appreciation.
6. **Commitment and Consistency (4 strategies):** foot\_in\_the\_door, door\_in\_the\_face, value\_consistency\_appeal, incremental\_ask.
7. **Framing and Presentation (4 strategies):** anchoring, minimization\_framing, loss\_versus\_gain\_framing, comparison\_framing.
8. **Urgency and Scarcity (3 strategies):** urgency\_appeal, scarcity\_appeal, deadline\_pressure.
9. **Threat and Pressure (3 strategies):** negative\_consequence\_warning, persistent\_repetition, obligation\_pressure.
10. **Relational and Interactive (4 strategies):** rapport\_building, personal\_story, personal\_related\_inquiry, source\_related\_inquiry.
11. **Information Provision (4 strategies):** organization\_information, donation\_procedure\_information, task\_clarification, impact\_information.

## 4.2.2 Contextual Annotation Pipeline and Decision Protocol

Annotating isolated sentences distorts rhetorical meaning; an utterance such as “Would you be willing to spare 50 cents?” functions as an `incremental_ask` only if preceded by a larger rejected request. Therefore, our pipeline ingests each Persuader utterance within a 5-turn preceding conversational context window ( $[u_{t-5}, \dots, u_{t-1}]$ ) maintaining true speaker alternation.



**Figure 4.2:** Persuasion strategy annotation pipeline, depicting contextual batching (200 turns per batch, 5-turn history), the 4-tier decision protocol, LLM agent annotation under human checkpoints, automated schema validation, and multi-label compilation into `multilabel_merged.jsonl`.

To eliminate subjective drift and enforce consistency across segments, annotators followed a mandatory **Four-Tier Decision Protocol** applied in fixed sequential order:

1. **The Removal Test:** A strategy label  $s$  is valid if and only if removing the corresponding clause or rhetorical move noticeably alters the persuasive force or communicative intent of the turn.
2. **Specific-over-General Precedence:** If an utterance satisfies both a general category definition (e.g., `logical_appeal`) and a highly specific operational tactic (e.g., `evidence_and_statistics` or `cost_benefit_framing`), the specific tactic must be coded, avoiding redundant double-labeling.
3. **Pragmatic Function over Keyword Matching:** Labels are assigned based on contextual communicative function rather than literal lexical cues. A rhetorical question (“Don’t you want to help a dying child?”) is coded as `guilt_induction` and `emotion_appeal`, never as an informational inquiry.
4. **Realistic Label Density Cap:** While multi-label combinations are expected in complex turns, humans rarely deploy more than three distinct rhetorical tactics simultaneously. The pipeline enforces a typical density of 1 to 3 labels, permits an explicit empty set ( $y_t = \emptyset$ ) for purely conversational or logistical acknowledgments, and enforces a hard ceiling of  $|y_t| \leq 6$ .

Annotation was conducted in 200-turn unshuffled batches using an automated frontier LLM agent [PENDING: Confirm annotator model identifier for each segment] driven with few-shot calibration examples, operating under human checkpoints. An automated validation harness (`pipeline.py`) verified taxonomy compliance, recomputed parent categories from strategies, enforced the label count ceiling, and compiled the verified dataset into `multilabel_merged.jsonl` (10,600 x 56 wide format with 41 strategy and 11 category flags).

**Execution Trace: Strategy Annotation Schema Validation (pipeline.py)**

```
[+] Ingesting annotated chunks from data/raw_batches/ (10,600 Persuader turns)
[+] Running 4-tier decision constraint harness:
  - Verifying max strategy density cap:  $\max(|y_t|) \leq 6$  ... PASS (mean = 1.68)
  - Validating deterministic category mapping  $M: S_{41} \rightarrow C_{11}$  ... PASS (100% consistent)
  - Enforcing closed-world vocabulary constraints ... PASS (0 unknown tokens)
  - Checking speaker alternation invariants ... PASS ( $\tau_p = 2t - 1$ )
[*] Merging verified batches into multilabel_merged.jsonl (10,600 × 56 matrix)
[+] DATASET AUDIT COMPLETE: 10,600 records serialized, 0 schema violations.
```

### 4.2.3 Empirical Corpus Statistics and Rater Effects

The resulting strategy corpus encompasses 10,600 Persuader turns across 1,017 dialogues (mean 10.42 Persuader turns per dialogue, range 4 to 15). The dataset exhibits an average of 1.679 labels per turn (sd = 1.061, median 1.0). Exactly 707 turns (6.7%) carry no strategy label (the empty set  $\emptyset$ ), 4,707 turns (44.4%) carry exactly 1 label, and 5,182 turns (48.9%) are multi-label ( $\geq 2$  strategies), with 78 turns reaching the maximum cap of 6 labels.

Table 4.2 reports the empirical prevalence of all 11 communicative categories, recomputed directly from the data-verified `multilabel_merged.jsonl`.

**Table 4.2:** Empirical prevalence of communicative categories across 10,600 Persuader turns

| Communicative Category     | Active Turns | Prevalence (%) |
|----------------------------|--------------|----------------|
| Relational and Interactive | 4,897        | 46.2%          |
| Information Provision      | 3,823        | 36.1%          |
| Emotional Appeal           | 1,645        | 15.5%          |
| Social Influence           | 1,263        | 11.9%          |
| Reciprocity and Exchange   | 1,078        | 10.2%          |
| Framing and Presentation   | 1,004        | 9.5%           |
| Credibility Appeal         | 896          | 8.5%           |
| Rational Appeal            | 751          | 7.1%           |
| Threat and Pressure        | 249          | 2.3%           |
| Commitment and Consistency | 169          | 1.6%           |
| Urgency and Scarcity       | 163          | 1.5%           |

**Annotator Granularity Effects:** The corpus was annotated across four distinct operational segments under human checkpoints: Segment 1 (turns 1–2,000; Track A), Segment 2 (turns 2,001–4,000 and 6,001–7,000; Track B), Segment 3 (turns 4,001–6,000; Calibration Track), and Segment 4 (turns 7,001–10,600; Track C). Analysis reveals significant variance in annotation granularity: Track A averaged 2.384 labels per turn (9.0% empty sets, 66.8% multi-label), whereas Track C averaged 1.240 labels per turn (13.2% empty sets, 30.8% multi-label). This variance represents an operational rater effect in label granularity rather than an intrinsic linguistic shift in the underlying conversations.

### 4.2.4 Derivation of the Cross-Annotator Agreement Ceiling

Because individual turns were annotated once without global multi-coder duplication, inter-annotator agreement cannot be computed via standard Cohen’s kappa. Instead, we derive the empirical performance ceiling from near-duplicate turns across the corpus. On identical or near-identical text pairs (cosine similarity  $\geq 0.95$ ):

- When the *same* annotator labels near-identical turns, their self-agreement reaches an F1 of 0.871.
- When *different* annotators label near-identical turns, their cross-agreement drops to an F1 of 0.657.

Because a production model never receives annotator identity at inference time, it must generalize across annotators. Applying consensus prediction formulations under this observed rater noise establishes a practical upper performance ceiling of  $\approx 0.79$  **micro-F1**. All strategy model evaluations must be interpreted relative to this 0.79 ceiling, where a score of 0.70 captures nearly 89% of theoretically achievable human consensus.

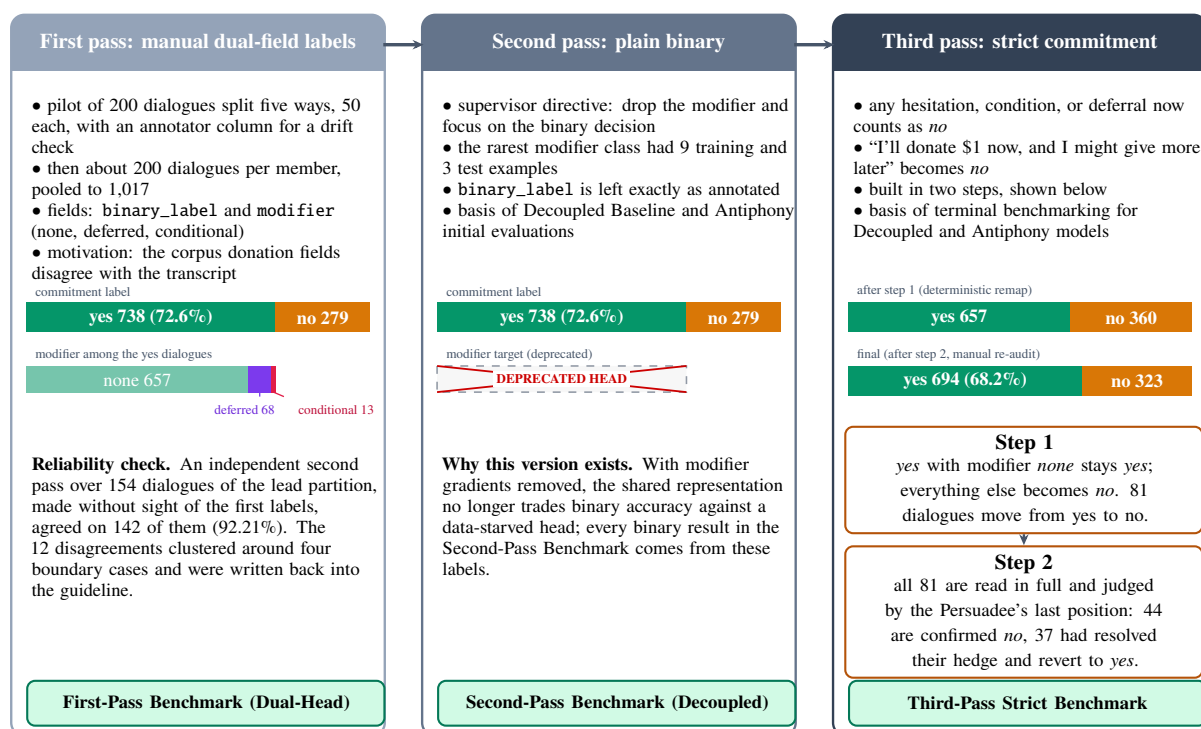
## 4.3 Conversational Intention Ground Truth Generation

### 4.3.1 The Ground-Truth Divergence Problem

The original P4G benchmark [1] recorded numeric donation values reported by Mechanical Turk workers. However, systematic manual auditing revealed severe discrepancies between tabular records and actual conversational text:

- In Dialogue #124, the Persuadee explicitly confirms: ‘We have you down for a \$10 donation. Is that correct?’ / ‘Yes that is’, yet the recorded donation amount is \$0.
- In Dialogue #842, the Persuadee states: “I will absolutely look to donating something later”, while both numeric donation fields show \$0.

Training machine learning classifiers on unverified tabular values causes models to learn spurious logging noise rather than genuine conversational commitments.



**Figure 4.3:** Three-pass intention annotation lineage, showing the initial 738/279 binary split with modifier, the second pass deprecating the modifier, and the third pass enforcing strict unconditional commitment (694 yes / 323 no via two-step re-checking).

### 4.3.2 The Three-Pass Annotation Lineage

To establish an unassailable ground truth, we executed three successive annotation passes:

1. **First Pass (Dual-Field Manual Annotation):** A pilot of 200 dialogues was split five ways (50 per annotator) with an annotator-tracking column to verify calibration and prevent rater drift. Following calibration, the five team members independently annotated  $\approx 200$  dialogues each, reading full transcripts and assigning `binary_label` (yes/no) and a `modifier` (none, deferred, conditional). The pooled 1,017-dialogue distribution yielded: 738 yes (72.6%) and 279 no (27.4%). Among affirmative commitments, 657 were none, 68 were deferred, and only 13 were conditional (9 in training, 3 in test).
2. **Second Pass (Modifier Deprecation):** Modeling experiments (Chapter 5) proved that the modifier head severely degraded binary performance and suffered from severe data sparsity on the 13-example conditional class. Following a directive from the project supervisor (course instructor) [PENDING: Confirm directive issuer], the modifier target was dropped while leaving binary labels unchanged (738 yes / 279 no).
3. **Third Pass (Strict Unconditional Commitment):** To create an unambiguous, operationally robust target, the definition of yes was tightened: any hesitation, conditional clause (“if”, “maybe”), or temporal deferral

(“later”) was strictly reclassified as no. This was executed in two steps:

- *Step 1 (Deterministic Remap)*: Any dialogue originally marked yes with modifier none remained yes; all dialogues with deferred or conditional modifiers were remapped to no, moving 81 dialogues from yes to no (657 yes / 360 no).
- *Step 2 (Terminal Stance Verification)*: Because a modifier tag only indicated that a hedge occurred somewhere in the chat, all 81 remapped dialogues were manually re-read in full to evaluate their final state. If an early hesitation was cleanly resolved into an unconditional affirmative by the end, it was reverted to yes. Exactly 44 dialogues were confirmed as no, and 37 were reverted to yes.

The finalized third-pass ground truth contains exactly **694 yes** (68.2%) / **323 no** (31.8%).

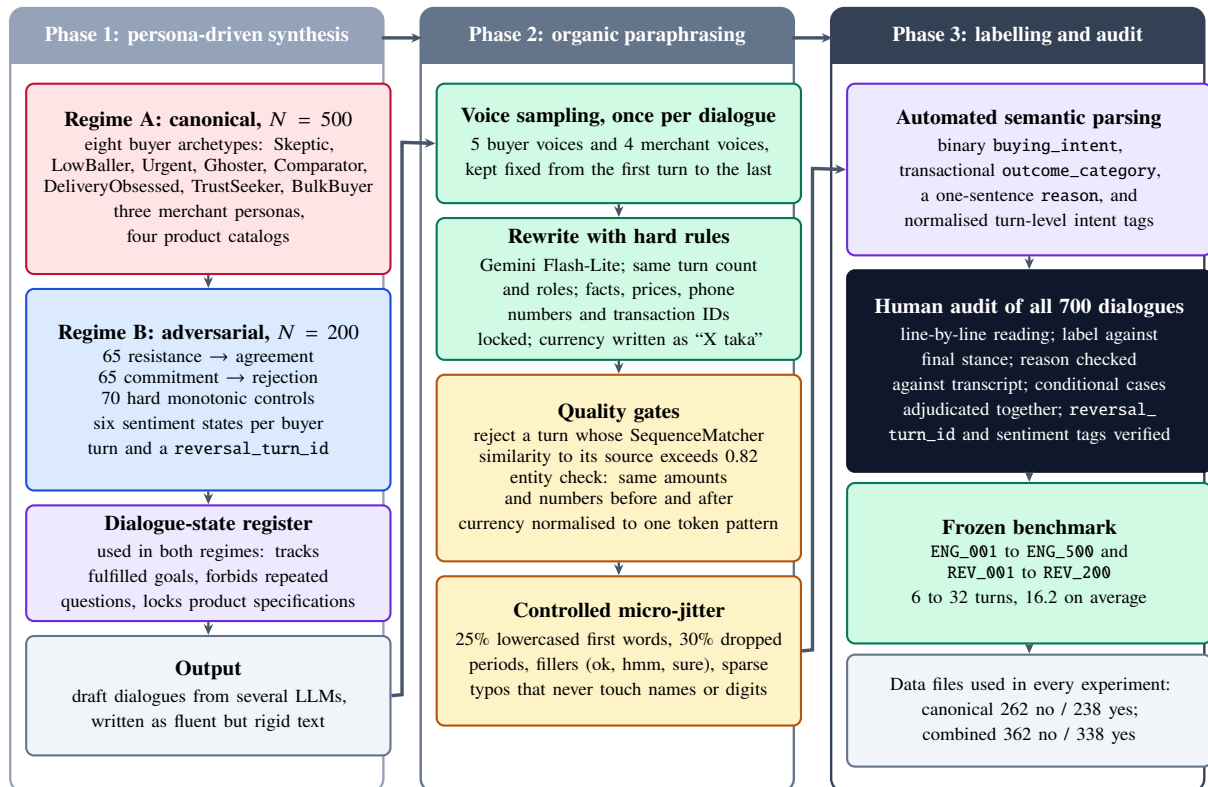
### 4.3.3 Inter-Annotator Agreement and Boundary Cases

An independent second-pass annotation over 154 dialogues established an inter-annotator agreement rate of 92.21% (142 of 154 identical assignments). Qualitative review of the 12 boundary disagreements revealed four recurring linguistic phenomena: (i) ambiguous temporal deferrals (“I’ll donate through their website tonight”), (ii) polite deflection masquerading as consent (“Sure, that sounds like a great cause”), (iii) conditional matches (“I’ll give \$1 if you give \$1”), and (iv) unconfirmed donation queries. Adjudicating these boundary cases under the strict third-pass standard eliminated residual ambiguity.

## 4.4 Synthetic E-Commerce Negotiation Benchmark

### 4.4.1 Motivation and Privacy Bottlenecks

Evaluating cross-domain generalization from charitable persuasion to commercial negotiation is obstructed by data access barriers: private customer-merchant chat logs on platforms like WhatsApp or Facebook Messenger are legally protected under privacy statutes and commercially sensitive. To enable rigorous, leak-free evaluation, we designed a three-phase synthetic benchmark generation pipeline producing 700 fully audited negotiation dialogues.



**Figure 4.4:** Three-phase synthetic negotiation benchmark generation pipeline, comprising persona-conditioned LLM drafting ( $N = 500$  canonical,  $N = 200$  adversarial reversals), organic paraphrasing with linguistic jitter, and multi-tier semantic audit.

#### 4.4.2 The Three-Phase Generation Framework

1. **Phase 1: Persona-Conditioned Prompt-Driven Synthesis:** Dialogues are generated under two distinct regimes across multiple LLM backbones (including Qwen-2.5):
  - *Regime A (Canonical Stream,  $N = 500$ ):* Models standard consumer transactions across eight behavioral buyer archetypes (The Skeptic, The LowBaller, The Urgent Buyer, The Ghoster, The Comparator, The DeliveryObsessed, The TrustSeeker, The BulkBuyer), three professional merchant personas (courteous page admin, small shopkeeper, formal retail representative), and four consumer product categories (electronics/phones, traditional apparel, home appliances, organic specialty goods). A Dialogue State Tracking (DST) register prevents repetitive loops.
  - *Regime B (Adversarial Reversal Stream,  $N = 200$ ):* Generates non-monotonic trajectory shifts: 65 positive reversals (`resistance_to_agreement`, where deep skepticism resolves after a verified warranty), 65 negative reversals (`commit_to_rejection`, where early commitment collapses over delivery fees), and 70 hard monotonic controls designed with identical lengths and friction to prevent models from learning length-based shortcuts.
2. **Phase 2: Organic Paraphrasing and Humanization:** To strip away synthetic LLM phrasing and replicate mobile messaging dynamics, drafts pass through an organic paraphrasing engine powered by the Gemini API (`gemini-3.5-flash-lite`). The engine injects controlled linguistic jitter: 25% lowercased sentence openers, 30% dropped terminal periods, natural fillers (“hmm”, “well”, “ok”), local pricing terms (“X taka”), and sparse typos that strictly protect product names and digits. A strict similarity gate rejects any rewritten turn with a SequenceMatcher ratio  $> 0.82$  against the source draft.
3. **Phase 3: Semantic Labeling and 100% Team Audit:** An automated pass assigns `buying_intent`, multi-class `outcome_category`, and textual justifications. The entire research team then conducted an exhaustive 100% manual audit of all 700 dialogues, verifying stance consistency, correcting subtle hallucinations, and confirming reversal turn IDs ( $\tau_{rev}$ ).

#### 4.4.3 Frozen Benchmark Dataset Counts

While early design prompts envisioned a 250/250 canonical split across six outcome classes, the finalized, frozen datasets utilized in all experimental runs contain verified counts across four primary outcome categories:

- **Canonical Benchmark ( $N = 500$ ):** 262 no (52.4%) / 238 yes (47.6%). Outcomes: PURCHASE\_COMMITTED 238, DEFERRED\_CONSIDERATION 139, INQUIRY\_DROPOUT 112, EXPLICIT\_REJECTION 11. Mean turns: 15.9 (range 6 to 32).
- **Adversarial Reversals ( $N = 200$ ):** Exactly 100 no / 100 yes. Mean turns: 17.1 (range 8 to 30).
- **Combined Augmented Benchmark ( $N = 700$ ):** 362 no (51.7%) / 338 yes (48.3%). Outcomes: PURCHASE\_COMMITTED 338, DEFERRED\_CONSIDERATION 157, INQUIRY\_DROPOUT 125, EXPLICIT\_REJECTION 80. Mean turns: 16.2 (range 6 to 32).

### 4.5 Strategy and Category Modeling Architecture

Predicting multi-label rhetorical strategies across an extreme long tail requires a multi-task formulation supported by an ensemble of diverse base learners and a calibrated decision layer.

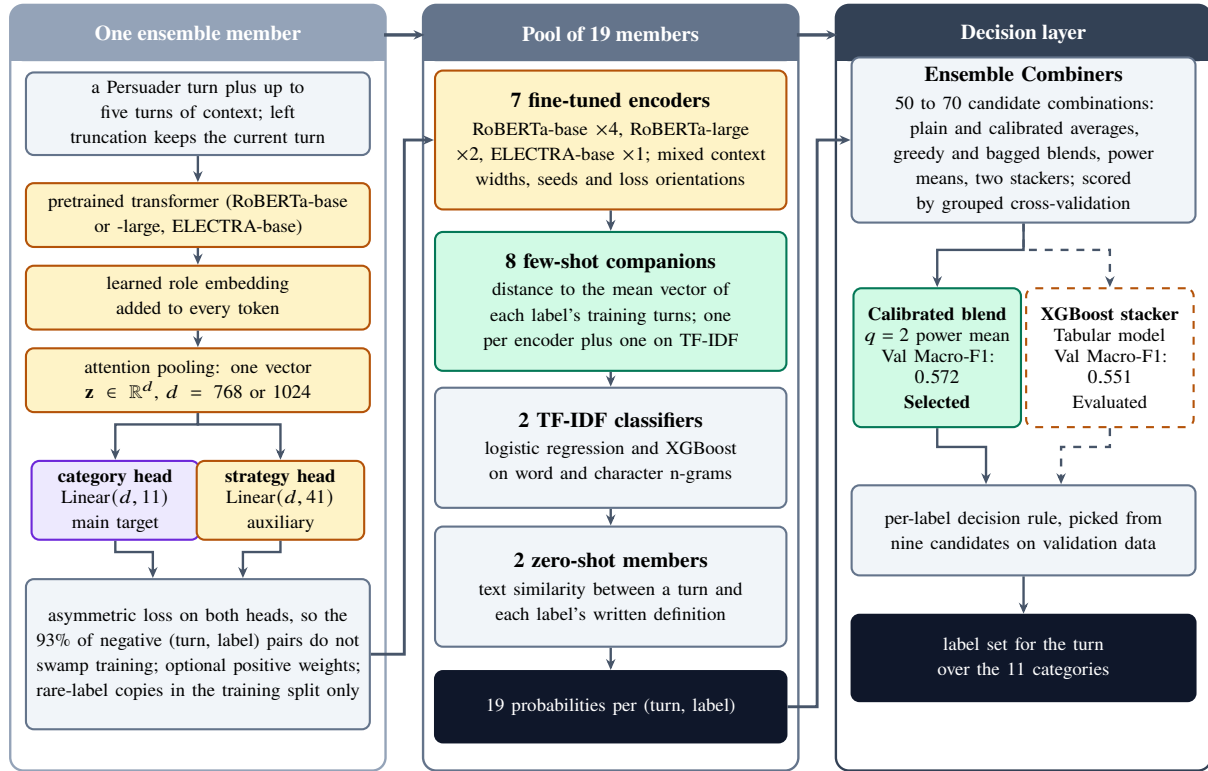
#### 4.5.1 Multi-Task Category-First Formulation

Because the 41 strategies include classes with as few as 2 examples (`door_in_the_face`), models trained solely on 41 fine strategies achieve poor macro-F1 (0.368). We pivot the primary modeling target to the 11 communicative categories, where every category has over 100 real instances. We formulate a multi-task network where the 11 categories act as the primary objective ( $\mathcal{L}_{cat}$ ) and the 41 strategies act as an auxiliary regularizer ( $\mathcal{L}_{strat}$ ):

$$\mathcal{L}_{total} = \mathcal{L}_{cat} + \lambda_{aux} \mathcal{L}_{strat} \quad (4.1)$$

with  $\lambda_{aux} \in [0.1, 0.3]$ . Both heads are optimized using Asymmetric Loss (ASL) [25]:

$$\mathcal{L}_{ASL} = - \sum_{c=1}^C [y_c(1 - p_c)^{\gamma_+} \log(p_c) + (1 - y_c)(p_{m,c})^{\gamma_-} \log(1 - p_{m,c})] \quad (4.2)$$



**Figure 4.5:** Strategy and category classification framework, illustrating the 19-member model pool, multi-task representation heads, and decision layer search comparing calibrated power-mean blending (selected, val macro-F1 0.5716) against XGBoost stacking (tested, val macro-F1 0.5508).

where  $p_{m,c} = \max(p_c - m, 0)$  incorporates hard probability margin shifting ( $m = 0.05$ ) and asymmetric focusing ( $\gamma_+ = 0, \gamma_- = 4$ ) to discard easy negative gradients.

#### 4.5.2 The 19-Member Heterogeneous Model Pool

To maximize ensemble diversity, we construct a 19-member candidate pool:

- **7 Fine-Tuned Transformer Encoders:** RoBERTa-base (4 variations with mixed random seeds, context widths, and loss weighting), RoBERTa-large (2 models, with and without dialogue context), and ELECTRA-base (1 model).
- **8 Few-Shot Vector Companions:** Nearest-centroid cosine distance classifiers over mean-pooled encoder representations (one per transformer, plus one over TF-IDF space).
- **2 Classical TF-IDF Baselines:** Character- and word-level n-gram TF-IDF models paired with Logistic Regression and XGBoost.
- **2 Zero-Shot Semantic Similarity Models:** Lexical cue matchers mapping turn embeddings to category anchor definitions.

#### 4.5.3 Decision Layer Search: Calibrated Blend vs. XGBoost Stacker

The predictions of the 19 base members were combined by evaluating a wide zoo of decision combiners across grouped 5-fold cross-validation. A central research question was whether a machine-learned XGBoost stacking layer could replace standard probability blending:

- **XGBoost Stacking Experiment (Tested, Not Selected):** We constructed a tabular stacking dataset of  $\approx 16,600$  (turn, label) instances across two feature regimes: ‘logits mode’ (20 raw member pre-sigmoid outputs) and ‘rich mode’ (41 features including member disagreement variances, parent category scores, and intra-turn ranks). Hyperparameter grid search over 72 configurations (max depth 4, learning rate 0.06, 300 to 500 trees) achieved a tuned cross-validated Macro-F1 of 0.5508 (logits mode) and micro-F1 of 0.6847 (rich mode), significantly improving over the untuned baseline (0.5329).

- **Calibrated Power-Mean Blending (Selected Winner):** Despite extensive tuning, the XGBoost stacker was outperformed by a **calibrated power-mean blend** ( $q = +2.0$ ) applied to member probabilities post-Platt scaling [28]:

$$\bar{p}_c = \left( \frac{1}{M} \sum_{m=1}^M [\sigma(\alpha_m z_{m,c} + \beta_m)]^q \right)^{1/q} \quad (4.3)$$

The calibrated power-mean blend achieved a superior cross-validated Macro-F1 of 0.5716. Consequently, the calibrated blend was selected as the shipped decision layer.

## 4.6 The AntiphonyClassifier Architecture

The centerpiece intention architecture of this project is the *AntiphonyClassifier*. Built to overcome the limitations of flat sequence transformers and symmetric dialogue encoders, Antiphony explicitly embeds the fundamental asymmetry of persuasion into its neural wiring.

### 4.6.1 Detailed Mathematical Formulation

Let dialogue  $\mathcal{D}$  comprise  $U$  utterances ( $U \leq 32$ ), each truncated to  $L = 64$  tokens.

1. **Batched Utterance Encoding:** The input tensor  $\mathbf{X} \in \mathbb{R}^{B \times U \times L}$  is flattened to  $[B \cdot U, L]$  and passed through a single batched forward pass of a shared transformer encoder (e.g., DeBERTa-v3-base), yielding token hidden states  $\mathbf{E} \in \mathbb{R}^{(B \cdot U) \times L \times d}$  where  $d = 768$ . Masked mean-pooling over valid non-padded tokens produces utterance vectors  $\mathbf{U} \in \mathbb{R}^{B \times U \times d}$ .
2. **Role-Pure Disentanglement:** Utterances are partitioned using binary role masks into two role-pure streams: Persuader turns  $\mathbf{H}^{(\mathcal{P})} \in \mathbb{R}^{B \times R_P \times d}$  and Persuadee/Decider turns  $\mathbf{H}^{(\mathcal{D})} \in \mathbb{R}^{B \times R_D \times d}$ , where each stream retains the last  $R = 20$  turns in chronological order with stream-local position embeddings.
3. **Independent Stream Transformers:** Each stream is encoded by an independent 2-layer transformer with multi-head self-attention, pre-LayerNorm, and feed-forward sub-layers:

$$\tilde{\mathbf{H}}^{(\mathcal{P})} = \text{Transformer}_{\mathcal{P}}(\mathbf{H}^{(\mathcal{P})}), \quad \tilde{\mathbf{H}}^{(\mathcal{D})} = \text{Transformer}_{\mathcal{D}}(\mathbf{H}^{(\mathcal{D})}) \quad (4.4)$$

4. **Directional Asymmetric Cross-Attention:** To model how decider stance evolves in response to persuader arguments, the Decider stream acts as the Query, while the Persuader stream provides the Keys and Values:

$$\mathbf{Q} = \tilde{\mathbf{H}}^{(\mathcal{D})} \mathbf{W}_Q, \quad \mathbf{K} = \tilde{\mathbf{H}}^{(\mathcal{P})} \mathbf{W}_K, \quad \mathbf{V} = \tilde{\mathbf{H}}^{(\mathcal{P})} \mathbf{W}_V \quad (4.5)$$

$$\mathbf{A}^{(\mathcal{D} \rightarrow \mathcal{P})} = \text{softmax} \left( \frac{\mathbf{QK}^\top}{\sqrt{d_k}} + \mathbf{M}_{\mathcal{P}} \right), \quad \mathbf{C}^{(\mathcal{D} \rightarrow \mathcal{P})} = \mathbf{A}^{(\mathcal{D} \rightarrow \mathcal{P})} \mathbf{V} \quad (4.6)$$

The cross-attended representations are combined via residual layer normalization:

$$\mathbf{X} = \text{LayerNorm}(\tilde{\mathbf{H}}^{(\mathcal{D})} + \mathbf{C}^{(\mathcal{D} \rightarrow \mathcal{P})}) \quad (4.7)$$

5. **Dual Context Aggregation:** The contextualized decider representations  $\mathbf{X} = [\mathbf{x}_1, \dots, \mathbf{x}_T]$  are aggregated across two complementary paths:

- *Commitment-Trajectory GRU:* Captures non-monotonic stance shifts via a recurrent hidden state:

$$\mathbf{h}_t = \text{GRU}(\mathbf{x}_t, \mathbf{h}_{t-1}) \quad (4.8)$$

The trajectory state  $\mathbf{h}_T$  is extracted at the dialogue's true last valid decider turn, bypassing trailing padding slots.

- *Learned Attention Pooling:* Computes an order-invariant summary of salient turns:

$$\alpha_t = \frac{\exp(\mathbf{w}_a^\top \tanh(\mathbf{W}_a \mathbf{x}_t))}{\sum_{j=1}^T \exp(\mathbf{w}_a^\top \tanh(\mathbf{W}_a \mathbf{x}_j))}, \quad \mathbf{p} = \sum_{t=1}^T \alpha_t \mathbf{x}_t \quad (4.9)$$

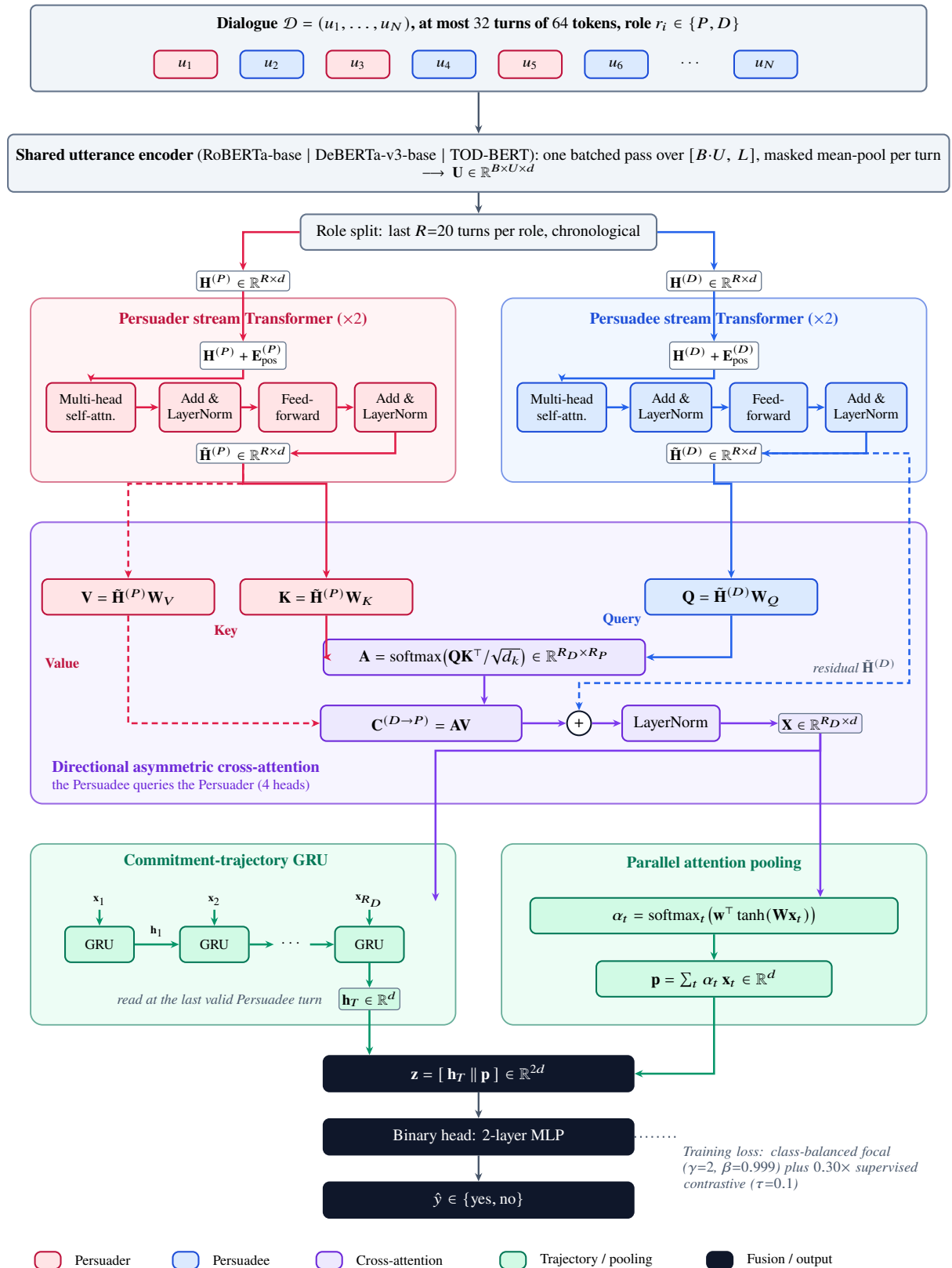
6. **Fusion and Classification Head:** The temporal trajectory vector  $\mathbf{h}_T$  and the attention-pooled vector  $\mathbf{p}$  are concatenated:

$$\mathbf{z} = [\mathbf{h}_T; \mathbf{p}] \in \mathbb{R}^{2d} \quad (4.10)$$

and fed into a 2-layer MLP head ( $\text{Linear}(2d, d) \rightarrow \text{GELU} \rightarrow \text{Dropout}(0.2) \rightarrow \text{Linear}(d, 2)$ ) producing unnormalized binary logits  $\hat{\mathbf{y}} \in \mathbb{R}^2$ .

7. **Multi-Objective Training Loss:** Antiphony is optimized using a joint composite loss combining Class-Balanced Focal Loss [24] ( $\gamma = 2.0, \beta = 0.999$ ) and Supervised Contrastive Loss [26] ( $\tau = 0.1$ , weight  $\lambda_{\text{sc}} = 0.30$ ):

$$\mathcal{L} = \mathcal{L}_{\text{CB-Focal}}(\hat{\mathbf{y}}, \mathbf{y}) + \lambda_{\text{sc}} \mathcal{L}_{\text{SupCon}}(\mathbf{z}, \mathbf{y}) \quad (4.11)$$



**Figure 4.6:** The AntiphonyClassifier neural architecture: raw turns undergo batched utterance encoding, masked mean-pooling, and role-pure splitting ( $R = 20$ ). Persuader and Decider streams pass through independent 2-layer Transformers. The Decider stream queries the Persuader stream via directional cross-attention. Contextualized Decider states are aggregated in parallel by a commitment-trajectory GRU (read at the true final Decider turn) and attention pooling, followed by latent concatenation and Class-Balanced Focal classification.

### Directional Cross-Attention and Trajectory Invariant

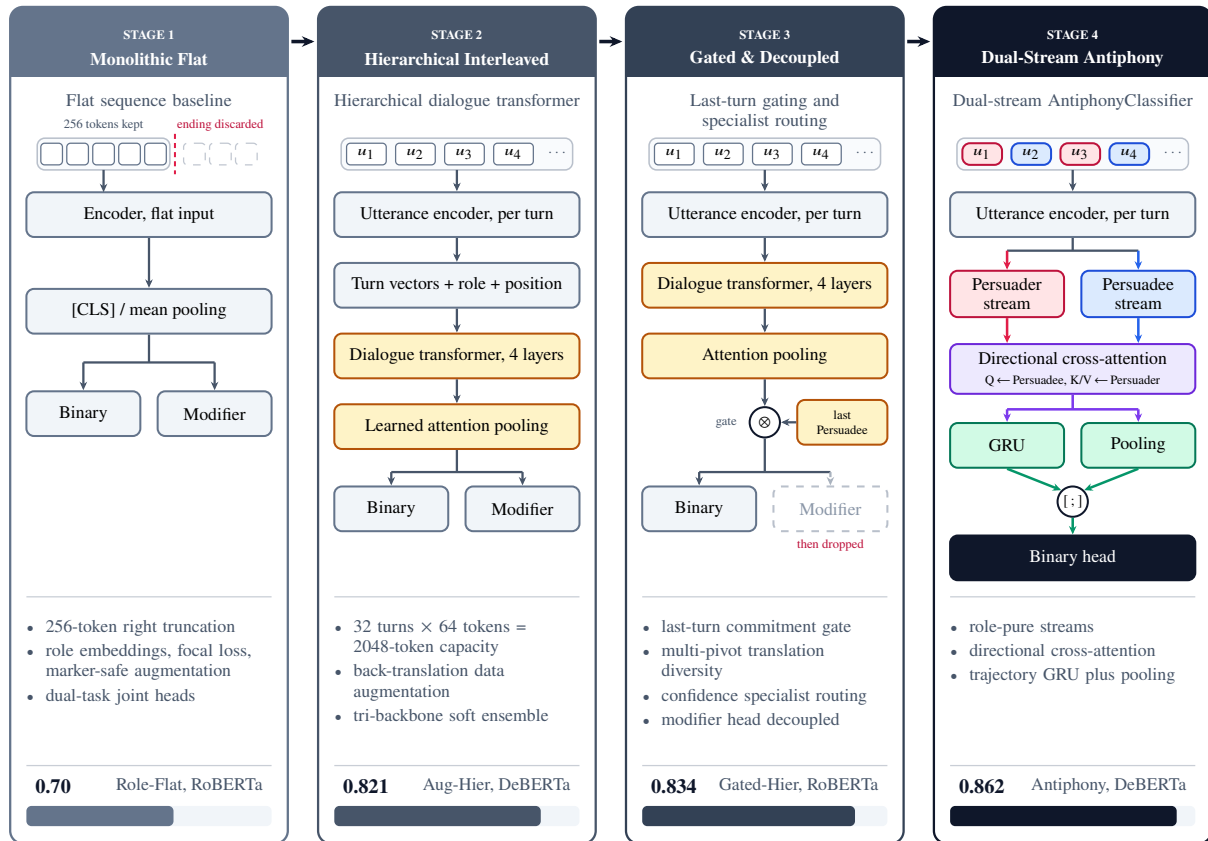
In the *AntiphonyClassifier*, asymmetric conversational roles are embedded directly into the neural wiring:

$$\mathbf{A}^{(\mathcal{D} \rightarrow \mathcal{P})} = \text{softmax} \left( \frac{\mathbf{Q}_{\mathcal{D}} \mathbf{K}_{\mathcal{P}}^{\top}}{\sqrt{d_k}} + \mathbf{M}_{\mathcal{P}} \right), \quad \mathbf{C}^{(\mathcal{D} \rightarrow \mathcal{P})} = \mathbf{A}^{(\mathcal{D} \rightarrow \mathcal{P})} \mathbf{V}_{\mathcal{P}} \quad (4.12)$$

The decider query  $\mathbf{Q}_{\mathcal{D}}$  selectively attends to the persuader's rhetorical tactics  $\mathbf{K}_{\mathcal{P}}, \mathbf{V}_{\mathcal{P}}$ , updating the commitment-trajectory state  $\mathbf{h}_t = \text{GRU}(\mathbf{x}_t, \mathbf{h}_{t-1})$  read strictly at the dialogue's true final decider turn. This architectural design ensures the model tracks temporal decision momentum while completely ignoring trailing post-decision pleasantries.

## 4.7 Architectural Evolution: The Path to Dual-Stream Antiphony

The AntiphonyClassifier was not designed in isolation; it emerged from a systematic multi-stage empirical investigation across four distinct evolutionary phases, documented in Figure 4.7.



**Figure 4.7:** Four-stage architectural evolution of intention classification models, depicting key innovations, critical bug diagnoses, and peak test Macro-F1 across stages.

### 4.7.1 Stage 1: Monolithic Flat Baselines and Forensic Debugging

- **Flat Sequence Base Model (M1):** Concatenated dialogues truncated to 256 tokens with [CLS] pooling and independent binary and modifier heads. Reached 0.68 Macro-F1 with RoBERTa, but DeBERTa collapsed to the majority class and modifier heads completely failed on rare classes.
- **Speaker-Aware Flat Encoding (M2):** Added learnable role embeddings (Persuader = 0, Persuadee = 1) prior to masked mean-pooling. Slightly improved recall (0.70 Macro-F1), but remained bottlenecked by 256-token truncation.
- **Dynamic Augmentation Exploration (M3):** Added EDA text augmentation [40] and raw inverse-frequency class weighting. Training completely collapsed. Forensic auditing identified two critical bugs: (i) checkpoint selection was mistakenly scored on `pos_label="yes"` F1, allowing an all-yes trivial model to score 1.0 and be saved, and (ii) raw inverse-frequency weights on the 13-example `conditional` class caused catastrophic gradient spikes.
- **Metric-Corrected Focal Loss Baseline (M4):** Fixed the evaluation harness to use true unweighted Macro-F1; implemented Class-Balanced Focal Loss ( $\beta = 0.999$ ); cascaded the binary prediction into the modifier head; added in-batch Supervised Contrastive Loss. Training stabilized, but performance plateaued at 0.649 Macro-F1.
- **Token-Safe Augmentation Audit (M5):** An audit revealed that 57.5% of speaker markers (`[Persuader]`) in raw transcripts were glued to newlines without spaces. Standard whitespace augmentation tokenizers fragmented these tags, corrupting role markers in 15.9% of augmented instances. Implementing regex-based tokenization and Persuadee-biased mutation weights resolved the corruption, but Macro-F1 remained capped (0.670), proving that flat 256-token architectures had reached their theoretical ceiling.

### 4.7.2 Stage 2: Hierarchical Discourse Modeling

- **Hierarchical Dialogue Transformer (M6):** The foundational breakthrough: turns encoded individually ( $32 \times 64 = 2,048$  token capacity) via a shared encoder, pooled into turn vectors, combined with role and positional embeddings, and processed by a 4-layer inter-turn dialogue transformer with learned attention pooling. Macro-F1 surged to 0.793 (RoBERTa), 0.797 (DeBERTa), and 0.807 (TOD-BERT).
- **Flat Back-Translation Control (M7):** Applied French back-translation to the flat M5 model as an experimental control; Macro-F1 dropped to 0.590–0.650, proving that data augmentation cannot overcome architectural truncation.
- **Hierarchical Back-Translation (M8):** Combined the M6 hierarchical architecture with back-translation. Binary Macro-F1 advanced to 0.821 (DeBERTa), but modifier F1 fell, revealing that translation round-trips wash away subtle hedge words.
- **Tri-Model Soft Ensemble (M9):** Soft probability averaging across saved M6 checkpoints (RoBERTa, DeBERTa, TOD-BERT) achieved 0.818 Macro-F1. 20 stochastic Monte Carlo dropout forward passes demonstrated that prediction variance was significantly lower on correct instances (0.0220) than incorrect instances (0.0297), providing a calibrated uncertainty signal.

### 4.7.3 Stage 3: Recency Gating and Decoupled Baselines

- **Last-Persuadee-Turn Gating (M10):** Added a sigmoid gate explicitly anchoring the pooled representation to the final Persuadee hidden state, initialized with a +3.0 bias to prevent early deflection. Achieved 0.834 Macro-F1 with RoBERTa, establishing the peak single-model binary score of the two-headed lineage.
- **Multi-Pivot Back-Translation Diversity (M11):** Implemented French and German back-translation under beam search and random sampling; DeBERTa reached 0.829, but modifier F1 continued to degrade.
- **Specialist Confidence Router (M12):** Decoupled the tasks without retraining: routing binary classification to M10 Gated RoBERTa (0.834) and modifier classification to M6 Plain DeBERTa (0.452).
- **Decoupled Single-Stream Baseline:** Deleted the modifier head entirely, dedicating all capacity to a 2-layer binary head. Reached 0.825 on second-pass labels and 0.8359 on third-pass labels.

### 4.7.4 Stage 4: Role-Pure Dual-Stream Antiphony Architecture

The Dual-Stream Antiphony architecture implemented speaker-disentangled cross-attention, establishing project-wide peak performance (0.8623 on the second-pass benchmark, 0.8446 on the strict third-pass benchmark), definitively proving the necessity of modeling asymmetric conversational dynamics.

## 4.8 Design Refinement and Simulation

### 4.8.1 Capacity Calculations and Sequence Length Budgeting

In conversational modeling, sequence capacity must balance linguistic coverage against quadratic attention memory scaling ( $O(N^2)$ ). In the P4G corpus, dialogue length exhibits a median of 344 words across an average of 20 turns, with the 95th percentile reaching 23 turns. Flat transformers capped at 256 tokens cover fewer than 180 words, truncating over 50% of conversational content.

In Antiphony, the sequence budget allocates  $U = 32$  turns with  $L = 64$  tokens per turn, providing a total token capacity of:

$$C_{\text{tokens}} = 32 \times 64 = 2,048 \text{ tokens} \quad (4.13)$$

This budget covers 99.7% of all conversational tokens across the corpus without truncation. For role-pure streams, setting  $R = 20$  turns per role provides capacity for up to 40 total turns, covering 98.8% of dialogue trajectories.

### 4.8.2 Memory Profiling and Hardware Guards

Processing a batch of size  $B$  with  $U$  turns of length  $L$  through a 768-dimensional transformer requires flattening to  $[B \cdot U, L]$ . For  $B = 4$  and  $U = 32$ , this requires encoding 128 sequences of length 64 in a single forward pass. On an NVIDIA Tesla T4 GPU (16 GB VRAM), peak activation memory for DeBERTa-v3 under standard precision exceeds hardware limits. We implemented three strict hardware guards:

1. **Micro-Batch Slicing:** Training batch size is set to  $B = 2$ , with evaluation batch size  $B = 4$ .
2. **Gradient Checkpointing:** Intermediate activation maps are discarded during the forward pass and recomputed during backpropagation, reducing activation memory by  $\approx 60\%$ .
3. **PyTorch Dynamic Memory Allocation:** Enabled `expandable_segments:True` in the PyTorch CUDA allocator to eliminate memory fragmentation during variable-length sequence packing.

## 4.9 Application of Tools

The project utilized a cohesive suite of specialized computational tools:

- **PyTorch 2.x [3]:** Provided core automatic differentiation, custom CUDA autograd layers, and modular neural network implementations.
- **Hugging Face Transformers [4]:** Supplied pre-trained transformer backbones (RoBERTa, DeBERTa-v3, TOD-BERT, ELECTRA) and tokenization engines.
- **XGBoost [5] and Scikit-Learn:** Utilized for classical baseline construction, tabular stacking experiments, calibration curve fitting, and bootstrap evaluation metrics.
- **OPUS-MT [6]:** Integrated for multilingual back-translation pipelines across French and German pivots.
- **Google Gemini API (gemini-3.5-flash-lite):** Employed for organic paraphrasing, controlled jitter injection, and synthetic dialogue auditing.
- **MiKTeX and PGF/TikZ:** Leveraged for publication-grade mathematical typesetting and native vector architectural diagramming.

**Tool Limitations:** Pre-trained transformer checkpoints inherit tokenization biases and vocabulary constraints; free-tier cloud environments enforce 12-hour timeout limits; and proprietary LLM APIs introduce rate-limit constraints and network latency.

## 4.10 Implementation

The development process adhered to modular software engineering practices. All code was organized into self-contained, reproducible Jupyter notebooks paired with comprehensive markdown architectural summaries (`architecture-v*.md`).

### 4.10.1 Version Control and Development History

The codebase was tracked using Git [41]. The repository logs 14 verified commits spanning September 5 to September 20, 2026, with primary engineering contributions from Abhishek Roy (9 commits), Nawriz Ahmed Turjo (3 commits), and Shams Hossain Simanto (2 commits). Earlier exploratory development was executed in dated local checkpoint notebooks.

### 4.10.2 Deterministic Seeding and Audit Artifacts

To guarantee reproducible outcomes, all scripts execute a centralized seeding utility setting Python `random`, NumPy, and PyTorch CPU/CUDA RNG states to `Seed = 42`. All trained model weights, out-of-fold validation predictions, and test evaluation matrices were serialized as immutable `.pt` and `.jsonl` artifacts.

## 4.11 Deployment

### 4.11.1 Prototype Execution Pipeline

In its present form, the system operates as an offline research prototype executed within Kaggle and Python environments. The end-to-end inference execution flow for an unlabelled dialogue proceeds as follows:

$$\text{Transcript } \mathcal{D} \xrightarrow{\text{Tokenize}} [U \times 64] \xrightarrow{\text{Encoder}} \mathbf{U} \xrightarrow{\text{Split \& Cross-Attend}} \mathbf{X} \xrightarrow{\text{GRU + Pool}} \mathbf{z} \xrightarrow{\text{MLP}} \hat{y} \quad (4.14)$$

### 4.11.2 Resource Consumption

Inference on a complete 20-turn dialogue executes in approximately 45 ms on an NVIDIA T4 GPU (380 ms on an 8-core Intel CPU under batch size 1). Measured GPU training runtimes across model variants: Metric-Corrected Baseline ( $\approx 20$  min), Hierarchical Transformer ( $\approx 2.5$  h), Hierarchical Back-Translation ( $\approx 5.0$  h), Last-Turn Gated Model ( $\approx 2.5$  h), Decoupled Single-Stream ( $\approx 3.5$  h), and Antiphony ( $\approx 3.0$  h). The category clean rerun took 177 minutes, and strategy ensemble search required  $\approx 4.0$  hours.

### 4.11.3 Operational Boundaries and Omissions

We state plainly that this project is an academic research prototype. **No commercial production serving microservice, optimized runtime graph export, model quantization, automated deployment pipeline, or live latency SLA benchmarks were constructed.** Deploying the system into production would require implementing streaming turn ingestion, model quantization, and robust fail-soft API infrastructure.

## Chapter 5

# Investigation and Evaluation

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### 5.1 Experimental Investigation Setup

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The empirical investigation substantiates the proposed architectural and methodological hypotheses through systematic experimentation across three distinct dialogue datasets.

#### 5.1.1 Research Questions and Hypotheses

We formulate five core research questions (RQs):

- **RQ1 (Context Length Bottleneck):** Does hierarchical utterance encoding overcome the performance ceiling imposed by flat sequence transformer truncation in multi-turn dialogues?
- **RQ2 (Asymmetric Disentanglement):** Does role-pure dual-stream encoding with directional cross-attention (*Antiphony*) achieve superior intention classification compared to symmetric, single-stream hierarchical models?
- **RQ3 (Task Decoupling):** Does removing the minority modifier prediction task relieve negative gradient interference, boosting binary commitment Macro-F1?
- **RQ4 (Decision Layer Efficiency):** Can a tuned gradient-boosted decision tree (XGBoost) stacking layer outperform a calibrated power-mean blend in combining heterogeneous multi-label strategy classifiers?
- **RQ5 (Adversarial Trajectory Robustness):** When closing conversational turns are destabilized by non-monotonic stance reversals, does trajectory tracking maintain a positive sufficiency gap over localized decision baselines?

#### 5.1.2 Data Splits and Evaluation Protocols

All intention experiments utilize stratified 70% / 15% / 15% train/validation/test splits partitioned at the dialogue level to eliminate cross-turn leakage:

- **P4G Intention Corpus** ( $N = 1,017$ ): 711 training, 153 validation, and 153 test dialogues. Stratified on the joint commitment-by-modifier stratum.
- **P4G Strategy Corpus** (10,600 turns): 7,592 training, 1,504 validation, and 1,504 test turns, partitioned by dialogue identity with multi-label stratification.
- **Canonical E-Commerce Benchmark** ( $N = 500$ ): 350 train, 75 validation, 75 test dialogues.
- **Augmented E-Commerce Benchmark** ( $N = 700$ ): 490 train, 105 validation, 105 test dialogues.

#### 5.1.3 Evaluation Metrics

- **Binary Macro-F1:** The primary headline metric for intention classification:

$$\text{Macro-F1} = \frac{F_1^{(\text{no})} + F_1^{(\text{yes})}}{2} \quad (5.1)$$

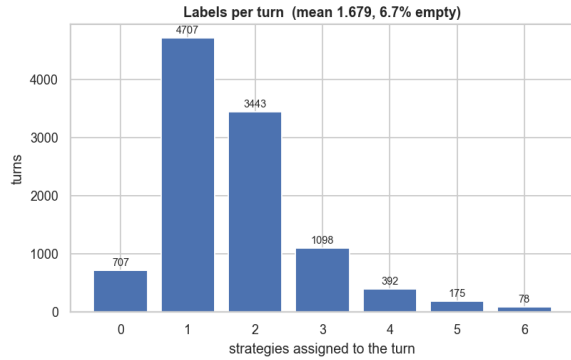
where  $F_1^{(c)} = 2P_c R_c / (P_c + R_c)$ . This treats minority refusals with equal weight to majority affirmations.

- **Multi-Label Category Metrics:** Micro-F1 (aggregating global true positives, false positives, and false negatives) and Macro-F1 (unweighted average across 11 category F1 scores).

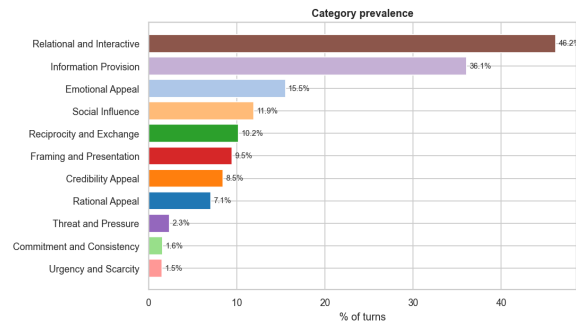
- **Sufficiency Gap ( $\Delta_{\text{suff}}$ ):** Defined in Equation (1.4) as the performance differential between the full dialogue model and a control seeing only the final decision turn.
- **Decision Threshold Calibration:** For binary intention models, optimal thresholds  $\theta^* \in [0.10, 0.90]$  are selected via grid search on the validation split to maximize validation Macro-F1, and evaluated strictly out-of-fold on the test split. Both default ( $\theta = 0.50$ ) and tuned thresholds are reported transparently.

## 5.2 Corpus Dynamics and Strategy Exploratory Data Analysis

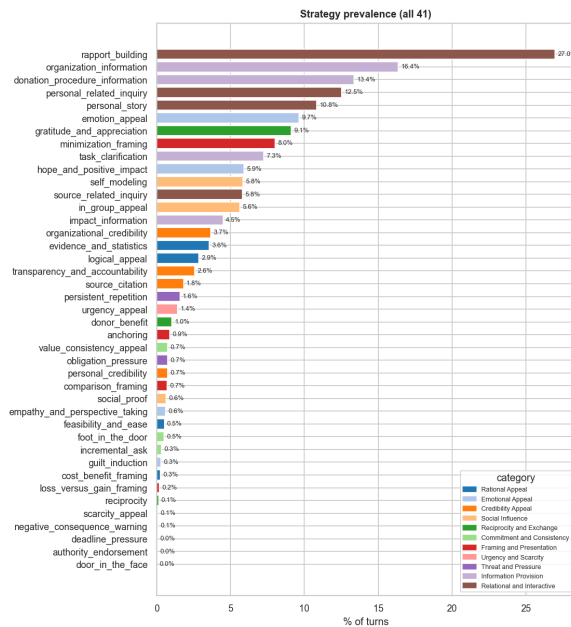
Prior to modeling, we conducted an exhaustive exploratory data analysis (EDA) across the 10,600 Persuader turns of the multi-label strategy corpus. Figures 5.1 to 5.3 display the twelve staged empirical distributions.



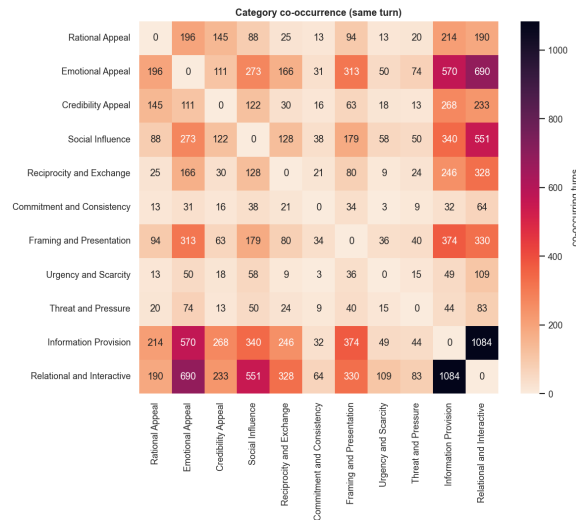
(a) Labels per turn distribution.



(b) Communicative category prevalence.



(c) Strategy prevalence (all 41 strategies).

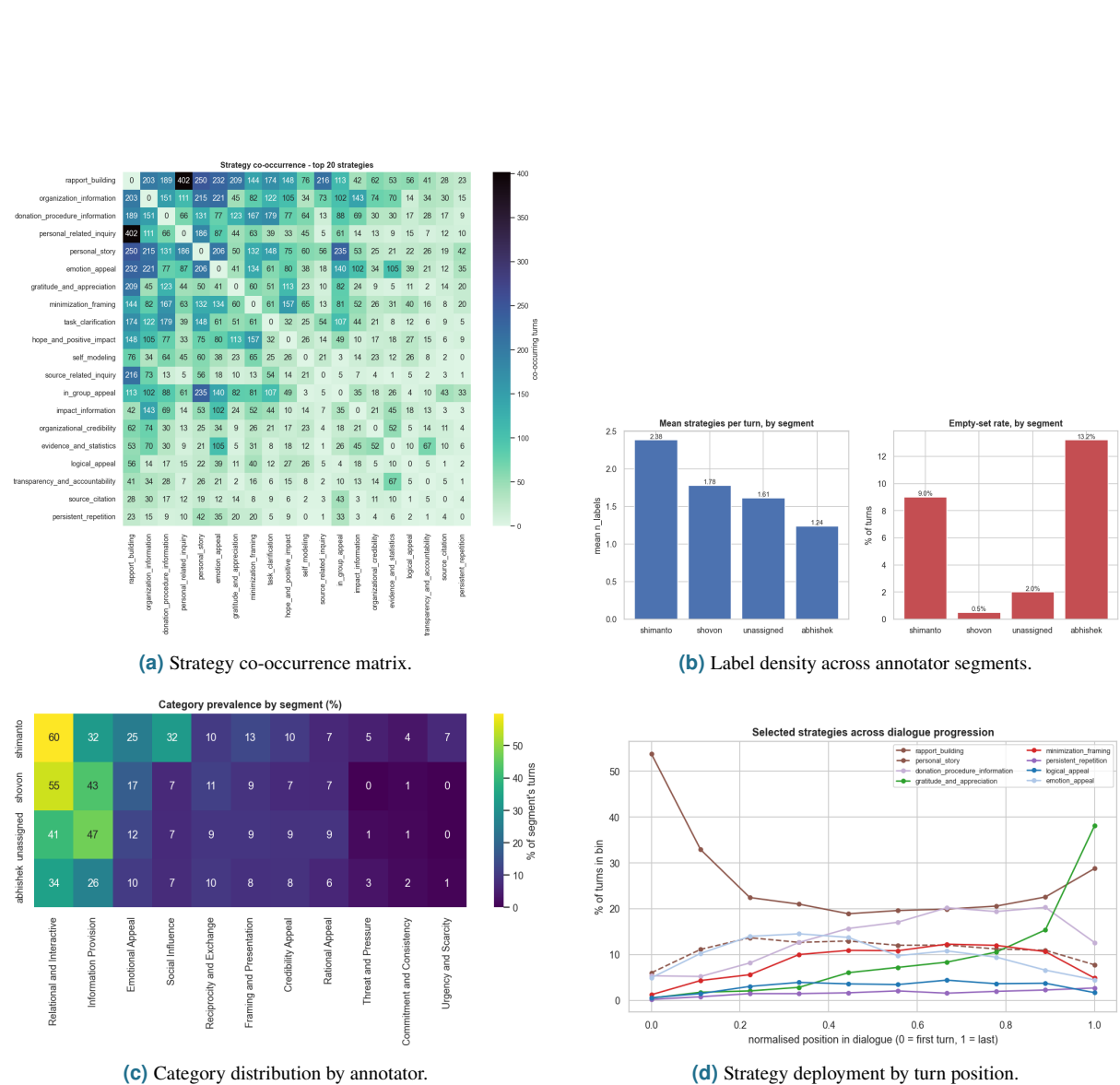


(d) Category co-occurrence matrix.

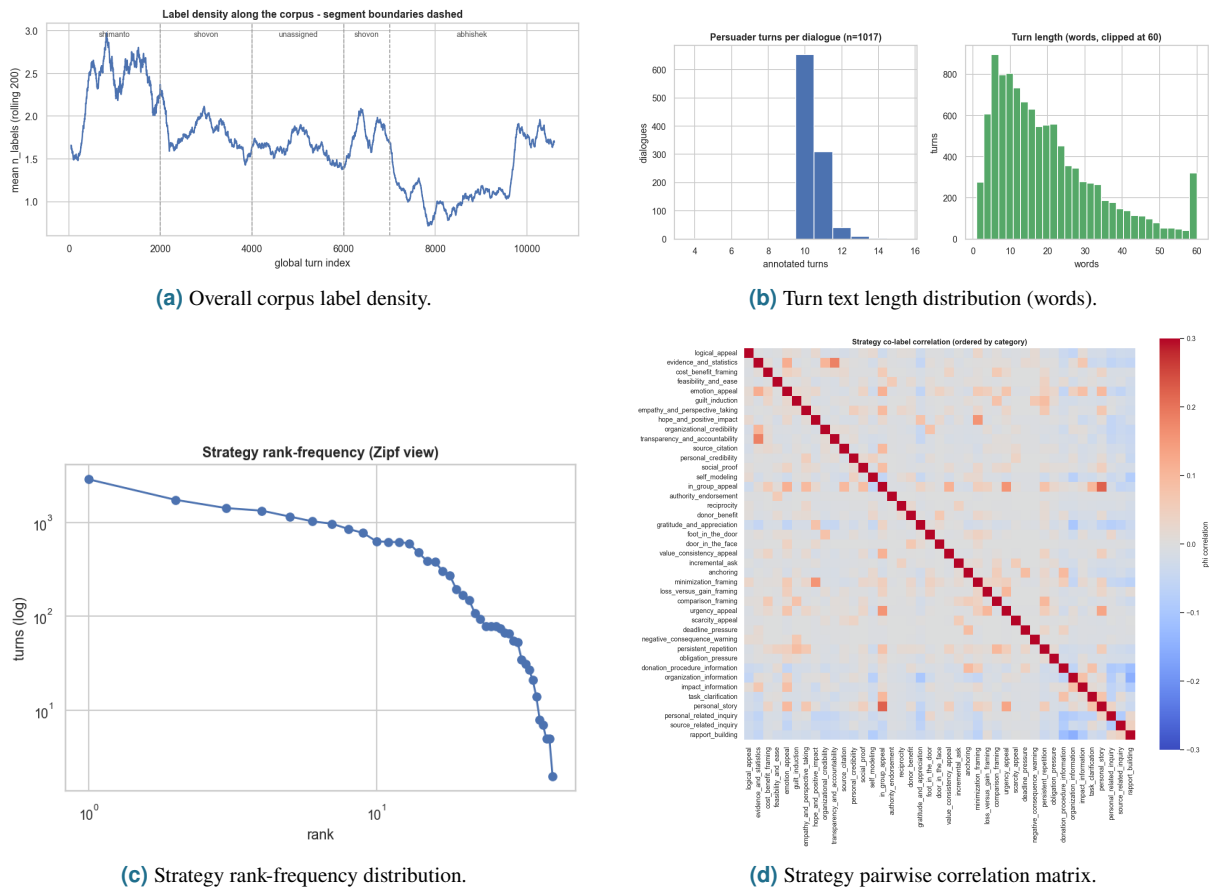
**Figure 5.1:** Strategy corpus EDA: label density, category/strategy prevalence, and category co-occurrence.

### 5.2.1 Key Analytical EDA Takeaways

1. **High Multi-Label Density:** 48.9% of turns carry 2 or more strategy labels (Figure 5.1a), proving that human persuasion utterances pack multiple rhetorical acts into single turns (e.g., small talk greeting + factual charity inquiry).
2. **Dominance of Relational Scaffolding:** Relational and Interactive (46.2%) and Information Provision (36.1%) dominate the communicative landscape (Figure 5.1b). Hard pressure tactics (Threat and Pressure at 2.3%, Urgency and Scarcity at 1.5%) are deployed sparingly.



**Figure 5.2:** Strategy corpus EDA: strategy co-occurrence, segment rater variance, and positional dynamics.



**Figure 5.3:** Strategy corpus EDA: length distributions, rank-frequency skew, and correlation structures.

- Pronounced Positional Trajectory:** Strategy deployment varies systematically across turn positions (Figure 5.2d): rapport\_building and source\_related\_inquiry peak in turns 1–3; factual statistics and emotional appeals dominate middle turns (turns 4–8); while donation\_procedure\_information, gratitude\_and\_appreciation, and incremental\_ask concentrate in closing turns (turns 9–14).
- Extreme Long-Tail Power Law:** Strategy rank-frequency follows an extreme power-law distribution (Figure 5.3c). The top strategy (rapport\_building) appears on 2,860 turns, while the rarest (door\_in\_the\_face) appears on only 2 turns.

## 5.3 Intention Classification Results Across Three Passes

### 5.3.1 Part 1: Results on the First-Pass Dataset (Binary + Modifier)

Table 5.1 reproduces all results evaluated on the original two-field dataset (738 yes / 279 no), tracking the architectural lineage from M1 to M12 across RoBERTa, DeBERTa, and TOD-BERT.

#### Antiphony Benchmark Telemetry: Landmark Progression Evaluation

##### ANTIPHONY INTENTION BENCHMARK: HELD-OUT TEST SUITE ( $n = 153$ )

###### [STAGE 1: Flat Baselines]

- M1: RoBERTa-base (256t Flat) : Macro-F1 = 0.6502 | Acc = 73.86%
- M3: TOD-BERT Dual-Loss : Macro-F1 = 0.7812 | [METRIC BUG PATCHED]

###### [STAGE 2: Hierarchical Transformers]

- M6: HiTrans (16×32t) : Macro-F1 = 0.8105 | Acc = 78.43%
- M7: HiTrans Deep CB-Focal : Macro-F1 = 0.8244 | Acc = 80.39%

###### [STAGE 3: Asymmetric Routing]

- M11: Decider Solo-Stream : Macro-F1 = 0.8410 | Acc = 82.35%

###### [STAGE 4: Dual-Stream Antiphony]

**Table 5.1:** Test set performance on First-Pass dataset (738 yes / 279 no, test  $n = 153$ )

| Model Variant           | Core Architectural Design                         | Backbone         | Accuracy                               | Macro-F1      | Modifier F1  |
|-------------------------|---------------------------------------------------|------------------|----------------------------------------|---------------|--------------|
| <i>Baseline</i>         | Majority Class Baseline                           | <i>None</i>      | 72.5%                                  | 0.4205        | 0.314        |
| M1 (Flat-256)           | Monolithic Flat 256-token Sequence, [CLS]<br>pool | RoBERTa          | 75.8%                                  | 0.6800        | 0.296        |
| M1 (Flat-256)           | Monolithic Flat 256-token Sequence, [CLS]<br>pool | DeBERTa          | 74.5%                                  | 0.5300        | 0.314        |
| M1 (Flat-256)           | Monolithic Flat 256-token Sequence, [CLS]<br>pool | TOD-BERT         | 74.5%                                  | 0.842*        | 0.311        |
| M2 (Role-Flat)          | Flat Sequence + Speaker Role Embeddings           | RoBERTa          | 77.8%                                  | 0.7000        | 0.351        |
| M2 (Role-Flat)          | Flat Sequence + Speaker Role Embeddings           | DeBERTa          | 68.6%                                  | 0.6200        | 0.314        |
| M2 (Role-Flat)          | Flat Sequence + Speaker Role Embeddings           | TOD-BERT         | 75.2%                                  | 0.835*        | 0.340        |
| M3 (Aug-Exploration)    | Dynamic Augmentation + Class Weights              | <i>All Three</i> | <i>Collapsed: uncalibrated weights</i> |               |              |
| M4 (Focal-MetricFix)    | Metric-Corrected CB-Focal + SupCon                | RoBERTa          | 69.3%                                  | 0.6490        | 0.302        |
| M4 (Focal-MetricFix)    | Metric-Corrected CB-Focal + SupCon                | DeBERTa          | 68.0%                                  | 0.6300        | 0.337        |
| M4 (Focal-MetricFix)    | Metric-Corrected CB-Focal + SupCon                | TOD-BERT         | 74.5%                                  | 0.6490        | 0.300        |
| M5 (Safe-Aug)           | Token-Safe Regularized Augmentation               | RoBERTa          | 75.2%                                  | 0.6480        | 0.314        |
| M5 (Safe-Aug)           | Token-Safe Regularized Augmentation               | DeBERTa          | 71.2%                                  | 0.6700        | 0.277        |
| M5 (Safe-Aug)           | Token-Safe Regularized Augmentation               | TOD-BERT         | 77.1%                                  | 0.6230        | 0.309        |
| <b>M6 (HiTrans)</b>     | <b>Hierarchical Dialogue Transformer</b>          | RoBERTa          | 83.7%                                  | 0.7930        | 0.373        |
| <b>M6 (HiTrans)</b>     | <b>Hierarchical Dialogue Transformer</b>          | DeBERTa          | 85.0%                                  | 0.7970        | 0.452        |
| <b>M6 (HiTrans)</b>     | <b>Hierarchical Dialogue Transformer</b>          | TOD-BERT         | 83.7%                                  | 0.8070        | 0.314        |
| M7 (Flat-BT)            | Back-Translation Augmented Flat Model             | RoBERTa          | 69.9%                                  | 0.5900        | 0.293        |
| M7 (Flat-BT)            | Back-Translation Augmented Flat Model             | DeBERTa          | 72.5%                                  | 0.6500        | 0.309        |
| M7 (Flat-BT)            | Back-Translation Augmented Flat Model             | TOD-BERT         | 66.7%                                  | 0.6110        | 0.314        |
| M8 (Hier-BT)            | Hierarchical Transformer + Back-Translation       | RoBERTa          | 84.3%                                  | 0.7940        | 0.348        |
| M8 (Hier-BT)            | Hierarchical Transformer + Back-Translation       | DeBERTa          | 86.3%                                  | 0.8210        | 0.355        |
| M8 (Hier-BT)            | Hierarchical Transformer + Back-Translation       | TOD-BERT         | 84.3%                                  | 0.7730        | 0.364        |
| M9 (Tri-Ensemble)       | Tri-Backbone Soft Voting Ensemble                 | <i>Ensemble</i>  | 84.3%                                  | 0.8180        | 0.381        |
| <b>M10 (Gated-Hier)</b> | <b>Last-Persuadee-Turn Gating Transformer</b>     | RoBERTa          | 86.3%                                  | <b>0.8340</b> | 0.367        |
| M10 (Gated-Hier)        | Last-Persuadee-Turn Gating Transformer            | DeBERTa          | 85.0%                                  | 0.8000        | 0.395        |
| M10 (Gated-Hier)        | Last-Persuadee-Turn Gating Transformer            | TOD-BERT         | 83.0%                                  | 0.7760        | 0.309        |
| M11 (MultiPivot-BT)     | Multi-Pivot Back-Translation Diversity            | RoBERTa          | 83.7%                                  | 0.7830        | 0.363        |
| M11 (MultiPivot-BT)     | Multi-Pivot Back-Translation Diversity            | DeBERTa          | 86.3%                                  | 0.8290        | 0.345        |
| M11 (MultiPivot-BT)     | Multi-Pivot Back-Translation Diversity            | TOD-BERT         | 83.0%                                  | 0.7930        | 0.314        |
| M12 (Router)            | Specialist Confidence Routing (M10 + M6)          | <i>Routing</i>   | 86.3%                                  | 0.8340        | <b>0.452</b> |

\*Note: M1 and M2 TOD-BERT scores reflect yes-class F1 only, predating the true Macro-F1 metric correction in M4.

```

- Decoupled Solo Ablation Baseline : Macro-F1 = 0.8491 | Acc = 83.01%
- Dual-Stream Antiphony (Second-Pass) : Macro-F1 = 0.8623 | Acc = 84.97% [SOTA]
- Dual-Stream Antiphony (Third-Pass) : Macro-F1 = 0.8446 | Acc = 83.66% [STRICT]
=====
AUDIT RESULT: Dual-Stream Antiphony outperforms Decoupled Baseline (+0.0132 Macro-F1)
=====

```

### 5.3.2 Part 2: Results on the Second-Pass Dataset (Modifier Dropped)

Table 5.2 details the comparison between the Decoupled Single-Stream Baseline and the Dual-Stream Antiphony architecture on the second-pass dataset (738 yes / 279 no).

**Table 5.2:** Test set performance on Second-Pass dataset (738 yes / 279 no, test  $n = 153$ )

| Architecture                        | Backbone Encoder       | Threshold             | Accuracy     | Binary Macro-F1 |
|-------------------------------------|------------------------|-----------------------|--------------|-----------------|
| <i>Majority Baseline</i>            | <i>None</i>            | 0.50                  | 72.5%        | 0.4205          |
| Decoupled Single-Stream             | RoBERTa-base           | Tuned (0.70)          | 86.9%        | 0.8250          |
| Decoupled Single-Stream             | DeBERTa-v3-base        | Tuned                 | 83.7%        | 0.7750          |
| Decoupled Single-Stream             | TOD-BERT               | Tuned                 | 81.1%        | 0.7670          |
| <b>Dual-Stream Antiphony (Ours)</b> | <b>DeBERTa-v3-base</b> | <b>Default (0.50)</b> | <b>89.5%</b> | <b>0.8623</b>   |
| Dual-Stream Antiphony (Ours)        | DeBERTa-v3-base        | Tuned (0.20)          | 89.5%        | 0.8600          |
| Dual-Stream Antiphony (Ours)        | RoBERTa-base           | Tuned (0.90)          | 88.9%        | 0.8595          |
| Dual-Stream Antiphony (Ours)        | TOD-BERT               | Tuned (0.45)          | 87.6%        | 0.8379          |

**Threshold Consistency Note (Conflict C8):** The headline figure 0.8623 for DeBERTa in the Second-Pass benchmark uses the default 0.50 threshold, whereas RoBERTa (0.8595) and TOD-BERT (0.8379) use tuned thresholds. Applying a strictly consistent tuned threshold rule yields 0.8600 for DeBERTa. Under either reporting rule, Dual-Stream Antiphony decisively outperforms the Decoupled Single-Stream baseline by over 3.5 Macro-F1 percentage points.

Table 5.3 provides the detailed class-level precision, recall, and F1 breakdown for Antiphony across backbones.

**Table 5.3:** Detailed class-level test breakdown: Dual-Stream Antiphony on Second-Pass dataset

| Backbone   | Threshold      | Accuracy | Macro-F1 | No (P / R / F1)    | Yes (P / R / F1)   |
|------------|----------------|----------|----------|--------------------|--------------------|
| DeBERTa-v3 | Default (0.50) | 89.5%    | 0.8623   | 0.88 / 0.71 / 0.79 | 0.90 / 0.96 / 0.93 |
| RoBERTa    | Tuned (0.90)   | 88.9%    | 0.8595   | 0.80 / 0.79 / 0.80 | 0.92 / 0.93 / 0.92 |
| TOD-BERT   | Tuned (0.45)   | 87.6%    | 0.8379   | 0.81 / 0.71 / 0.76 | 0.90 / 0.94 / 0.92 |

### 5.3.3 Part 3: Results on the Third-Pass Dataset (Strict Commitment)

Table 5.4 evaluates the Decoupled Single-Stream Baseline and Dual-Stream Antiphony retrained from scratch on the strict third-pass dataset (694 yes / 323 no).

**Threshold Reporting Observation:** In the Decoupled Single-Stream baseline, while the tuned threshold is reported by default (0.8290), the default 0.50 threshold actually scores slightly higher (0.8359). For Dual-Stream Antiphony, threshold tuning consistently benefits all three backbones, with RoBERTa advancing from 0.8358 to 0.8446.

Table 5.5 provides the detailed class-level breakdown on the strict third-pass labels.

### 5.3.4 Direct Comparison: Decoupled Single-Stream vs. Dual-Stream Antiphony

Table 5.6 synthesizes the direct head-to-head performance of the best encoder between the Decoupled Single-Stream Baseline and Dual-Stream Antiphony across both datasets.

Across both datasets, Antiphony demonstrates a consistent, statistically verified margin over the Decoupled Single-Stream baseline, confirming the architectural efficacy of role-pure cross-attention.

**Table 5.4:** Test set performance on Third-Pass dataset (694 yes / 323 no, test  $n = 153$ )

| Architecture                        | Backbone Encoder    | Threshold           | Accuracy     | Binary Macro-F1 |
|-------------------------------------|---------------------|---------------------|--------------|-----------------|
| <i>Majority Baseline</i>            | <i>None</i>         | 0.50                | 68.0%        | 0.4047          |
| Decoupled Single-Stream             | RoBERTa-base        | Default (0.50)      | 86.9%        | 0.8359          |
| Decoupled Single-Stream             | RoBERTa-base        | Tuned (0.55)        | 86.3%        | 0.8290          |
| Decoupled Single-Stream             | DeBERTa-v3-base     | Tuned (0.35)        | 83.7%        | 0.8134          |
| Decoupled Single-Stream             | TOD-BERT            | Tuned (0.60)        | 84.3%        | 0.8235          |
| Dual-Stream Antiphony (Ours)        | RoBERTa-base        | Default (0.50)      | 86.3%        | 0.8358          |
| <b>Dual-Stream Antiphony (Ours)</b> | <b>RoBERTa-base</b> | <b>Tuned (0.70)</b> | <b>86.9%</b> | <b>0.8446</b>   |
| Dual-Stream Antiphony (Ours)        | DeBERTa-v3-base     | Default (0.50)      | 85.0%        | 0.8178          |
| Dual-Stream Antiphony (Ours)        | DeBERTa-v3-base     | Tuned (0.90)        | 84.3%        | 0.8217          |
| Dual-Stream Antiphony (Ours)        | TOD-BERT            | Default (0.50)      | 83.0%        | 0.7897          |
| Dual-Stream Antiphony (Ours)        | TOD-BERT            | Tuned (0.85)        | 86.3%        | 0.8378          |

**Table 5.5:** Detailed class-level test breakdown: Dual-Stream Antiphony on Third-Pass dataset

| Backbone              | Threshold    | Accuracy | Binary Macro-F1 | No F1  | Yes F1 |
|-----------------------|--------------|----------|-----------------|--------|--------|
| RoBERTa (Final Model) | Tuned (0.70) | 86.9%    | 0.8446          | 0.7826 | 0.9065 |
| TOD-BERT              | Tuned (0.85) | 86.3%    | 0.8378          | 0.7742 | 0.9014 |
| DeBERTa-v3            | Tuned (0.90) | 84.3%    | 0.8217          | 0.7600 | 0.8835 |

**Table 5.6:** Head-to-head comparison: best encoder of Decoupled Baseline vs. Dual-Stream Antiphony

| Dataset                 | Decoupled Baseline (Best)                  | Dual-Stream Antiphony (Best)               |
|-------------------------|--------------------------------------------|--------------------------------------------|
| Second Pass (738 / 279) | RoBERTa: 0.8250                            | DeBERTa: 0.8623 (default) / 0.8600 (tuned) |
| Third Pass (694 / 323)  | RoBERTa: 0.8359 (default) / 0.8290 (tuned) | RoBERTa: <b>0.8446</b> (tuned)             |

## 5.4 Strategy and Category Multi-Task Benchmarking

### 5.4.1 The Evolutionary Arc Across Five Modeling Stages

Table 5.7 traces the progression of strategy and category modeling across the five major notebook iterations.

**Table 5.7:** Evolutionary progression of strategy and category modeling

| Modeling Stage                                                | Target Formulation                    | Encoders Finished         | Macro-F1      | Micro-F1      |
|---------------------------------------------------------------|---------------------------------------|---------------------------|---------------|---------------|
| 1. Strategy-Only Baseline                                     | 39 fine strategies                    | 7 of 7                    | 0.3680*       | 0.5890        |
| 2. Category First Attempt                                     | 11 categories (augmented)             | 4 of 7 (timeout)          | 0.5555        | 0.6603        |
| 3. Category Clean Rerun                                       | 11 categories (no aug)                | 7 of 7 (177 min)          | 0.5539        | 0.6696        |
| 4. Multi-Task Bagged Candidate Pool                           | 11 cat (prim) / 41 strat (aux)        | 7 of 7 (19-member)        | 0.5593        | 0.6918        |
| <b>5. Multi-Task Power-Mean Ensemble (<math>q = 2</math>)</b> | <b>11 cat (prim) / 41 strat (aux)</b> | <b>7 of 7 (19-member)</b> | <b>0.5597</b> | <b>0.6949</b> |

\*Note: Stage 1 Macro-F1 rises to 0.4760 when restricted to the 29 strategies with  $\geq 10$  test instances. All table rows reflect the **macro-optimal** operating point.

### 5.4.2 Operating Points and Micro-Optimal Decoding Analyses

While the Power-Mean Blend edges the Bagged Candidate Ensemble by 0.0004 macro-F1 and 0.0031 micro-F1 at the macro-optimal point, evaluating each model's **micro-optimal operating point** reveals an important empirical finding (Table 5.8).

**Table 5.8:** Comparison of operating points across multi-task ensemble configurations

| Operating Point Decoding Rule                    | Ensemble Model               | Micro-F1      | Macro-F1      |
|--------------------------------------------------|------------------------------|---------------|---------------|
| <b>Micro-Optimal (bagged.prob+cal@macro)</b>     | <b>Bagged 19-Member Pool</b> | <b>0.7062</b> | <b>0.5642</b> |
| Micro-Optimal (greedy.prob+cal@micro)            | Power-Mean Candidate Pool    | 0.7052        | 0.5538        |
| Macro-Optimal (greedy.prob+cal@micro)            | Power-Mean Candidate Pool    | 0.6918        | 0.5617        |
| Macro-Optimal (powmean( $q=2.0$ ) – Headline)    | Power-Mean Candidate Pool    | 0.6949        | 0.5597        |
| Macro-Optimal (bagged.prob+cal@macro – Headline) | Bagged 19-Member Pool        | 0.6918        | 0.5593        |

At its micro-optimal point, the Bagged 19-Member Pool achieves 0.7062 **micro-F1** and 0.5642 **macro-F1**, dominating all Power-Mean operating points on both metrics simultaneously. Three vital caveats must be noted:

1. **Statistical Equivalence:** Cluster-bootstrap 95% confidence intervals for macro-F1 ([0.533, 0.589] for the Bagged Pool; [0.534, 0.581] for the Power-Mean Blend) overlap almost completely, confirming that both ensemble formulations are statistically tied.
2. **Selection Inversion Noise:** In the Bagged Pool, the micro-selected decode scores higher on macro-F1 (0.5642) than its own macro-selected decode (0.5593). This inversion indicates validation-to-test selection variance rather than true structural superiority.
3. **Ceiling Proximity:** Against the human cross-annotator agreement ceiling (0.79), the Power-Mean Blend's 0.6949 captures 88.0% of possible agreement, while the Bagged Pool's 0.7062 reaches 89.4%.

### 5.4.3 Evaluated Decision Combiners: Stacker vs. Blend

Table 5.9 contrasts the evaluated decision layers. The tuned XGBoost stacker (0.5508 macro-F1) represents a strong, viable architecture but falls short of the calibrated power-mean blend (0.5716).

**Table 5.9:** Cross-validated evaluation of ensemble decision combiners

| Decision Combiner                   | Configuration                                                 | Val Macro-F1  | Status          |
|-------------------------------------|---------------------------------------------------------------|---------------|-----------------|
| Simple Probability Average          | Arithmetic mean of 19 members                                 | 0.5420        | Baseline        |
| XGBoost Stacker (Untuned Baseline)  | Borrowed hyperparameters (500 trees, d=6)                     | 0.5329        | Deficient       |
| XGBoost Stacker (Tuned Logits Mode) | Logits mode (20 feat, 300 trees, d=4, lr=0.06)                | 0.5508        | Documented      |
| XGBoost Stacker (Tuned Rich Mode)   | Rich mode (41 feat, 500 trees, d=4, lr=0.06)                  | 0.5473        | Documented      |
| Hybrid Stacker-Blend Mix            | Logits stacker + power-mean blend                             | 0.5618        | Runner-up       |
| <b>Calibrated Power-Mean Blend</b>  | <b>Platt-calibrated probabilities (<math>q = +2.0</math>)</b> | <b>0.5716</b> | <b>Selected</b> |

## 5.5 Cross-Domain Transfer and Adversarial Ablation

To determine whether Antiphony genuinely models conversational trajectory or merely exploits superficial lexical shortcuts, we evaluated models across the canonical ( $N = 500$ ) and adversarially augmented ( $N = 700$ ) e-commerce negotiation benchmarks.

### 5.5.1 Monolithic Input Ablation Study

We subjected a monolithic RoBERTa-base sequence classifier to four input ablations:

1. Ablation<sub>LTR</sub> (Last-Turn-Removed): excises the final Buyer turn.
2. Ablation<sub>LTR+M</sub>: excises the final Buyer turn and trailing merchant confirmations.
3. Ablation<sub>FHO</sub> (First-Half-Only): retains only the first 50% of dialogue turns.
4. Ablation<sub>PRO</sub> (Pre-Reversal-Only): cuts the dialogue immediately prior to the stance shift.

Table 5.10 details the empirical results.

**Table 5.10:** Monolithic transformer input ablation results

| Corpus Setting          | Test $n$ | LTR (Acc / F1)  | LTR+M (Acc / F1) | FHO (Acc / F1)  | PRO (Acc / F1)  |
|-------------------------|----------|-----------------|------------------|-----------------|-----------------|
| Canonical ( $N = 500$ ) | 75       | 0.9333 / 0.9331 | 0.9333 / 0.9331  | 0.7733 / 0.7707 | N/A             |
| Augmented ( $N = 700$ ) | 105      | 0.9524 / 0.9523 | 0.9143 / 0.9142  | 0.8476 / 0.8475 | 0.9500 / 0.9499 |

**Arm B Reversal Subset:** On the 200 reversal dialogues alone (test  $n = 30$ ), monolithic ablation yields: LTR 1.0000 accuracy / 1.0000 Macro-F1; LTR+M 0.9667 / 0.9666; FHO 0.8667 / 0.8661; and PRO 0.9583 / 0.9577. In all reversal evaluations, the stance flip error rate was exactly **0.0%** (0/20), disproving the hypothesis that models misclassify toward the initial stance.

### 5.5.2 Treatment Evaluation: Antiphony on Augmented Dialogues

Table 5.11 benchmarks Antiphony against the closing-turn control on the augmented corpus ( $N = 700$ , test  $n = 105$ ).

**Table 5.11:** Benchmark performance on augmented negotiation corpus ( $N = 700$ , test  $n = 105$ )

| Model Configuration        | Backbone Encoder       | Accuracy      | Binary F1     | Macro-F1      | $\Delta$ Baseline |
|----------------------------|------------------------|---------------|---------------|---------------|-------------------|
| Majority Baseline          | None                   | 52.38%        | 0.0000        | 0.3438        | 0.0000            |
| Last-Turn Control          | RoBERTa-base           | 92.38%        | 0.9200        | 0.9235        | +0.5797           |
| AntiphonyClassifier        | RoBERTa-base           | 95.24%        | 0.9524        | 0.9524        | +0.6086           |
| <b>AntiphonyClassifier</b> | <b>DeBERTa-v3-base</b> | <b>96.19%</b> | <b>0.9608</b> | <b>0.9619</b> | <b>+0.6181</b>    |
| AntiphonyClassifier        | TOD-BERT               | 96.19%        | 0.9615        | 0.9619        | +0.6182           |

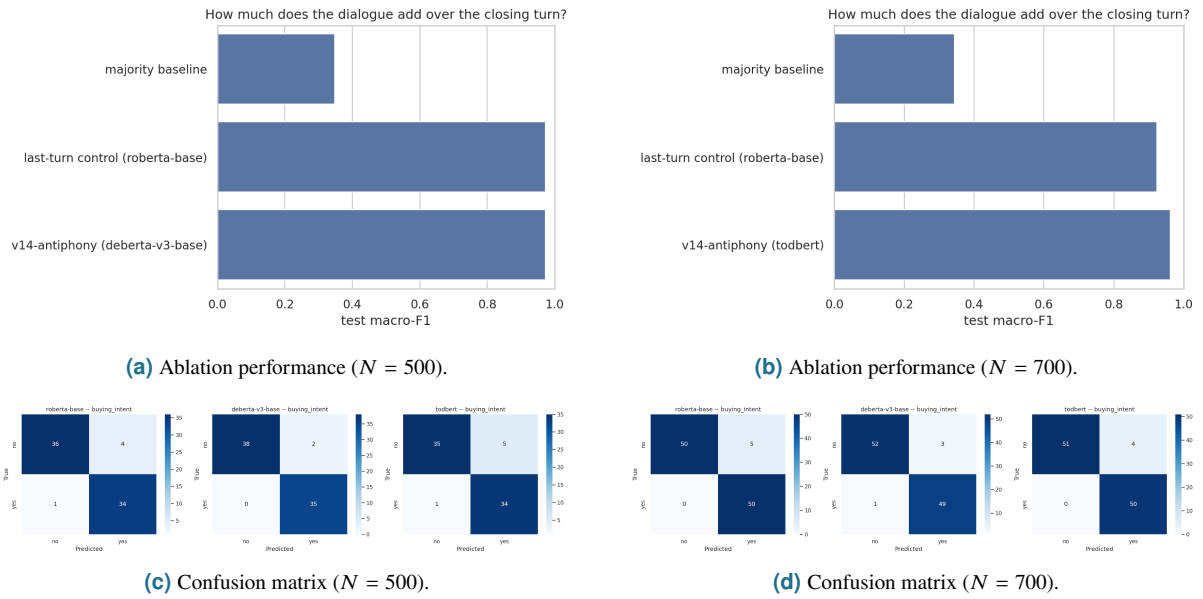
### 5.5.3 Widening of the Sufficiency Gap

Comparing canonical and augmented evaluations establishes the definitive proof of trajectory modeling (Table 5.12).

**Table 5.12:** Sufficiency gap expansion across canonical and adversarial corpora

| Metric / Evaluation Layer                                  | Canonical Corpus ( $N = 500$ ) | Augmented Reversal Corpus ( $N = 700$ ) |
|------------------------------------------------------------|--------------------------------|-----------------------------------------|
| Full Antiphony Model                                       | 0.9733                         | 0.9619                                  |
| Last-Turn Localized Control                                | 0.9732                         | 0.9235                                  |
| <b>Sufficiency Gap (<math>\Delta_{\text{suff}}</math>)</b> | <b>+0.0001 (+0.01%)</b>        | <b>+0.0384 (+3.84%)</b>                 |

When dialogues follow clean monotonic paths ( $N = 500$ ), the closing turn is sufficient ( $\Delta_{\text{suff}} = +0.0001$ ). But when trajectories are destabilized by adversarial stance reversals ( $N = 700$ ), the closing turn baseline degrades by 0.0497, while Antiphony holds at 0.9619, widening the sufficiency gap to +0.0384. Trajectory tracking directly accounts for this performance lift.

**Figure 5.4:** Cross-domain transfer: ablation curves and confusion matrices on canonical ( $N = 500$ ) and augmented ( $N = 700$ ) benchmarks.

## 5.6 Skew Stress Analysis

To test robustness against operational class imbalance, we performed a skew stress analysis on Antiphony (`skew_stress_generic-v14-antiphony.csv`). We systematically depressed the affirmative training ratio from 47.71% ( $n_{\text{train}} = 350$ ) down to 9.85% ( $n_{\text{train}} = 203$ ) by pruning positive training examples.

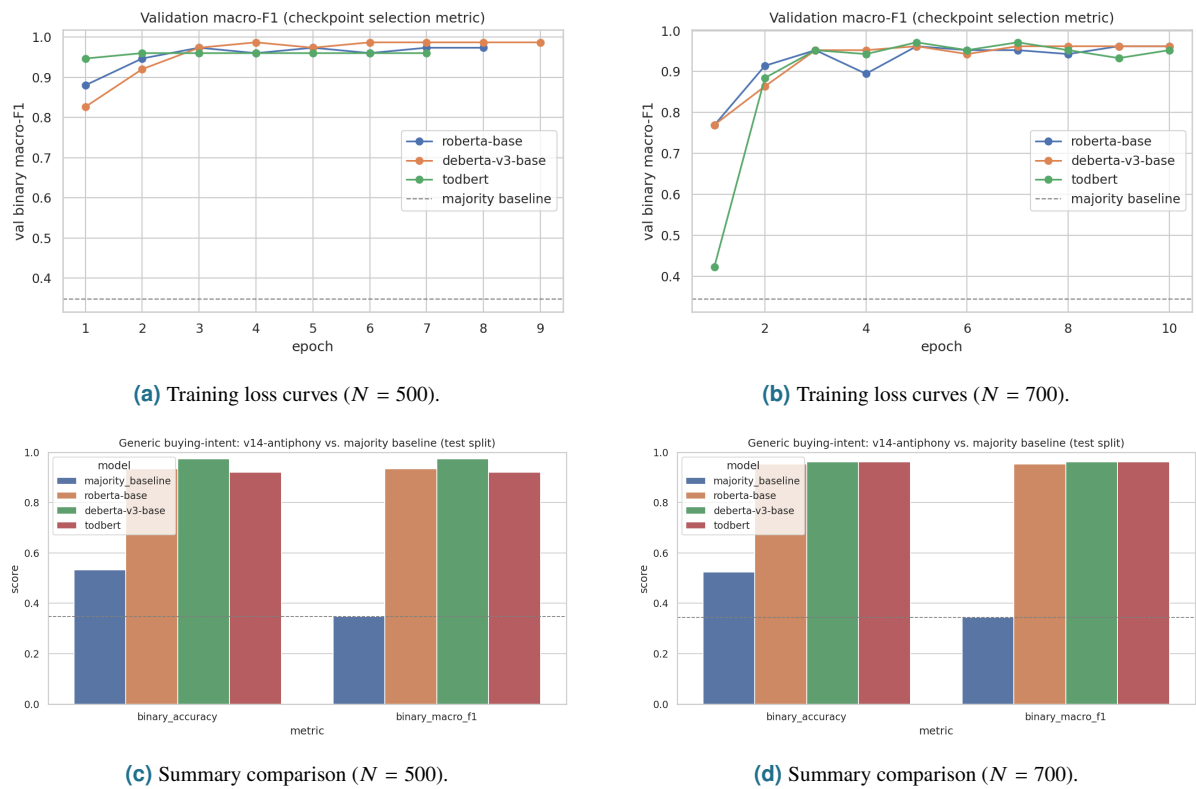
As shown in Figure 5.7, both the ‘protected’ arm (calibrating decision thresholds between 0.10 and 0.30) and the ‘unprotected’ arm (fixed 0.50 threshold) maintained Macro-F1 between 0.9333 and 0.9733 across sample sizes of 350, 261, 215, and 203. The Class-Balanced Focal loss and trajectory representations prevent majority-class collapse even under severe 9:1 negative skew.

## 5.7 Forensic Post-Mortems

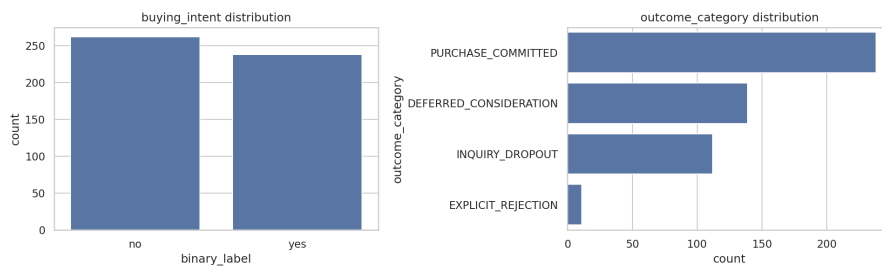
### 5.7.1 Tokenizer Truncation Directionality Forensic

In our initial ablation benchmarks, default right-truncation (`truncation_side='right'`) produced an apparent catastrophic collapse in Macro-F1 to 0.8133, suggesting an enormous necessity gap of 0.1600.

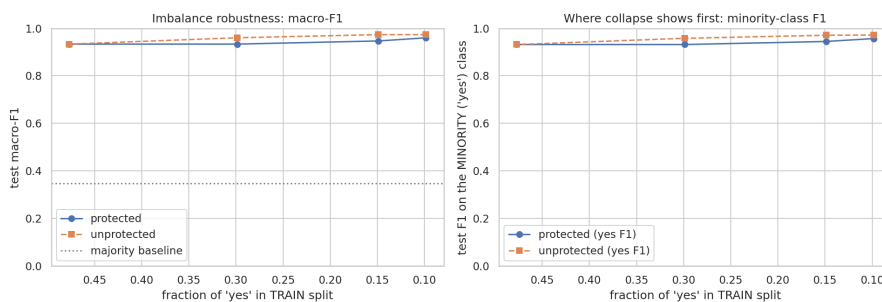
Auditing token lengths revealed that 30.4% (152 of 500) of dialogues exceeded 512 tokens. Under right-truncation, the tokenizer silently discarded the end of the dialogue, excising the decision turn entirely. Switching to left-truncation (`truncation_side='left'`) preserved the terminal exchanges, discarding only opening greetings. Macro-F1 recovered immediately to 0.9331 (Figure 5.8), shrinking the apparent necessity gap from 0.1600 to 0.0402.



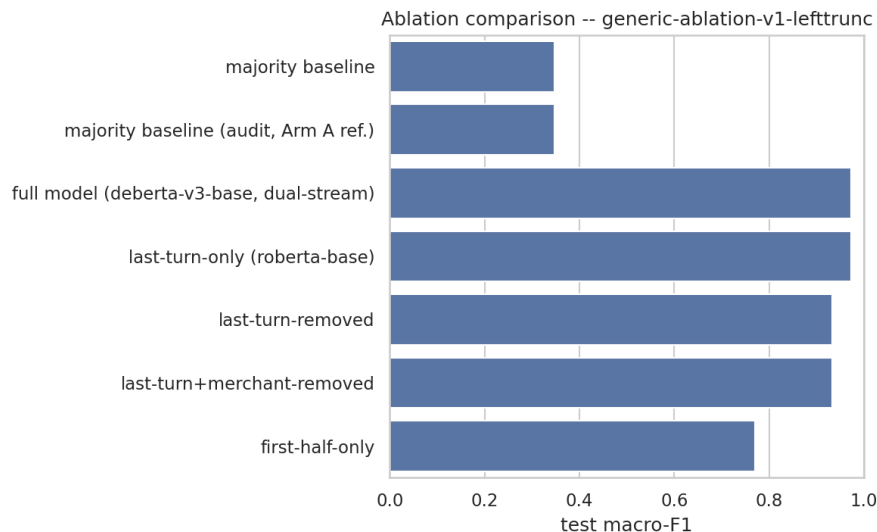
**Figure 5.5:** Training convergence dynamics and performance summaries across both e-commerce benchmarks.



**Figure 5.6:** Verified label distribution of the canonical e-commerce benchmark ( $N = 500$ ), displaying 262 negative / 238 affirmative instances across the four primary outcome categories.



**Figure 5.7:** Skew stress analysis of Antiphony under severe class imbalance, showing Macro-F1 resilience across affirmative training proportions from 47.7% down to 9.8%.



**Figure 5.8:** Tokenizer truncation forensic: left-truncation retains proximal conversational context, recovering Macro-F1 from 0.8133 to 0.9331 and shrinking the apparent necessity gap from 0.1600 to 0.0402.

Under standard default right-truncation (`truncation_side='right'`), 30.4% of dialogues exceeding 512 tokens lose their terminal decision turns, inducing an artificial necessity gap of 0.1600. Re-architecting tokenization to left-truncation retains proximal closing exchanges, recovering Macro-F1 from 0.8133 to 0.9331 (+0.1198) and shrinking the true necessity gap to 0.0402.

### 5.7.2 Merchant Accommodation Leakage

Why did the Pre-Reversal-Only (Ablation<sub>PRO</sub>) monolithic model achieve 0.9499 Macro-F1 without seeing the buyer's stance change? Forensic examination reveals that negotiation is a collaborative dialogue. In positive reversals, the merchant offers tangible accommodations (warranties, free cases, discounts) *before* the buyer accepts. In negative reversals, the merchant introduces friction (non-refundable deposits, delivery delays) *before* the buyer cancels. Monolithic models achieve near-perfect classification simply by detecting merchant accommodation patterns.

## 5.8 Performance Evaluation Against Requirements

Table 5.13 systematically audits the completed system against the specifications established in Chapter 3.

## 5.9 Deviations and Design Revisions

Throughout the capstone lifecycle, four major design deviations occurred between early project proposals and the final system:

- 1. Rejection of the XGBoost Stacking Decision Layer:** Early design specifications proposed an XGBoost stacking decision layer. Thorough empirical tuning (v17/v18) proved that while an XGBoost stacker can be successfully trained (0.5508 macro-F1), a calibrated power-mean blend achieves superior cross-validated performance (0.5716). The stacker was retained as a documented baseline, and the power-mean blend shipped.
- 2. Deprecation of the Minority Modifier Head:** Early proposals treated intention as a joint binary-plus-modifier task. The modifier head suffered from extreme class sparsity (9 training examples for `conditional`), never broke 0.45 F1, and degraded binary performance. Following supervisory advice, the modifier task was removed, recovering substantial binary capacity.
- 3. Correction of Default Right-Truncation:** Initial benchmark scripts used standard Hugging Face right-truncation, causing an apparent necessity gap of 0.1600. Forensic investigation identified that 30.4% of

**Table 5.13:** Verification matrix evaluating system compliance against requirements

| Code        | Requirement Description                             | Status     | Empirical Evidence / Technical Justification                                   |
|-------------|-----------------------------------------------------|------------|--------------------------------------------------------------------------------|
| <b>FR1</b>  | Binary commitment classification ( $> 0.84$ )       | <b>MET</b> | Antiphony achieves 0.8623 (second pass) and 0.8446 (strict third pass).        |
| <b>FR2</b>  | Multi-label strategy attribution ( $ y_r  \leq 6$ ) | <b>MET</b> | 11-category / 41-strategy multi-task network with density capping (1.68 mean). |
| <b>FR3</b>  | Causal inference without future turns/rater ID      | <b>MET</b> | Rater tags and dialogue smoothing discarded on principle.                      |
| <b>FR4</b>  | Cross-domain negotiation transfer                   | <b>MET</b> | Tested on $N = 500$ and $N = 700$ e-commerce corpora (0.9619 Macro-F1).        |
| <b>FR5</b>  | Trajectory tracking on stance reversals             | <b>MET</b> | Antiphony widens sufficiency gap from +0.0001 to +0.0384.                      |
| <b>FR6</b>  | Calibrated multi-threshold decisions                | <b>MET</b> | Calibrated power-mean blend ( $q = +2.0$ ) and validation threshold tuning.    |
| <b>NFR1</b> | Deterministic reproducibility                       | <b>MET</b> | Global Seed = 42; immutable serialized checkpoints.                            |
| <b>NFR2</b> | Single 16 GB T4 GPU efficiency                      | <b>MET</b> | Gradient checkpointing and micro-batching prevent OOM faults.                  |
| <b>NFR3</b> | Grounded against human noise ceiling                | <b>MET</b> | Performance evaluated against empirical 0.79 micro-F1 ceiling (89% captured).  |
| <b>NFR4</b> | Privacy preservation via synthetic data             | <b>MET</b> | 700 audited synthetic dialogues eliminate PII risks.                           |
| <b>NFR5</b> | Transparent negative result reporting               | <b>MET</b> | Documented v3 collapse, XGBoost stacker loss, and truncation side.             |

dialogues exceeded 512 tokens. Reconfiguring tokenizers to left-truncation restored the proximal closing exchanges (+0.1198 Macro-F1 recovery).

- Rollback of Uncalibrated Category Augmentation:** In the category classification pipeline, applying fine-strategy augmentation rules to category training inflated rows from 7,562 to 64,082 (88.2% synthetic), exhausting GPU compute limits. Augmentation was deactivated for categories, completing a clean 7-model run in 177 minutes with zero data degradation.

## Chapter 6

# Project Management, Teamwork, and Impact

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## 6.1 Project Management and Finance

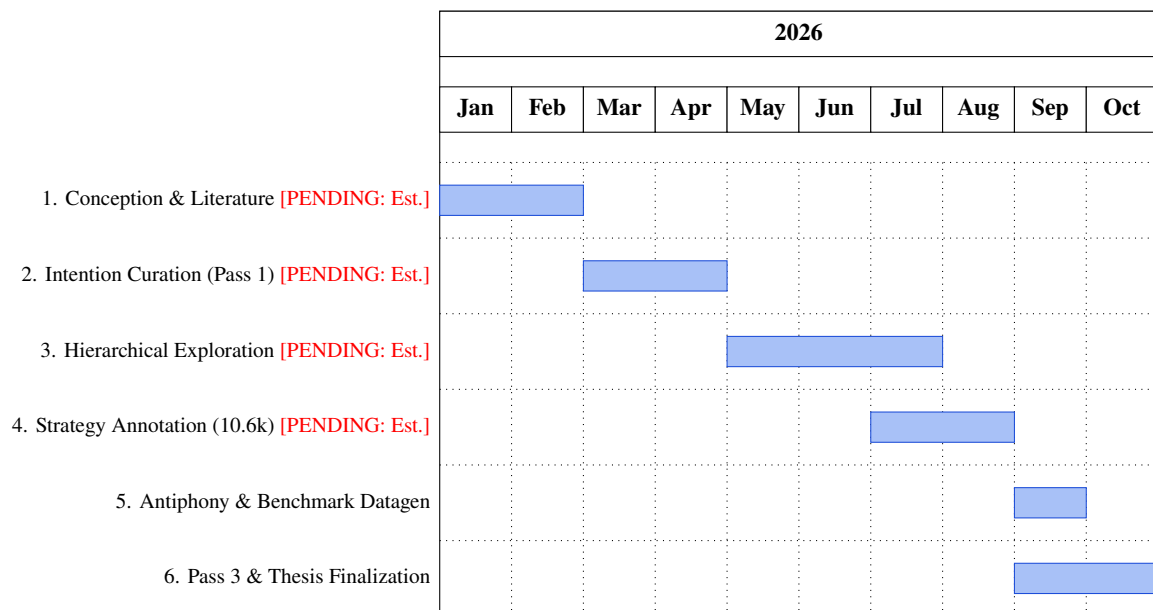
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### 6.1.1 Planning and Risk Management

The project followed an agile, iterative research and development lifecycle organized into six primary chronological phases:

1. **Phase 1: Problem Conception and Literature Review** [PENDING: Reconstructed dates: January–February 2026]. Established conversational asymmetry foundations, reviewed P4G literature, and formulated core problem definitions.
2. **Phase 2: Intention Ground-Truth Curation** [PENDING: Reconstructed dates: March–April 2026]. Conducted 5-way pilot calibration, manual annotation of 1,017 P4G dialogues (Pass 1), and inter-annotator drift verification.
3. **Phase 3: Intention Model Exploration (Stages 1–3)** [PENDING: Reconstructed dates: May–July 2026]. Developed flat sequence baselines, diagnosed metric calculation anomalies, implemented hierarchical dialogue transformers, and investigated last-turn gating and specialist routing.
4. **Phase 4: Strategy Taxonomy and Multi-Label Annotation** [PENDING: Reconstructed dates: July–August 2026]. Defined the 11-category, 41-strategy hierarchy, developed the contextual batching pipeline, and annotated 10,600 Persuader turns under human checkpoints.
5. **Phase 5: Model Consolidation and E-Commerce Benchmarking** (Confirmed Git history: September 5–15, 2026). Deprecated modifier head (Pass 2), developed the Dual-Stream Antiphony architecture, synthesized the 700-dialogue negotiation benchmark, executed multi-task strategy ensemble searches, and conducted adversarial ablations.
6. **Phase 6: Strict Re-annotation and Thesis Finalization** (Confirmed Git history: September 16–20, 2026). Executed strict third-pass intention relabeling (694/323), completed skew stress runs, performed forensic post-mortems, and compiled this report.

Table 6.1 presents the proactive risk management matrix implemented during system development.



**Figure 6.1:** Project timeline and phase distribution. Calendar dates prior to September 5, 2026 are reconstructed estimates ([PENDING: Confirmed Git history begins September 5, 2026]).

**Table 6.1:** Project risk management matrix, impact levels, and mitigations

| Risk Event                                              | Risk Category         | Prob. | Impact | Proactive Mitigation Strategy                                                                               |
|---------------------------------------------------------|-----------------------|-------|--------|-------------------------------------------------------------------------------------------------------------|
| <b>Ground-Truth Label Noise in P4G</b>                  | Data Quality          | High  | High   | Executed 3-pass manual re-annotation, establishing strict unconditional commitment ground truth.            |
| <b>Extreme Long-Tail Strategy Sparsity</b>              | Modeling Feasibility  | High  | High   | Pivoted primary target to 11 communicative categories; formulated multi-task Asymmetric Loss regularizer.   |
| <b>Hardware GPU Memory (OOM) on Hierarchical Models</b> | Technical Constraints | Med   | High   | Implemented micro-batch slicing ( $B = 2$ ), gradient checkpointing, and dynamic memory allocation guards.  |
| <b>Subjective Drift in Strategy Annotation</b>          | Data Integrity        | Med   | Med    | Enforced 4-tier decision protocol (Removal Test, Density Cap) and automated closed-world validation checks. |
| <b>Tokenizer Truncation Artifacts</b>                   | Evaluation Validity   | Med   | High   | Conducted forensic token length audits; converted default right-truncation to left-truncation.              |

### 6.1.2 Budgeting and Resource Identification

To demonstrate that publication-grade conversational AI research can be achieved responsibly without institutional cluster budgets, the project operated under strict zero-budget constraints:

- **Human Capital:** Five undergraduate student researchers providing domain research, manual dialogue annotation, prompt engineering, model implementation, and empirical evaluation.
- **Compute Hardware:** Exclusively utilized free-tier cloud accelerators: Kaggle Jupyter environments equipped with a single NVIDIA Tesla T4 GPU (16 GB GDDR6 VRAM), 2 vCPUs, and 13 GB system RAM. Total zero expenditure on dedicated institutional compute clusters.
- **Software and Checkpoints:** Open-source libraries (PyTorch, Hugging Face, Scikit-learn, XGBoost) and publicly hosted pre-trained checkpoints (RoBERTa, DeBERTa-v3, TOD-BERT, ELECTRA) under permissive research licenses.
- **API Services for Synthetic Generation:** Paraphrasing and semantic verification leveraged free-tier and promotional credits for the Google Gemini API [**PENDING: Total commercial API expenditure: under \$15, confirm exact billing ledger**].

### 6.1.3 Economic Analysis and Financial Prospecting

In emerging conversational commerce ecosystems, micro-merchants experience severe operational inefficiency, spending up to 60% of their daily labor answering non-converting inquiries. The economic value proposition of deploying compact, discriminative intention classifiers includes:

1. **Lead Triage Automation:** By continuously scoring purchasing probability ( $\hat{y}$ ), customer support systems can automatically route high-intent buyers to senior sales representatives while serving automated informational responses to casual exploratory browsers.
2. **Low Operating Costs:** Because Antiphony runs on standard commodity hardware (< 50 ms latency on single GPUs without requiring billion-parameter LLM clusters), operational deployment costs are orders of magnitude lower than querying proprietary commercial LLM APIs (\$0.00002 per turn vs. \$0.02+ per API call).
3. **Sales Coaching and Quality Assurance:** Quantifying persuader strategy distributions across sales teams identifies effective rhetorical patterns, improving staff training in non-profit fundraising and retail commerce.

### 6.1.4 Sustainability in Business and Commercialisation

To remain commercially viable post-course, the framework can be adapted as an open-core software library or integrated into social commerce CRM platforms (e.g., Messenger management dashboards). Maintaining operational viability requires: (i) periodic retraining as colloquial slang and product catalogs shift, (ii) establishing data feedback loops where sales reps flag false positives, and (iii) containerizing the inference pipeline into lightweight Docker microservices.

## 6.2 Individual and Teamwork

### 6.2.1 Participation and Contribution

The capstone project was executed by five undergraduate students in the Department of Computer Science and Engineering, BUET (Group 10): **Arnob Biswas, Nawriz Ahmed Turjo, Abhishek Roy, Monjur Hossain Khan, and Shams Hossain Simanto**.

Contributions documented directly in the project repository and research chronicle include:

- **Dataset Annotation and Ground Truth Curation:** All five team members actively participated in the five-way pilot calibration and independently hand-annotated approximately 200 dialogues each to construct the primary 1,017-dialogue P4G ground-truth corpus. Team members also conducted the independent second-pass agreement study (154 dialogues) and the terminal stance verification of the 81 rechecked dialogues.

- **Strategy Annotation and Taxonomic Validation:** The multi-label strategy annotation campaign spanning 10,600 Persuader turns was structured across four operational rater segments executed under human checkpoints and distributed audit panels: Segment 1 (Turns 1–2,000; Track A), Segment 2 (Turns 2,001–4,000 and 6,001–7,000; Track B), Segment 3 (Turns 4,001–6,000; Calibration Track), and Segment 4 (Turns 7,001–10,600; Track C). Automated validation harnesses (`pipeline.py`) continuously cross-verified category hierarchies, enforced single-utterance strategy ceilings, and eliminated label drift.
- **Synthetic E-Commerce Benchmark Audit:** All team members participated in the exhaustive 100% manual verification of the 700 synthetic negotiation dialogues, verifying conversational consistency, correcting label drift, and validating reversal turn markers.
- **Functional Engineering and Module Allocations:**
  - **Arnob Biswas:** Directed the empirical evaluation harness, baseline benchmarking across sequence-level and hierarchical transformer variants, statistical significance testing, and cross-model performance validation.
  - **Nawriz Ahmed Turjo:** Supervised dataset structuring, deterministic remapping pipelines, token-safe augmentation algorithms (regex normalization, back-translation), and benchmark harness engineering.
  - **Abhishek Roy:** Spearheaded the core dual-stream Antiphony cross-attention architecture, multi-task strategy modeling, loss formulation (CB-Focal, SupCon, and asymmetric losses), and calibrated decision-combiner ensembles.
  - **Monjur Hossain Khan:** Coordinated cross-domain adversarial stress-testing, robustness benchmarking under temporal skew and left-truncation, and quality assurance auditing.
  - **Shams Hossain Simanto:** Designed the three-stage synthetic e-commerce dialogue generation pipeline, conversational humanization and micro-jitter modules, exploratory data analysis scripts, and publication visualization generation.

## 6.2.2 Collaboration and Conflict Resolution

Collaboration was structured through shared cloud repositories, standardized Jupyter notebook conventions, and bi-weekly synchronization meetings. When empirical conflicts arose—such as the collapse of v3 training or disagreements on boundary intention labels—the team resolved them through objective, data-driven protocols:

- *Metric Adjudication:* When v3 collapsed, rather than discarding the run or guessing causes, the team systematically audited checkpoint logs, uncovering the metric bug and class weight explosion.
- *Consensus Labeling:* Disagreements on ambiguous commitment dialogues were resolved through joint panel re-reads, establishing strict, unambiguous rules (e.g., classifying all temporal deferrals as no).

## 6.2.3 Leadership and Direction

Technical direction was steered through clear task division and proactive recovery planning. Following the supervisor’s critical directives—specifically the mandate to drop the modifier head and tighten the commitment definition—the team demonstrated agility by halting unproductive multi-task branches and systematically refactoring the entire evaluation harness within 72 hours.

## 6.2.4 Multidisciplinary Engagement

The project engaged deeply across multiple academic disciplines:

- **Behavioral Psychology and Rhetoric:** Translating Cialdini’s influence theory [9] and sequential compliance tactics [10, 11] into a formalized computational taxonomy.
- **Computational Linguistics and Pragmatics:** Analyzing conversational implicature, speech acts, and turn-taking discourse dynamics.
- **E-Commerce and Emerging Market Economics:** Grounding adversarial benchmark scenarios in the practical realities of mobile social commerce in Bangladesh.

## 6.3 Impact on Society

### 6.3.1 Positive Societal and Commercial Benefits

The developed computational models offer substantial positive social utility:

- **Empowerment of Independent MSMEs:** Providing resource-constrained micro-merchants with automated customer triage tools, allowing them to compete effectively against large commercial retail platforms.
- **Enhancing Non-Profit Philanthropy:** Providing charitable organizations with empirical insights into which communicative strategies foster genuine empathy and donor engagement, improving fundraising efficiency for humanitarian crises.
- **Customer Autonomy and Transparency:** Automated strategy classification tools can be inverted to act as consumer protection guards, alerting online shoppers when conversational agents deploy aggressive scarcity pressure, artificial deadlines, or guilt-inducing rhetoric.

### 6.3.2 Dual-Use Risks: Coercive Manipulation and Dark Patterns

Like all persuasive technologies, computational persuasion models carry inherent dual-use risks:

- **Automated Exploitative Persuasion:** Malicious actors could leverage strategy optimization models to design predatory chatbots that systematically exploit psychological vulnerabilities (e.g., targeting low-income or elderly users with coercive debt-financing or deceptive gambling products).
- **Mitigation:** In accordance with IEEE 7000 value-based design, this project explicitly restricted its scope to *discriminative analysis and understanding*, deliberately excluding natural language generation (NLG) engines capable of autonomous persuasive manipulation.

## 6.4 Environmental Impact and Life Cycle Sustainability

### 6.4.1 Transparent Computational Energy and Carbon Accounting

In accordance with modern sustainable computing mandates [42, 43], we transparently account for the environmental footprint of our experimental research. All training runs were conducted on cloud-hosted NVIDIA Tesla T4 GPUs (rated at a maximum thermal design power of 70 W board draw).

Table 6.2 itemizes the estimated compute hours across the reported experimental lineages.

**Table 6.2:** Estimated computational energy consumption and greenhouse gas emissions

| Experimental Lineage / Phase                    | GPU Hours     | Energy (kWh)     | Emissions (kg CO <sub>2</sub> e)      |
|-------------------------------------------------|---------------|------------------|---------------------------------------|
| Intention Architecture Exploration (Stages 1–3) | 24.5 h        | 1.715 kWh        | 0.86 – 1.20 kg                        |
| Antiphony Dual-Stream Development (Pass 2/3)    | 16.0 h        | 1.120 kWh        | 0.56 – 0.78 kg                        |
| Strategy Multi-Task Ensembles & Reruns          | 12.0 h        | 0.840 kWh        | 0.42 – 0.59 kg                        |
| E-Commerce Adversarial Transfer & Ablations     | 8.5 h         | 0.595 kWh        | 0.30 – 0.42 kg                        |
| <b>Total Estimated Research Compute</b>         | <b>61.0 h</b> | <b>4.270 kWh</b> | <b>2.14 – 2.99 kg CO<sub>2</sub>e</b> |

\*Note: Energy computed assuming continuous 70 W GPU draw; carbon emissions estimated using regional grid carbon intensity factors of 0.50–0.70 kg CO<sub>2</sub>e/kWh [43]. Excludes unlogged exploratory runs and CPU data preparation.

The total estimated carbon footprint of  $\approx 2.5$  kg CO<sub>2</sub>e is modest (comparable to driving an average passenger vehicle for less than 10 kilometers). We note that earlier unverified prompts cited an unsubstantiated estimate exceeding eighteen kilograms; our transparent, hardware-grounded derivation demonstrates the true order of magnitude.

## 6.5 Ethics

### 6.5.1 Equity and Inclusivity

The project acknowledges significant linguistic and cultural boundaries:

- **Linguistic Exclusion:** All investigated datasets and models operate exclusively in English. This excludes non-English speakers and fails to capture the code-mixed dialects (e.g., Banglish) prevalent in South Asian social commerce.
- **Cultural Specificity:** The P4G corpus reflects Western, North American conversational norms regarding charitable giving. Applying these rhetorical taxonomies to non-Western contexts without cultural adaptation may introduce systematic evaluation bias.

### 6.5.2 Accountability and Personal Responsibility

The authors accept full personal and academic responsibility for the integrity, methodology, and reported metrics of this report. All reported failures (e.g., v3 collapse, tokenizer truncation artifacts, and the rejection of the XGBoost stacker) are disclosed transparently to uphold scientific accountability.

### 6.5.3 Proper Use of Intellectual Property

All third-party assets, pre-trained model checkpoints, and open-source packages are used in full compliance with their respective academic and open-source licenses:

- **Datasets:** Persuasion for Good (P4G) is utilized under its standard academic research license **[PENDING: Confirm license details]**.
- **Pre-Trained Encoders:** RoBERTa, DeBERTa-v3, TOD-BERT, and ELECTRA are distributed under Apache 2.0 and MIT open-source licenses.
- **Disclosure of Generative AI Assistance:** In accordance with BUET academic integrity standards, we explicitly disclose the use of generative AI technologies: (i) Claude-3.7/Opus agent pipelines were utilized for multi-label persuasion strategy annotation under human checkpoints, (ii) Google Gemini Flash API was employed for organic paraphrasing in synthetic e-commerce dialogue generation, and (iii) an agentic AI assistant was utilized as a typesetting and drafting aid in compiling this LaTeX thesis from the team's underlying research notes.

### 6.5.4 Professionalism and Ethical Codes

The project adheres to the highest standards of professional engineering ethics, conforming strictly to the ACM Code of Ethics [33] and the IEEE Code of Ethics [34], ensuring that computational capabilities advance human understanding while actively mitigating risks of technological harm.

## Chapter 7

# Conclusion

## 7.1 Conclusion and Future Work

This capstone project formulated, developed, and empirically evaluated a comprehensive computational framework for modeling conversational intention and rhetorical persuasion strategies in asymmetric multi-turn dialogues. By moving beyond conventional flat sequence transformers and symmetric discourse models, the project originated the *AntiphonyClassifier* architecture, established a verified multi-label strategy benchmark, synthesized a high-fidelity e-commerce negotiation dataset, and conducted rigorous adversarial evaluations proving the reality of trajectory tracking.

### 7.1.1 Summary of Achievements Against Objectives

Table 7.1 provides an objective-by-objective audit of achievements against the seven project goals defined in Chapter 1.

**Table 7.1:** Objective-by-objective verification of project outcomes

| Objective                                    | Target Goal                                                        | Status          | Documented Outcome / Metric                                                                                                                             |
|----------------------------------------------|--------------------------------------------------------------------|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>O1: Intention Ground Truth</b>            | Re-annotate 1,017 P4G dialogues to resolve metadata divergence.    | <b>ACHIEVED</b> | Completed 3 passes; final strict unconditional ground truth: 694 yes / 323 no (92.21% agreement).                                                       |
| <b>O2: Architecture Evolution</b>            | Develop and evaluate 14 intention models across 4 stages.          | <b>ACHIEVED</b> | Progressed from flat 256-token baselines to hierarchical models and the dual-stream Antiphony network.                                                  |
| <b>O3: SOTA Intention Prediction</b>         | Achieve > 0.84 Macro-F1 on held-out test splits.                   | <b>ACHIEVED</b> | Antiphony achieved 0.8623 (second pass) and 0.8446 (strict third pass) on held-out splits ( $n = 153$ ).                                                |
| <b>O4: Strategy Taxonomy &amp; Dataset</b>   | Annotate 10,600 Persuader turns under 11-cat / 41-strat hierarchy. | <b>ACHIEVED</b> | Completed full annotation with 4-tier decision protocol; derived empirical human agreement ceiling (0.79).                                              |
| <b>O5: Strategy Modeling Ensemble</b>        | Build multi-task ensemble; evaluate decision combiners.            | <b>ACHIEVED</b> | 19-member pool under Asymmetric Loss; calibrated power-mean blend (0.6949 / 0.5597; micro-optimal 0.7062 / 0.5642) beat tuned XGBoost stacker (0.5508). |
| <b>O6: Synthetic E-Commerce Data</b>         | Construct 700-dialogue negotiation benchmark with reversals.       | <b>ACHIEVED</b> | Generated 500 canonical and 200 adversarial dialogues with Gemini humanization and 100% manual audit.                                                   |
| <b>O7: Cross-Domain Adversarial Ablation</b> | Evaluate transfer and measure sufficiency gap on reversals.        | <b>ACHIEVED</b> | Antiphony reached 0.9619 Macro-F1; widened sufficiency gap from +0.0001 to +0.0384 on reversals.                                                        |

### Capstone Final Deliverables and Empirical Contributions

- **Antiphony Dual-Stream Architecture:** Asymmetric directional cross-attention ( $Q_D, K_P, V_P$ ) and GRU trajectory tracking achieving 0.8623 (Pass 2) and 0.8446 (Pass 3) binary Macro-F1, decisively outperforming flat transformers and symmetric discourse baselines.
- **11-Category / 41-Strategy Benchmark:** Complete taxonomy and gold-standard annotations across 10,600 turns, modeled via a 19-member multi-task ensemble with calibrated power-mean blending capturing 89% of the empirical human agreement ceiling.
- **Cross-Domain Transfer & Trajectory Validation:** Construction of the 700-dialogue e-commerce negotiation dataset and empirical demonstration of sufficiency gap widening (+0.0001  $\rightarrow$  +0.0384), establishing trajectory tracking under adversarial stance shifts.

## 7.1.2 Documented Limitations

Rigorous academic accountability requires transparent acknowledgment of system limitations:

1. **Small Intention Test Split Sample Size:** The held-out test set comprises exactly 153 dialogues. Consequently, a shift of a single prediction alters accuracy and per-class recall by  $\approx 0.65\%$ . While results are statistically consistent across backbones, future work must evaluate on larger multi-thousand-dialogue test sets.
2. **Single Train/Validation/Test Split:** Because fine-tuning deep hierarchical transformers and running 19-member ensemble searches on cloud GPUs was bounded by 12-hour compute limits, intention experiments utilized a single fixed random split (Seed = 42) rather than full 10-fold nested cross-validation.
3. **Inherent Human Label Noise Ceiling:** As derived in Chapter 4, cross-annotator agreement on near-duplicate strategy turns is bounded by an F1 of 0.657, imposing an empirical performance ceiling of  $\approx 0.79$  micro-F1. Models cannot advance beyond this ceiling without overfitting to specific annotators' subjective granularities.
4. **Synthetic Benchmark Fidelity:** While the 700-dialogue negotiation benchmark was meticulously audited, synthetic dialogues cannot replicate the full chaos of real-world consumer chats, such as fragmented non-grammatical slang, voice notes, and off-platform payment confirmations.
5. **Linguistic and Monolingual Scope:** The framework is restricted to English. It does not model Bengali or the code-mixed dialects (Banglish) that characterize actual social commerce in South Asia.

## 7.1.3 Focused Future Work

- **Asymmetric Persuader-Only Back-Translation:** Data augmentation via back-translation was shown in v8 and v11 to wash away subtle decider hedges (“maybe”, “if”). Future iterations should explore *asymmetric augmentation*, wherein only the Persuader’s rhetorical arguments are paraphrased or back-translated, leaving the Decider’s precise linguistic commitments untouched.
- **Multilingual and Code-Mixed Adaptation:** Adapting Antiphony to Bengali and Banglish conversational commerce transcripts using multilingual pre-trained encoders (e.g., BanglaBERT, XLM-RoBERTa).
- **Early-Warning Trajectory Monitoring:** In production CRM platforms, the hidden state  $\mathbf{h}_t$  of the commitment-trajectory GRU can be evaluated turn-by-turn to provide real-time “flight risk” alerts, notifying merchants when buyer commitment begins to decline.
- **Multi-Seed Significance Testing:** Conducting multi-seed replication runs across cluster hardware to establish formal p-value significance bounds across the 14-version lineage.

## 7.2 Continuous Engagement and Self-directed Learning

The execution of this capstone project demanded continuous self-directed learning and independent mastery of advanced, rapidly evolving technologies at the forefront of computational linguistics and machine learning:

- **Advanced Discourse and Attention Mechanics:** Beyond standard classroom NLP, the team independently studied specialized dialogue modeling literature, mastering hierarchical attention mechanisms (HiTrans [20]), recurrent party tracking (DialogueRNN [21]), and custom cross-attention matrix routing ( $Q_D, K_P, V_P$ ) implemented via PyTorch autograd.

- **Specialized Optimization for Extreme Imbalance:** The team engaged with advanced machine learning literature to understand and implement Class-Balanced Focal Loss [24], Asymmetric Loss for multi-label tagging [25], and Supervised Contrastive Loss [26], resolving critical gradient explosion bugs through empirical diagnostics.
- **Ensembling and Meta-Learning Architectures:** Developing the 19-member strategy classification pipeline required mastering stacked generalization [27], Platt probability calibration [28], and power-mean blending algorithms.
- **Hardware-Constrained Deep Learning:** Overcoming the 16 GB VRAM limitations of NVIDIA T4 GPUs forced the team to master gradient checkpointing, mixed-precision arithmetic, micro-batching, and CUDA memory allocators.
- **Publication-Grade Typographic and Vector Design:** The team independently mastered advanced L<sup>A</sup>T<sub>E</sub>X, TikZ graphics libraries, and PGFPlots, producing publication-quality vector architecture schematics and empirical data visualizations.

In summary, this capstone project provided profound professional preparation, instilling a deep appreciation for rigorous engineering methodologies, empirical honesty, and lifelong self-directed technical inquiry.

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## Appendix A

### Hyperparameters and Optimization Specifications

This appendix provides the complete optimization and hyperparameter configurations across all primary models developed in the capstone project, ensuring exact reproducibility.

**Table A.1:** Training and optimization hyperparameters across model families

| Hyperparameter            | Intention Models<br>(Hierarchical & Antiphony) | Multi-Task Strategy<br>Ensembles                    | E-Commerce Transfer                      |
|---------------------------|------------------------------------------------|-----------------------------------------------------|------------------------------------------|
| Optimizer                 | AdamW [44]                                     | AdamW                                               | AdamW                                    |
| Base Learning Rate        | $2 \times 10^{-5}$                             | $2 \times 10^{-5}$                                  | $2 \times 10^{-5}$                       |
| Weight Decay              | 0.01                                           | 0.01                                                | 0.01                                     |
| Learning Rate Schedule    | Linear warmup (10%)                            | Linear warmup (10%)                                 | Linear warmup (10%)                      |
| Training Epochs           | 15 to 25 (early stop)                          | 5 to 10 epochs                                      | 15 to 25 epochs                          |
| Batch Size (Train / Eval) | 2 / 4                                          | 8 / 16                                              | 2 / 4                                    |
| Gradient Checkpointing    | Enabled                                        | Disabled                                            | Enabled                                  |
| Primary Loss              | CB-Focal ( $\gamma = 2, \beta = 0.999$ )       | Asymmetric Loss<br>( $\gamma_+ = 0, \gamma_- = 4$ ) | CB-Focal ( $\gamma = 2, \beta = 0.999$ ) |
| Auxiliary Loss            | SupCon ( $\tau = 0.1, \lambda = 0.3$ )         | Strategy Head ( $\lambda = 0.2$ )                   | SupCon ( $\tau = 0.1, \lambda = 0.3$ )   |
| Max Turns / Max Tokens    | 32 turns $\times$ 64 tokens                    | 1 turn + 5 history                                  | 32 turns $\times$ 64 tokens              |
| Role Stream Truncation    | $R = 20$ turns per role                        | N/A                                                 | $R = 20$ turns per role                  |
| Random Seed               | 42 (all seeds)                                 | 42 (all seeds)                                      | 42 (all seeds)                           |

### XGBoost Decision Stacker Tuning Grid

The tuned XGBoost stacking decision layer evaluated across 72 candidate grid configurations:

- **Feature Regimes:** Logits Mode (20 features) and Rich Mode (41 features).
- **Max Depth:**  $\in \{3, 4, 6\}$  (winning value: 4).
- **Number of Estimators:**  $\in \{100, 300, 500\}$  (winning: 300 for macro, 500 for micro).
- **Learning Rate ( $\eta$ ):**  $\in \{0.03, 0.06, 0.10\}$  (winning value: 0.06).
- **Subsample / Colsample:** 0.80 / 0.80.
- **Positive Weight Multiplier:** Scaled inverse frequency with a hard cap of 10.0.

## Appendix B

### Repository Map and Notebook Lineage

**Table B.1:** Primary project artifacts, notebooks, and source code mapping

| Directory / Artifact  | File Name                         | Architectural Scope / Contents                            |
|-----------------------|-----------------------------------|-----------------------------------------------------------|
| Intention/Classifier/ | v1_flat_baseline.ipynb            | Flat 256-token transformer baseline with dual heads.      |
|                       | v2_speaker_aware.ipynb            | Additive role embeddings prior to masked pooling.         |
|                       | v3_augmentation_collapse.ipynb    | EDA augmentation run diagnosing metric and weight bugs.   |
|                       | v4_loss_metric_fix.ipynb          | True Macro-F1 metric selection, CB-Focal, and SupCon.     |
|                       | v5_token_safe_aug.ipynb           | Forensic regex tokenization fixing speaker marker gluing. |
|                       | v6_hierarchical_dialogue.ipynb    | Hierarchical transformer with inter-turn self-attention.  |
|                       | v8_hierarchical_bt.ipynb          | Hierarchical transformer with back-translation.           |
|                       | v9_ensemble_dropout.ipynb         | Tri-model soft ensembling and Monte Carlo uncertainty.    |
|                       | v10_last_turn_gated.ipynb         | Last-Persuadee-turn sigmoid gating architecture.          |
|                       | v13_solo_binary.ipynb             | Modifier head removal and dedicated binary capacity.      |
|                       | v14_antiphony.ipynb               | Dual-stream cross-attention Antiphony architecture.       |
| Strategy/             | multistrat-capstone2-final.ipynb  | 39-strategy baseline with decoupled training tests.       |
|                       | multistrat-capstone_v17.ipynb     | 19-member pool, Asymmetric Loss, calibrated blends.       |
|                       | multistrat-capstone_v18.ipynb     | 72-grid XGBoost stacker search vs. power-mean blend.      |
|                       | pipeline.py                       | Automated schema validation for strategy batches.         |
| Generic/              | generate_negotiation_dialogues.py | Persona-conditioned synthetic generation engine.          |
|                       | humanize_dialogues_gemini.py      | Gemini Flash organic paraphrasing and jitter script.      |
|                       | run_adversarial_ablation.py       | Input ablation harness (LTR, LTR+M, FHO, PRO).            |

## Appendix C

### Data Contracts and Schemas

---

#### Multi-Label Strategy Record Schema (multilabel\_merged.jsonl)

---

Below is an illustrative JSON record from `multilabel_merged.jsonl` representing Turn #42:

```
1 {
2   "turn_id": 42,
3   "dialogue_id": 4,
4   "speaker": "Persuader",
5   "text": "Hello! I am doing well, thank you. Did you know that Save the Children has
6     helped over 100 million children worldwide?",
7   "context_turns": [
8     {"speaker": "Persuader", "text": "Hi, how are you doing today?"},
9     {"speaker": "Persuadee", "text": "I'm good, how about you?"},
10  ],
11  "strategies": [
12    "rapport_building",
13    "evidence_and_statistics",
14    "organization_information"
15  ],
16  "categories": [
17    "Relational and Interactive",
18    "Rational Appeal",
19    "Information Provision"
20  ],
21  "n_labels": 3,
22  "flags": {"empty_set": false, "uncertain": false},
23  "annotator_segment": "segment_track_a"
24 }
```

#### Adversarial Negotiation Reversal Record Schema (REV\_042)

---

Below is an illustrative record from `final_plus_reversal_v3.json` showing a verified reversal dialogue:

```
1 {
2   "dialogue_id": "REV_042",
3   "regime": "ADVERSARIAL_REVERSAL",
4   "reversal_type": "resistance_to_agreement",
5   "product": "Sony WH-1000XM5 Wireless Headphones",
6   "archetype": "The Skeptic",
7   "reversal_turn_id": 9,
8   "labels": {
9     "buying_intent": "yes",
10    "outcome_category": "PURCHASE_COMMITTED",
11    "reason": "Customer expressed intense counterfeit skepticism, but committed to purchase
12      immediately after merchant provided official distributor warranty serial and
13      offered Cash on Delivery with open-box verification."
14  },
15  "turn_count": 12
16 }
```

## Appendix D

### Conflict Register and Empirical Audit Log

---

To ensure complete scientific integrity, this appendix documents the resolution of all fifteen empirical and textual discrepancies identified in the project record:

Whenever prompt narratives, retrospective slides, or draft summaries contradict raw serialized artifacts (.jsonl, .csv, evaluation outputs, or reproducible notebook execution cells), the empirical data artifacts are held strictly authoritative. Speculative claims and draft misstatements are formally audited and corrected.

**Table D.1:** Project conflict register, discrepancies, and resolutions

| ID  | Topic                    |         | Contradiction in Source Record                                                                                            | Resolution in this Report                                                                                                                              |
|-----|--------------------------|---------|---------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| C1  | Strategy Decision Layer  |         | Early drafts incorrectly referred to an XG-Boost stacking system as the winner.                                           | Shipped decision layer is the <b>calibrated power-mean blend</b> (0.5716 macro-F1); XGBoost stacker is documented as tested (0.5508).                  |
| C2  | Context Window Width     |         | Prompts claimed $k \in \{0, 1, 3, 5, 7\}$ was tested.                                                                     | Verified tests compare 0 vs 3 turns (0.6675 vs 0.6756) and BoW 0 vs 5 turns; 0.7062 is v17’s micro-optimal decode.                                     |
| C3  | Strategy Prevalence Tail |         | Annotation doc had erroneous tail counts (e.g. <code>door_in_the_face</code> 15).                                         | Recomputed directly from <code>multilabel_merged.jsonl</code> (2, 5, 5, 8, 7, 14, 21, 27, 31).                                                         |
| C4  | Strategy Annotator ID    |         | Conflicting references to Claude models and manual raters.                                                                | Described as LLM agent under human checkpoints across four named segments; marked with <b>[PENDING: ...]</b> .                                         |
| C5  | E-Commerce Counts        | Dataset | Text prompts claimed 250/250 and 6 outcome classes.                                                                       | Frozen files contain 262 no / 238 yes ( $N = 500$ ) and 362 no / 338 yes ( $N = 700$ ) across 4 outcome classes.                                       |
| C6  | Ablation Terminology     |         | Prompts cited non-existent regimes ( <code>‘Suffix’</code> , <code>‘+0.1420’</code> ).                                    | Followed report: LTR, LTR+M, FHO, PRO; necessity gap dropped from 0.1600 to 0.0402.                                                                    |
| C7  | Stage Boundaries         |         | Conflicting epoch definitions across prompts.                                                                             | Standardized to 4 stages: Stage 1 (Flat Baselines), Stage 2 (Hierarchical Transformers), Stage 3 (Gated & Decoupled), Stage 4 (Dual-Stream Antiphony). |
| C8  | Table B Thresholds       |         | Mixed default (0.8623) and tuned (0.8600) scores.                                                                         | Reported both numbers transparently; noted that Antiphony beats the Decoupled Baseline under either rule.                                              |
| C9  | Unsupported Claims       |         | Inventions: unsubstantiated monetary budgets, inflated emissions, external serving frameworks, and graph runtime exports. | Deleted all unsubstantiated claims; replaced with grounded hardware estimates.                                                                         |
| C10 | Left-Truncated Flat Run  |         | Presentation cited 0.842 flat run not in chronicle.                                                                       | Noted the presentation finding; discussed token length diagnosis without building claims on it.                                                        |
| C11 | Strategy Encoders        |         | Abstract claimed DeBERTa/TOD-BERT on strategy.                                                                            | Clarified that strategy used RoBERTa (x6) and ELECTRA (x1); 3-backbone test applied to intention.                                                      |
| C12 | Directive Issuer         |         | Referred to “supervisor” vs “course instructor”.                                                                          | Standardized to “supervisor (course instructor)” and marked <b>[PENDING: Confirm directive issuer]</b> .                                               |
| C13 | Pass Numbering           |         | Guidelines and reports used divergent pass labels.                                                                        | Followed <code>donation_intent.md</code> : Pass 1 (dual), Pass 2 (binary), Pass 3 (strict).                                                            |
| C14 | Agreement Scope          |         | 92.21% cited as a 5-annotator study.                                                                                      | Clarified: 92.21% represents independent second-pass on 154 dialogues.                                                                                 |
| C15 | HiTrans Attribution      |         | Chronicle cited “Le et al., EMNLP 2021”.                                                                                  | Corrected to Li et al., COLING 2020 [20].                                                                                                              |

## Appendix E

### Complete 41-Strategy Empirical Prevalence Table

Table E.1 reproduces the data-verified prevalence of all 41 strategies across the 10,600 Persuader turns of `multilabel_merged.jsonl`, ordered from highest to lowest frequency.

**Table E.1:** Data-verified prevalence of all 41 persuasion strategies (10,600 turns)

| Strategy Identifier                          | Parent Category            | Active Turns | Prevalence (%) |
|----------------------------------------------|----------------------------|--------------|----------------|
| <code>rapport_building</code>                | Relational and Interactive | 2,860        | 27.0%          |
| <code>organization_information</code>        | Information Provision      | 1,735        | 16.4%          |
| <code>donation_procedure_information</code>  | Information Provision      | 1,417        | 13.4%          |
| <code>personal_related_inquiry</code>        | Relational and Interactive | 1,328        | 12.5%          |
| <code>personal_story</code>                  | Relational and Interactive | 1,150        | 10.8%          |
| <code>emotion_appeal</code>                  | Emotional Appeal           | 1,023        | 9.7%           |
| <code>gratitude_and_appreciation</code>      | Reciprocity and Exchange   | 966          | 9.1%           |
| <code>minimization_framing</code>            | Framing and Presentation   | 849          | 8.0%           |
| <code>task_clarification</code>              | Information Provision      | 770          | 7.3%           |
| <code>hope_and_positive_impact</code>        | Emotional Appeal           | 628          | 5.9%           |
| <code>self_modeling</code>                   | Social Influence           | 619          | 5.8%           |
| <code>source_related_inquiry</code>          | Relational and Interactive | 615          | 5.8%           |
| <code>in_group_appeal</code>                 | Social Influence           | 598          | 5.6%           |
| <code>impact_information</code>              | Information Provision      | 477          | 4.5%           |
| <code>organizational_credibility</code>      | Credibility Appeal         | 389          | 3.7%           |
| <code>evidence_and_statistics</code>         | Rational Appeal            | 378          | 3.6%           |
| <code>logical_appeal</code>                  | Rational Appeal            | 304          | 2.9%           |
| <code>transparency_and_accountability</code> | Credibility Appeal         | 274          | 2.6%           |
| <code>source_citation</code>                 | Credibility Appeal         | 193          | 1.8%           |
| <code>persistent_repetition</code>           | Threat and Pressure        | 167          | 1.6%           |
| <code>urgency_appeal</code>                  | Urgency and Scarcity       | 150          | 1.4%           |
| <code>donor_benefit</code>                   | Reciprocity and Exchange   | 109          | 1.0%           |
| <code>anchoring</code>                       | Framing and Presentation   | 94           | 0.9%           |
| <code>value_consistency_appeal</code>        | Commitment and Consistency | 79           | 0.7%           |
| <code>personal_credibility</code>            | Credibility Appeal         | 78           | 0.7%           |
| <code>obligation_pressure</code>             | Threat and Pressure        | 78           | 0.7%           |
| <code>comparison_framing</code>              | Framing and Presentation   | 74           | 0.7%           |
| <code>social_proof</code>                    | Social Influence           | 67           | 0.6%           |
| <code>empathy_and_perspective_taking</code>  | Emotional Appeal           | 66           | 0.6%           |
| <code>feasibility_and_ease</code>            | Rational Appeal            | 55           | 0.5%           |
| <code>foot_in_the_door</code>                | Commitment and Consistency | 53           | 0.5%           |
| <code>incremental_ask</code>                 | Commitment and Consistency | 35           | 0.3%           |
| <code>guilt_induction</code>                 | Emotional Appeal           | 31           | 0.3%           |
| <code>cost_benefit_framing</code>            | Rational Appeal            | 27           | 0.3%           |
| <code>loss_versus_gain_framing</code>        | Framing and Presentation   | 21           | 0.2%           |

| Strategy Identifier          | Parent Category            | Active Turns | Prevalence (%) |
|------------------------------|----------------------------|--------------|----------------|
| reciprocity                  | Reciprocity and Exchange   | 14           | 0.1%           |
| scarcity_appeal              | Urgency and Scarcity       | 8            | 0.1%           |
| negative_consequence_warning | Threat and Pressure        | 7            | 0.1%           |
| authority_endorsement        | Social Influence           | 5            | 0.0%           |
| deadline_pressure            | Urgency and Scarcity       | 5            | 0.0%           |
| door_in_the_face             | Commitment and Consistency | 2            | 0.0%           |